

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #11

March 30 – April 3, 2026

Instructor: J. Michael Ruohoniemi

Topics (1/2)

- Digital signal transmission
- Baseband data link
- Bandwidth factors
- Inter-Symbol Interference (ISI)
- Zero ISI waveform / Design principles
- Nyquist Transfer Function
- RRC filters

Topics (2/2)

- Digital radio link (RF)
- Symbol rate and Bit rate
- Baseband link - Equalization
- RF Link - Equalization
- Implementing Zero ISI designs
- Course project update
- Digital phase modulation: BPSK and QPSK

Artemis II Update – Launch schedule for 6:24 PM ET, April 1



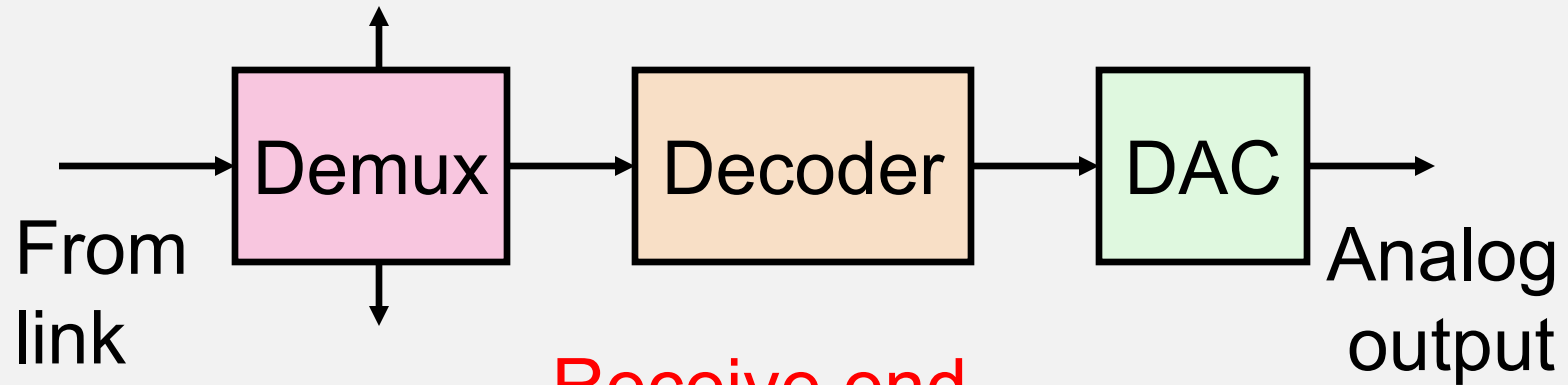
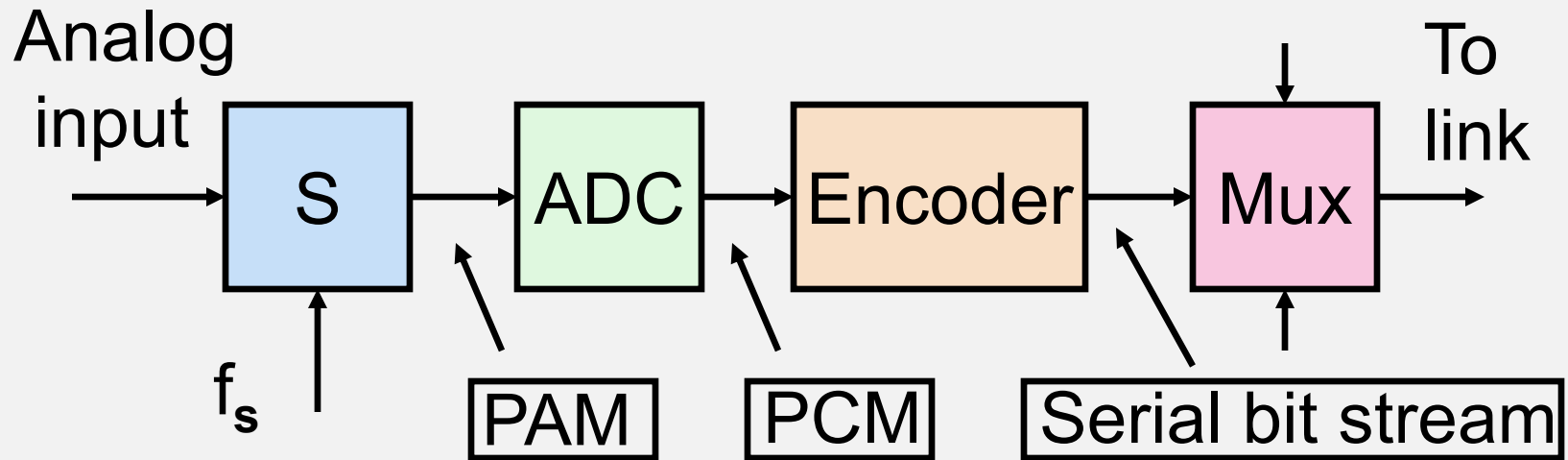
NASA's Artemis Space Launch System (SLS) rocket and Orion spacecraft on a mobile launcher at Launch Complex 39B

Ten day mission:
April 1-2: Testing in orbit
April 6: Lunar flyby
April 10:
Splashdown

Analog to Digital Conversion - Review

- First step in digital transmission of analog signal is sampling at a high enough sampling rate
- Next step in is to create a digital word from the samples
- Number of bits in the word (N) is important
- Digital words have discrete values
- The word is a *quantized* version of sample

Transmit end

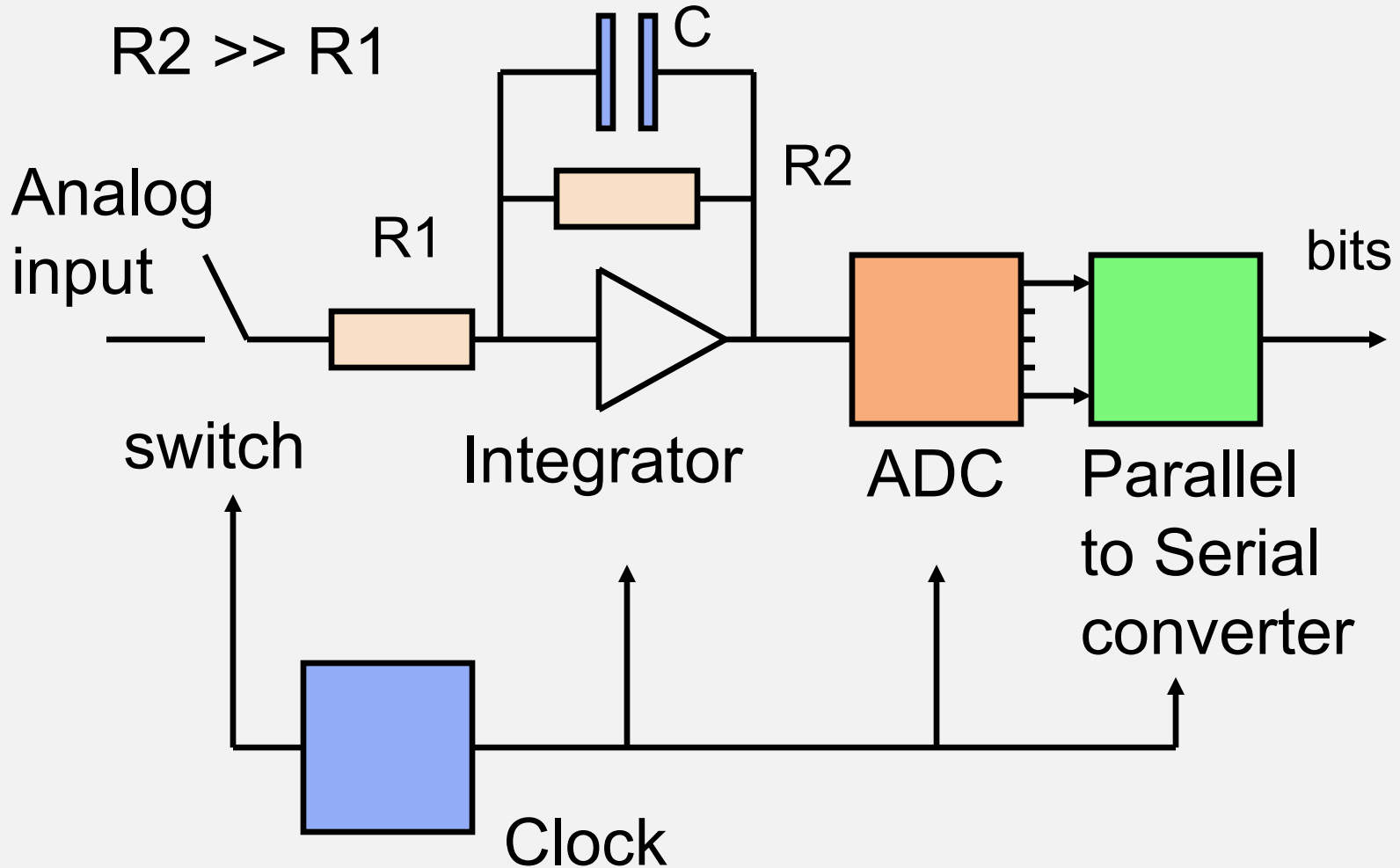


Receive end

Analog to Digital Conversion

- Samples are made with *Sample and Hold* circuit
- Fast charge / slow discharge integrator
- ADC is a specialized circuit that outputs a digital word corresponding to sample value
- ADCs are characterized by speed and number of bits

Sample and Hold and ADC



Bit Rates

- Suppose we digitize to N bits per sample
- The bit rate at the output of the parallel to serial converter is r_b where

$$r_b = f_s \times N$$

- Bit rate sets the bandwidth required of the communication channel
- More bits gives better S/N ratio but requires more bandwidth

Bit Rates

- Example #1:
- Sample rate for telephony is 8 kHz

$$\text{Bits/sample} = N = 8$$

$$\text{Transmission rate} = f_s \times N = 64 \text{ kbps}$$

- International standard of all telephone traffic

Bit Rates

- Example #2

Sample rate for CD recording = 44 kHz

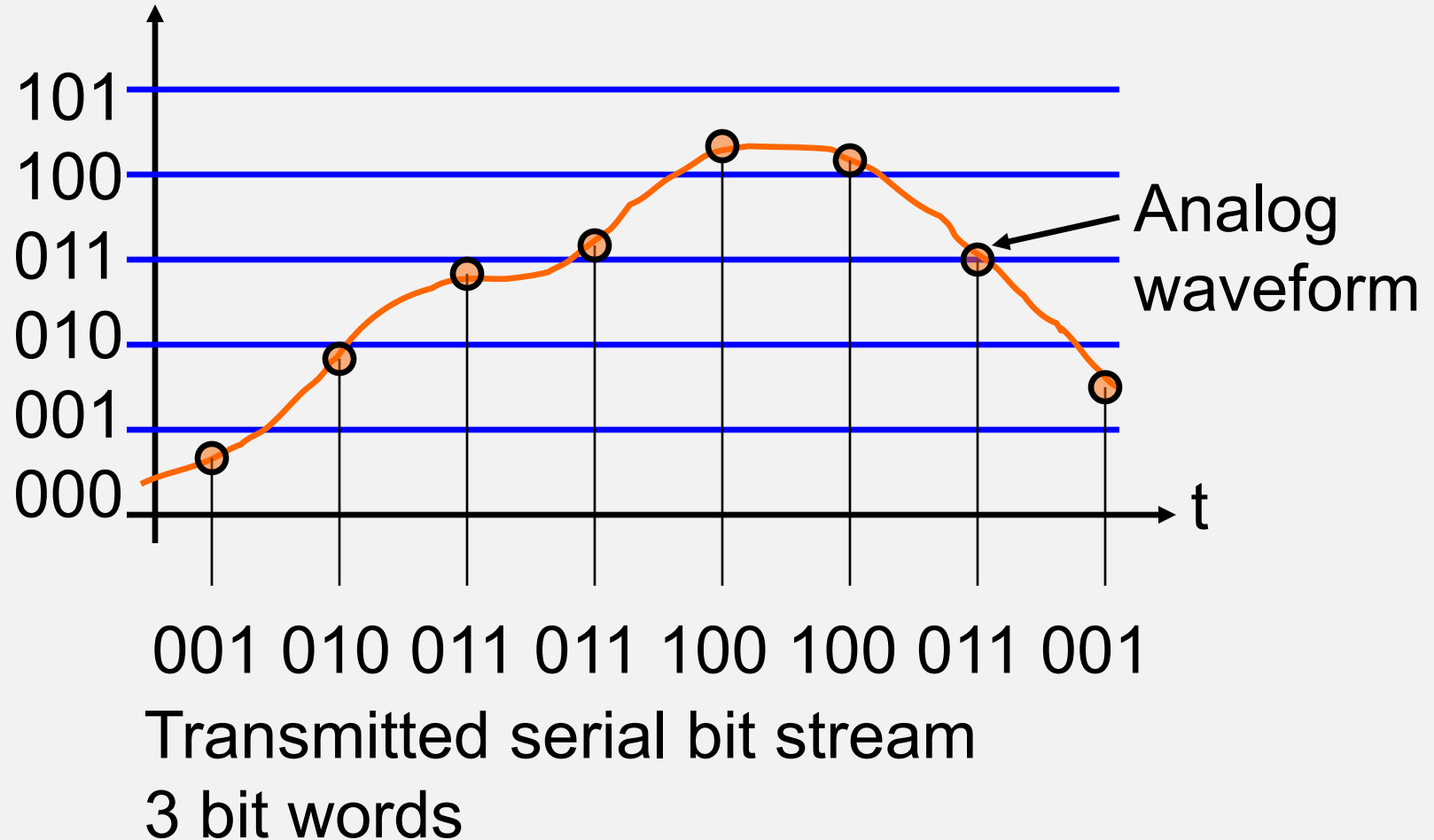
$N = 16$ bits, $r_b = 704$ kbps / channel

- CD records two channels (stereo) in encoded form
- *Message data rate* is 1408 kbps
- Actual bit rate of recording is ~ 3.6 Mbps
- Difference is in coding, packetizing, etc

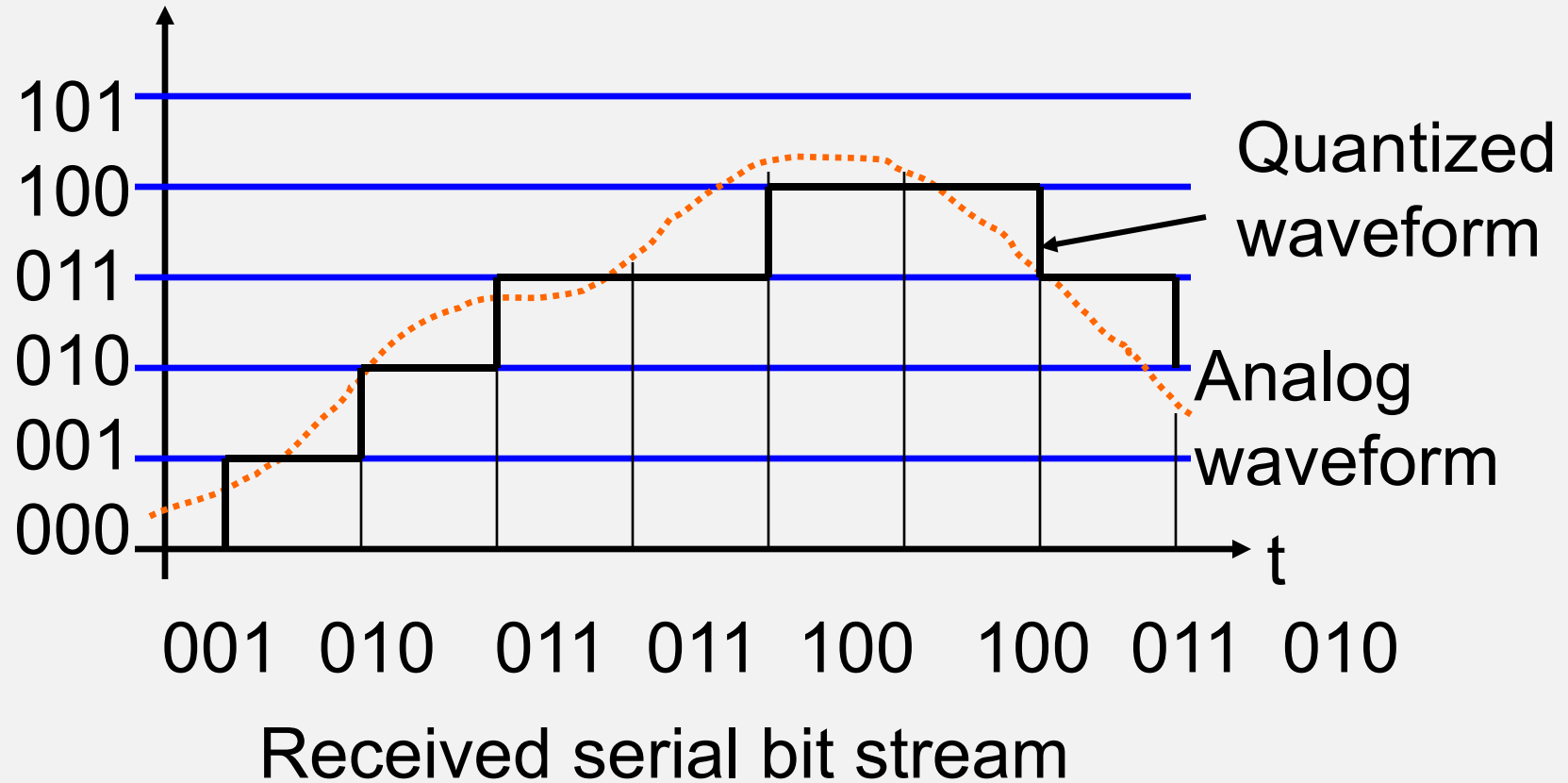
Quantization

- The ADC converts a voltage sample to a digital word
- Each digital word corresponds to a specific voltage level at the analog input to the ADC
- There are $2^N - 1$ voltage levels
- The ADC outputs the word that has a voltage level closest to the input signal
- This process is called quantization

Six level quantization



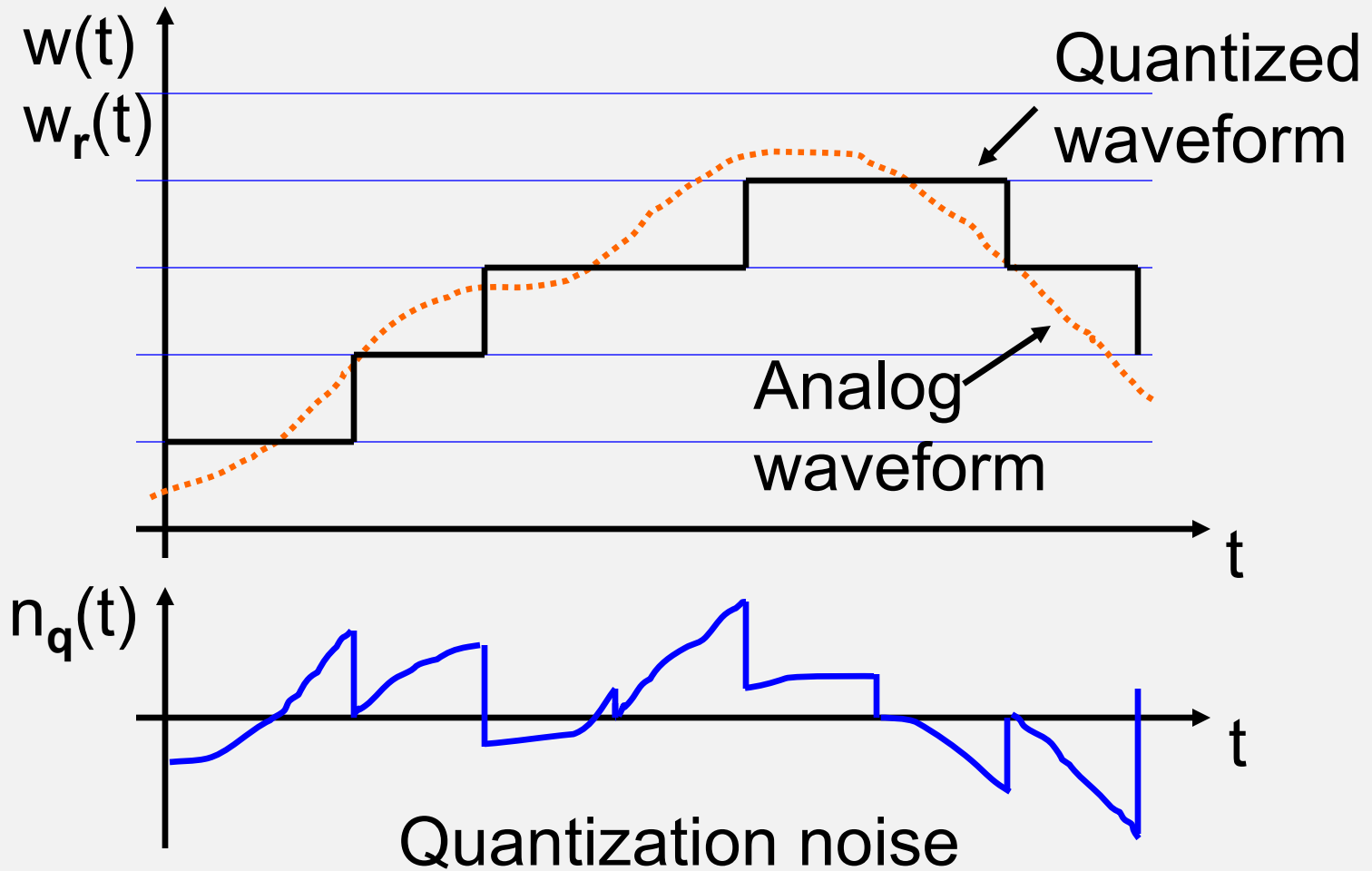
Receiver waveform with 3 bit words



Quantization Noise

- The quantized waveform is different from the analog waveform
- There is an error because the quantized waveform, $w_r(t)$, is not the same as the original signal $w(t)$
- We describe the error as $n_q(t)$ where
$$w_r(t) = w(t) + n_q(t)$$
- $n_q(t)$ is called *quantization noise*

Quantization noise with 3 bit words



Quantization Noise

- Quantization noise adds to the recovered analog signal and generates noise in the baseband signal (what is heard, seen)
- For uniform quantization and no bias in voltage level, baseband signal to (quantization) noise ratio is

$$(S/N)_Q \approx 6 N \text{ dB}$$

- E.g., for linear quantization with 8 bits

$$(S/N)_Q \approx 6 \times 8 = 48 \text{ dB}$$

Summary: Analog to Digital Conversion

- Many sources of signal are inherently analog - digital communication techniques require an ADC followed by a DAC
- PCM entails digital encoding of signals with quantization
- High bits rates from digitization require large bandwidths
- Quantization noise generates noise in the baseband signal

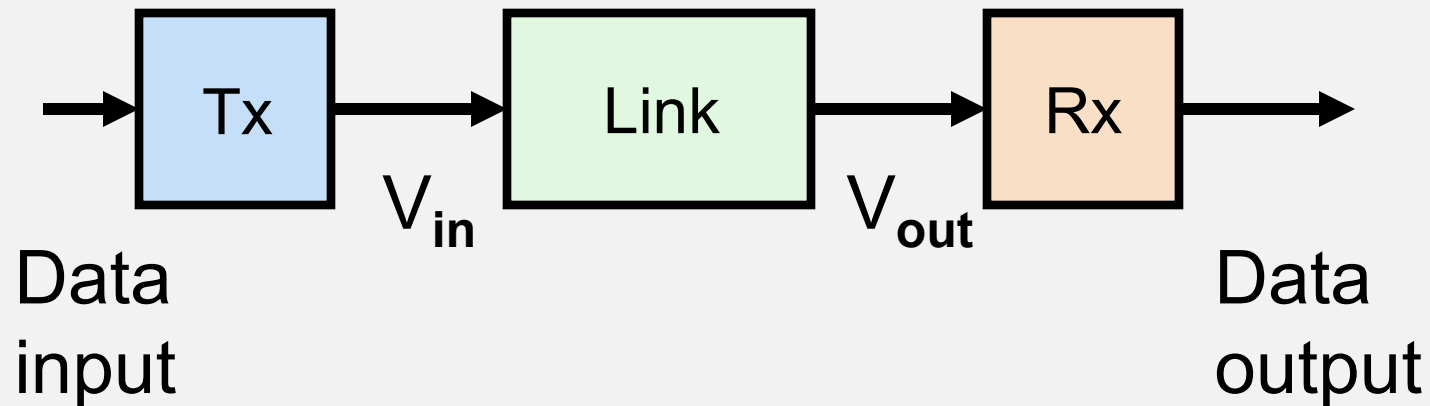
Digital Transmission

- We assume the baseband is digital or has been converted from analog to digital
- The bandwidth of a communications link must always be limited
- The Question:

What happens when we send digital data over a communications link with limited bandwidth?

Generic Digital Transmission System

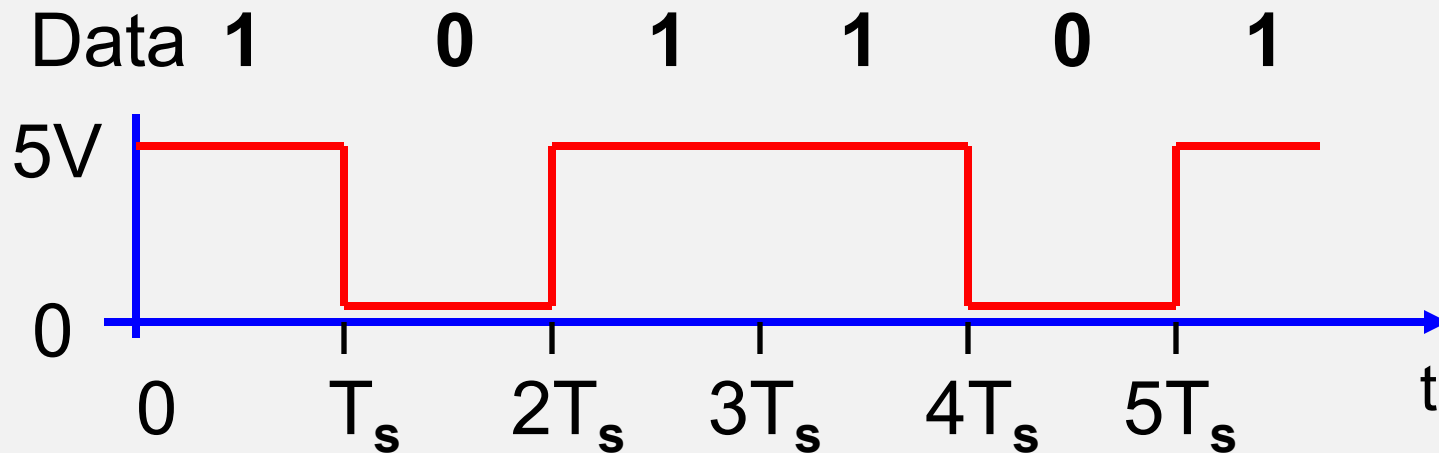
- Block diagram



Analyze as a *baseband data link* first, i.e., no carrier wave – think pulses carried by a wire

Digital Signal Transmission

- Digital signals consist of pulses representing the binary states logical **ones** and logical **zeros**
- Digital signals are square pulses with voltage levels of 0 V and 5 V (or 3.3 V)

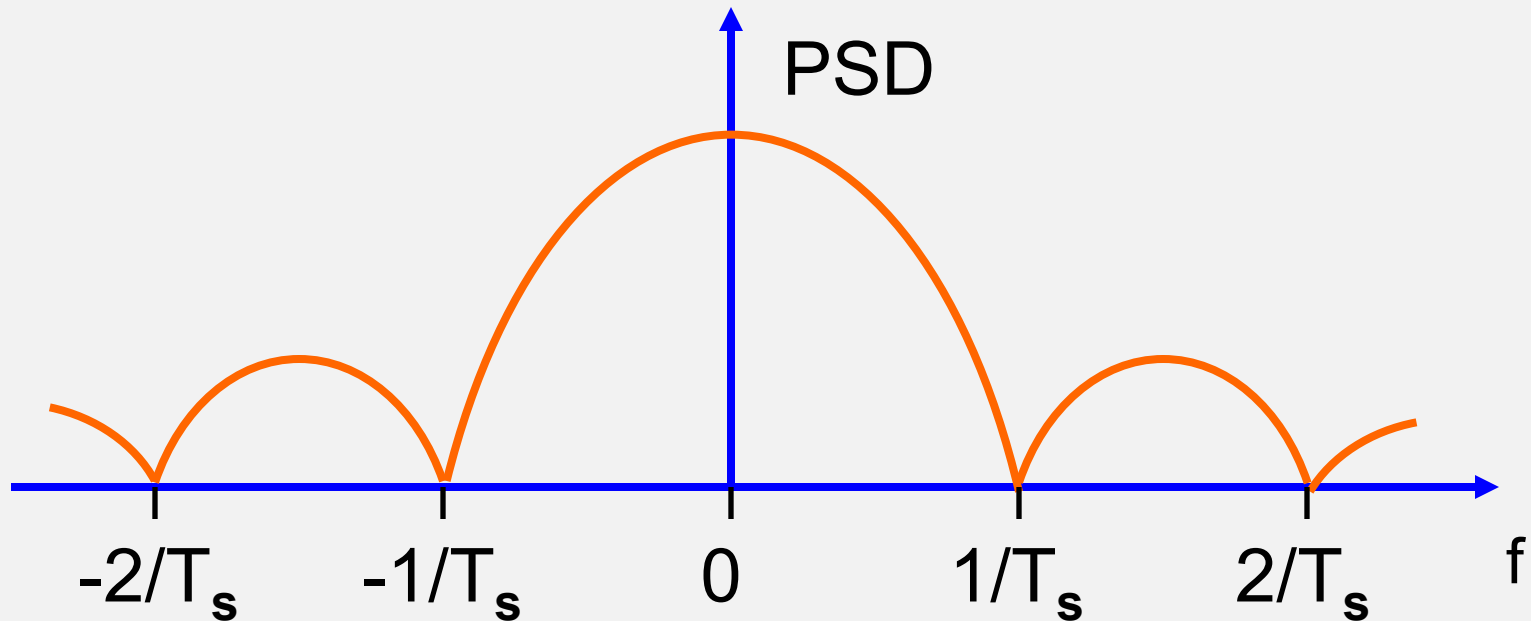


Digital Signal Transmission

- With a baseband data link, data are transmitted in the form of polar pulses (with no modulation onto a carrier wave)
- The duration of a pulse is the *symbol period*, T_s
- To understand the bandwidth issue, we Fourier transform from the time domain to the frequency domain

Digital Signal Transmission

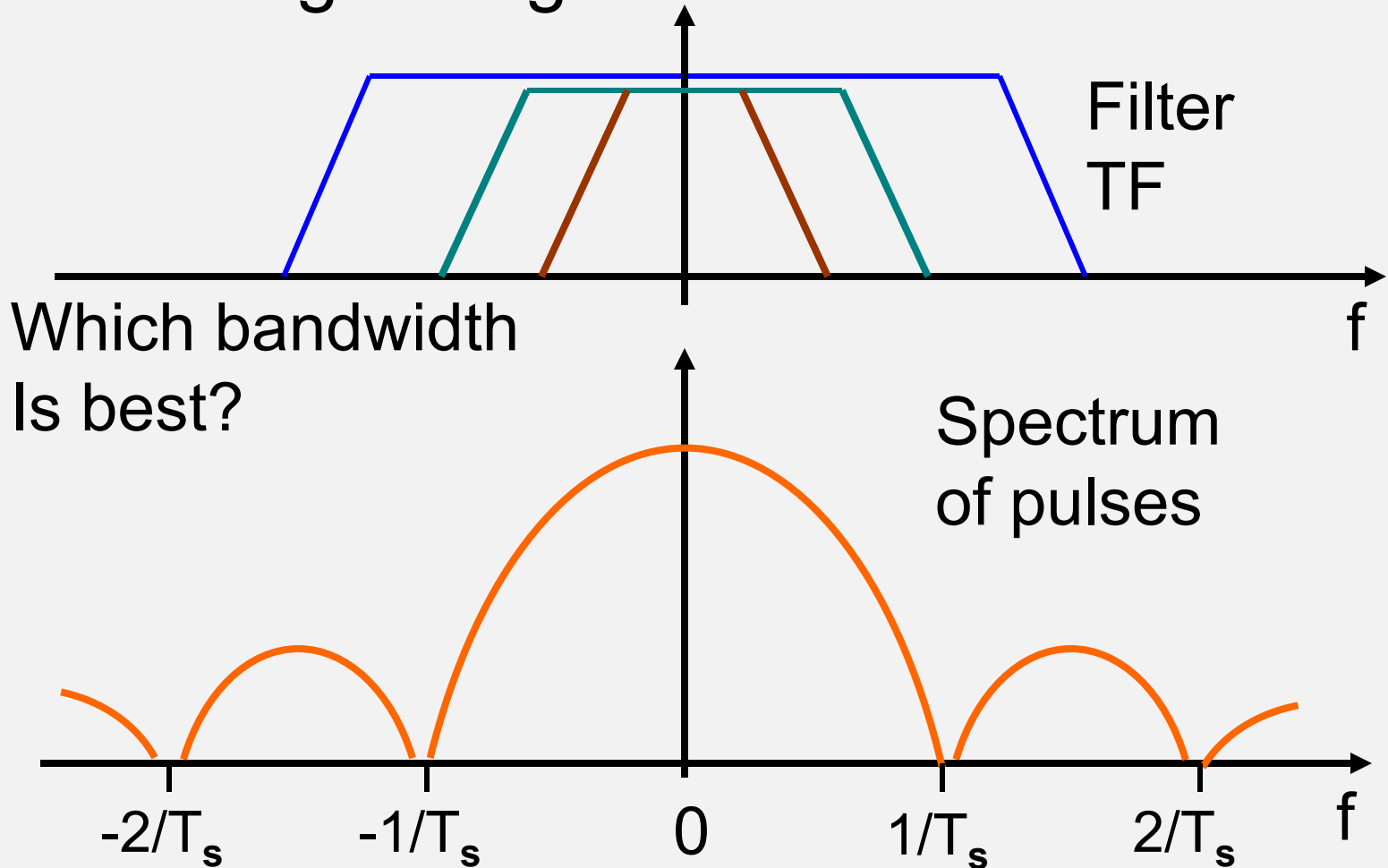
- Spectrum of train of rectangular pulses has $\sin X / X$ shape, infinite bandwidth with $X = \pi f T_s$



Bandwidth

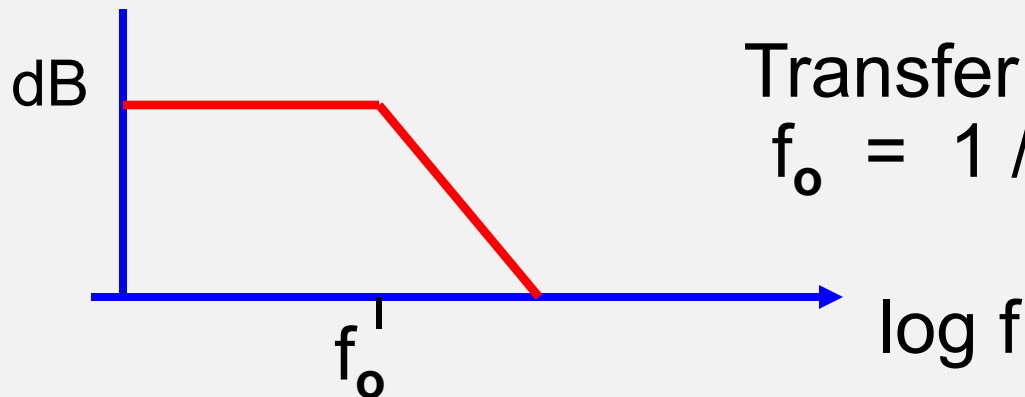
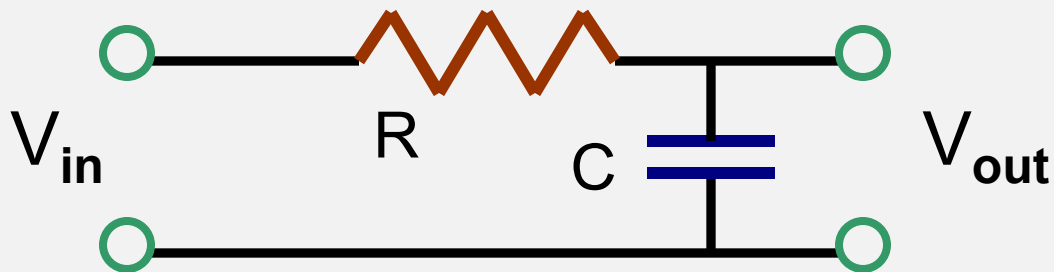
- $\sin X / X$ spectrum has infinite spectral width
- Requires infinite bandwidth to retain pulse shape
- Links must operate within finite bandwidth (as small as possible)
- Therefore, we cannot send / receive rectangular pulses
- We will limit the bandwidth by passing the rectangular pulse through a low pass filter

Digital Signal Transmission



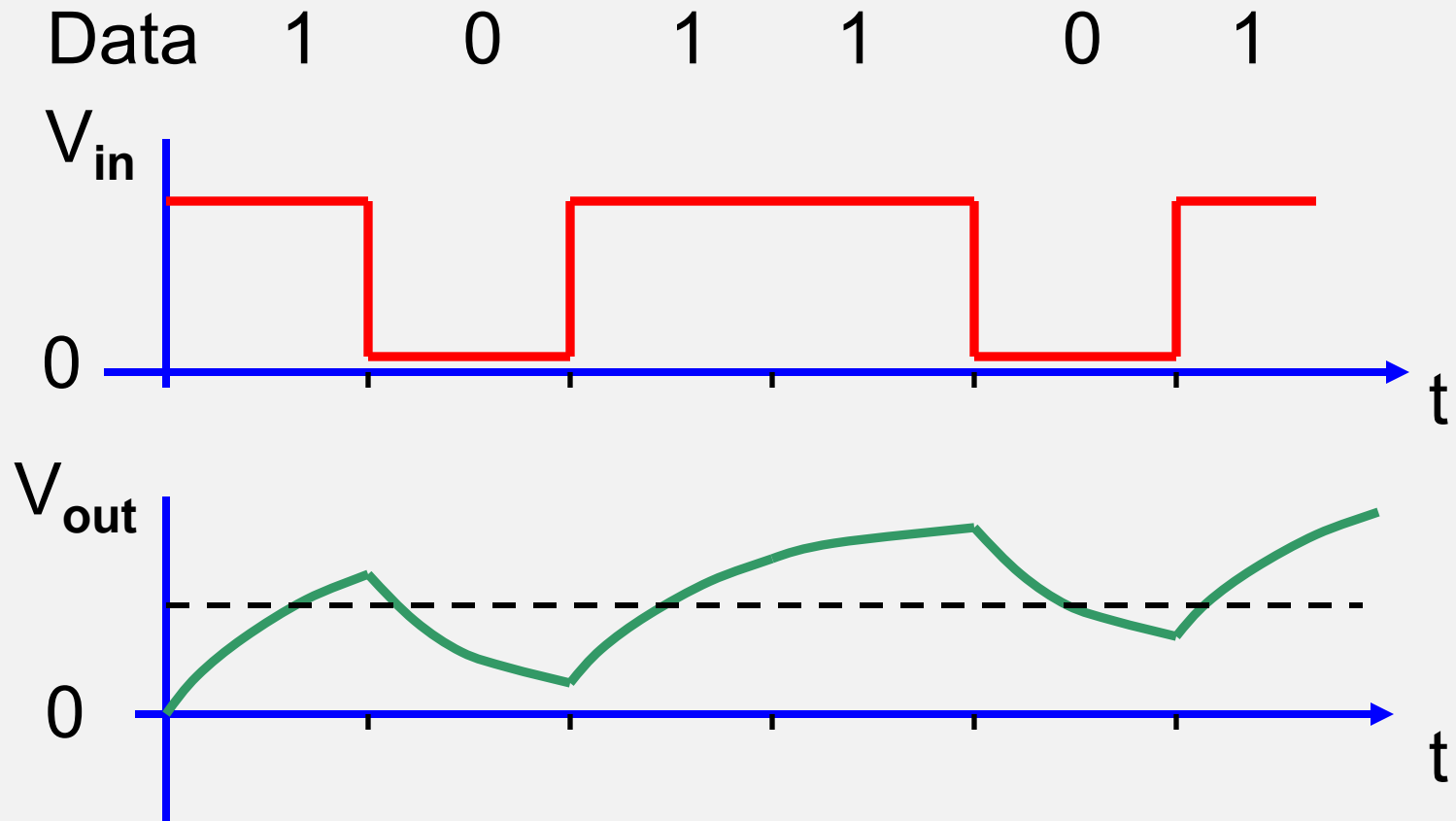
Band-limited channels

- Model transmission link as simple low pass R-C section



Transfer function
 $f_o = 1 / 2\pi RC$

Bandlimited channel waveforms



RC time constant effect – capacitor charges and discharges

Band limiting with a low pass filter

- The logical ones and zeros of baseband are carried by rectangular pulses
- We apply a band-limiting filter
- After passing through the filter:
 - the pulse shape is altered
 - the resulting waveform is delayed
- Each decaying pulse 'bleeds' into the next and distorts the signal level

Transmission over Bandlimited Channel

- First issue: Output of low pass channel has DC bias
- We must send signal with average (DC) level of 0 volts
- Consider having the rectangular pulses switch between levels $+V$ and $-V$
- Patterns of voltage or current used to represent digital data are called *Line Codes*

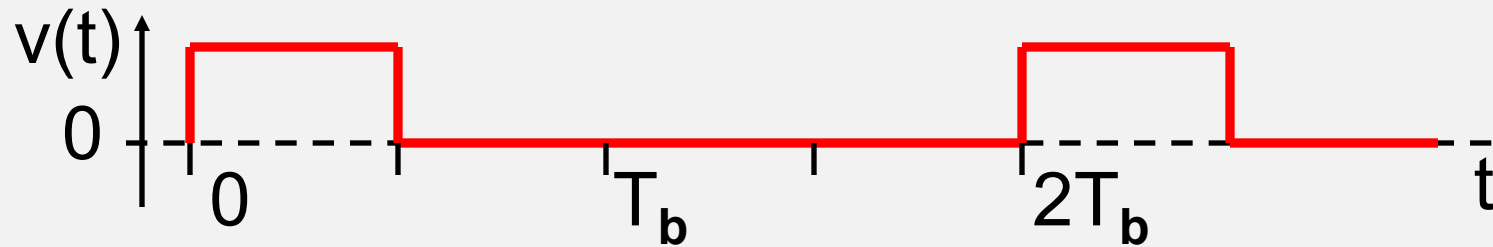
Unipolar, Polar, Return to Zero (RZ)

Data 1

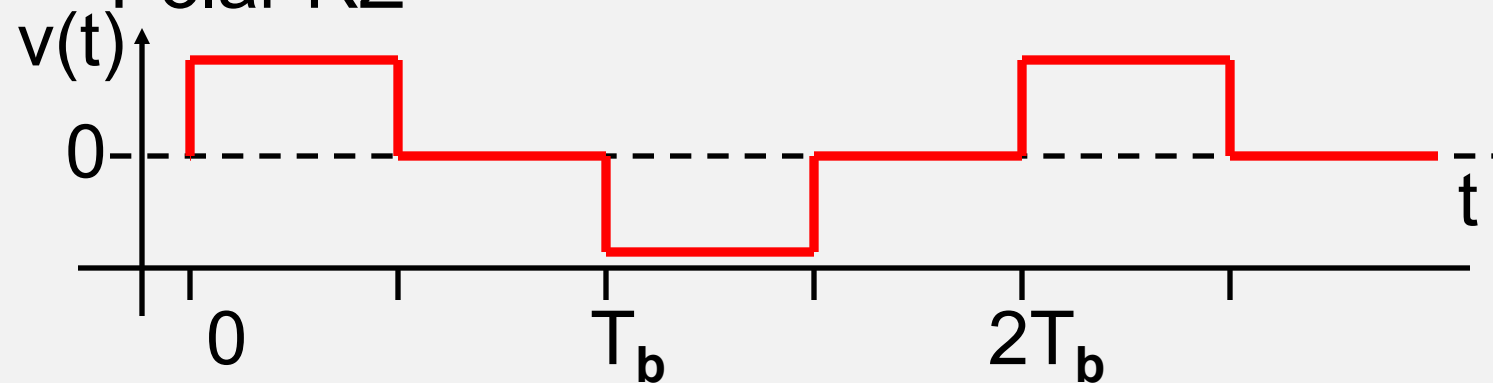
0

1

- Unipolar RZ



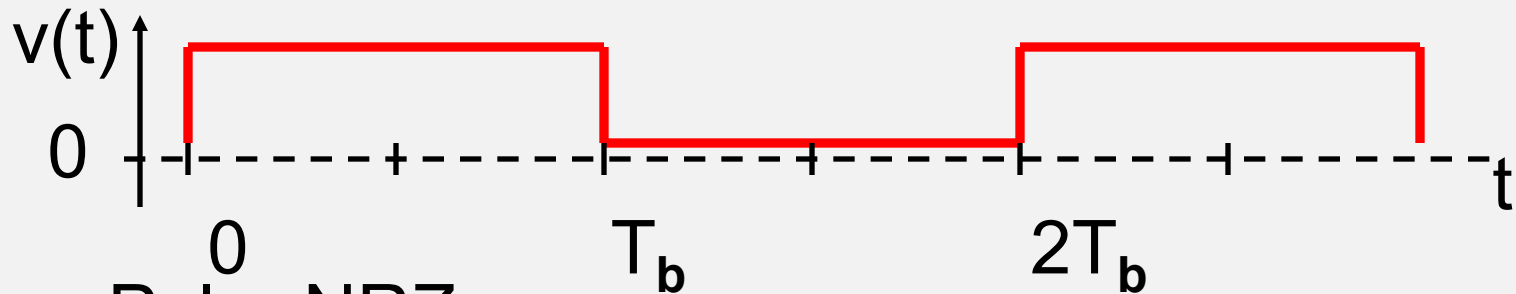
- Polar RZ



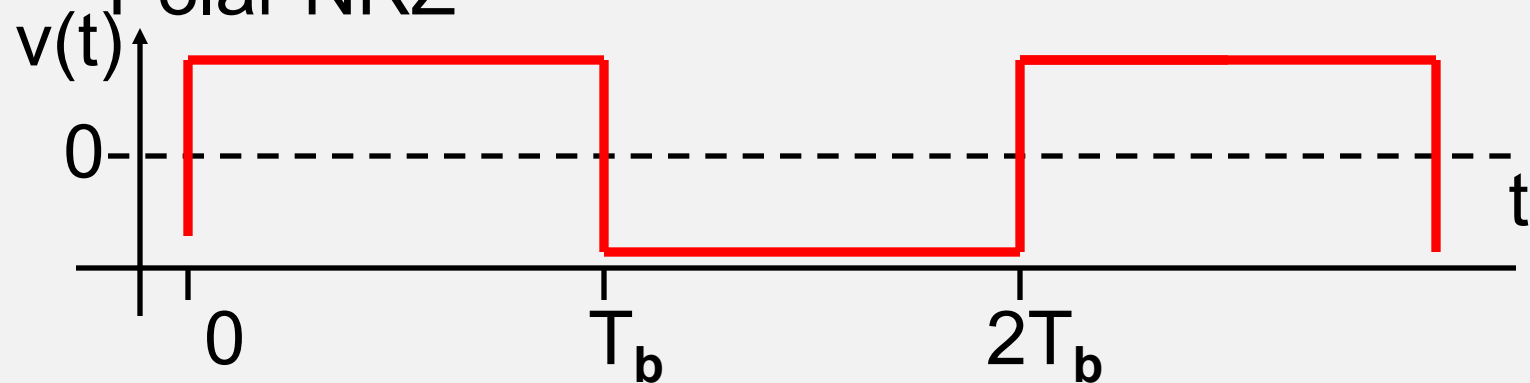
Unipolar, Polar, Non-Return to Zero (NRZ)

Data 1 0 1

- Unipolar NRZ



- Polar NRZ

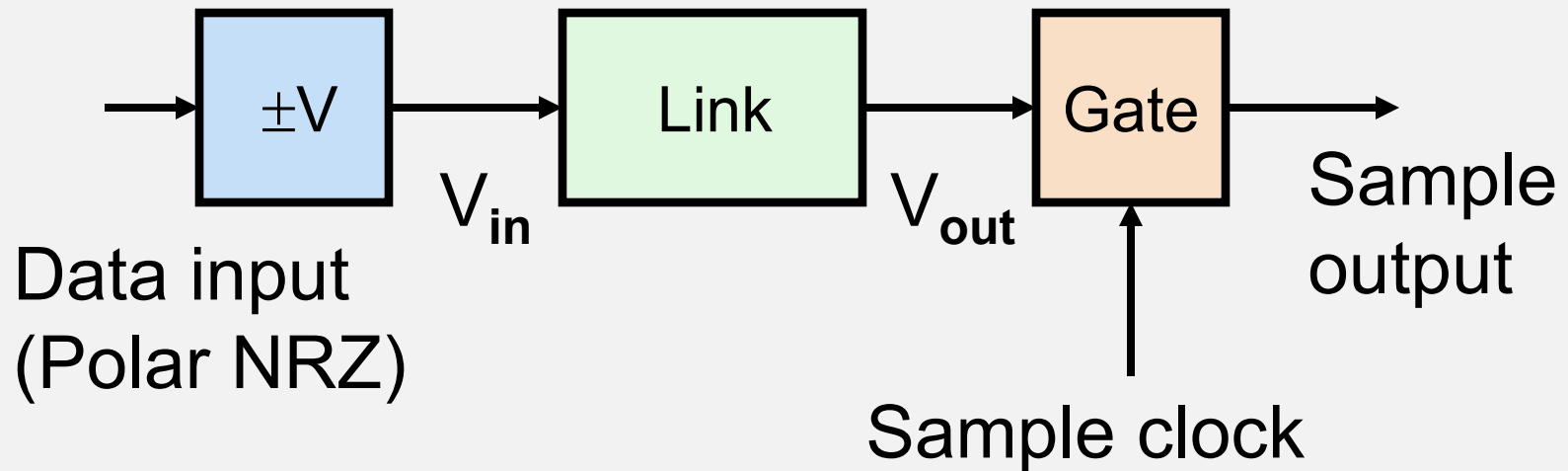


Line Codes

- Radio transmissions always use *NRZ* format and polar pulses: $1 = +V$, $0 = -V$
- NRZ = Non-return to zero
- Make sure we transmit equal number of ones and zeroes
- Then NRZ polar has no DC bias

Digital Transmission System

- Block diagram with sampling on Rx

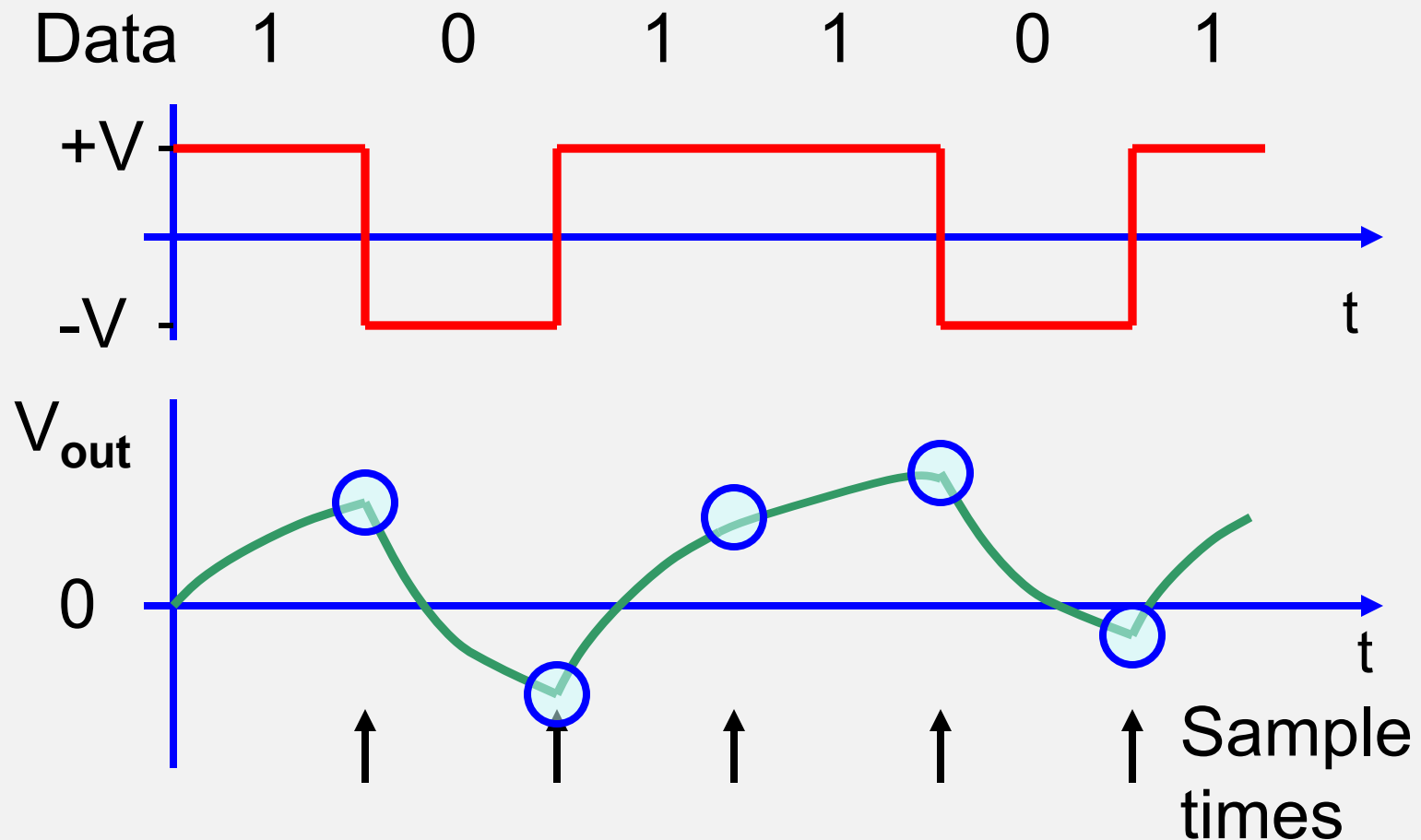


Baseband data link

Sampling

- We will sample the output of the link once per pulse period
- We sample at the end of each pulse period
- How well can we infer the voltage levels of the original pulses?
- Signal distortion might cause errors
- Noise will cause more ‘bad’ reads and more errors

Bandlimited Channel Transmission

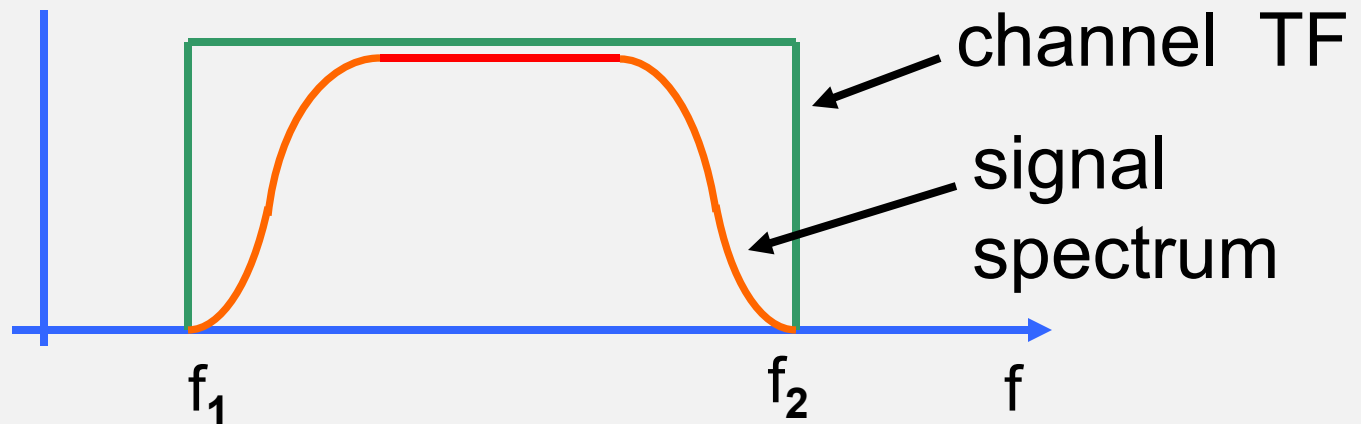


Inter-symbol Interference: ISI

- Sending pulses through a band-limited channel smears the pulses in time leading to waveform distortion
- Part of one pulse trails into the next pulse period causing *Inter-Symbol Interference* (ISI)
- ISI reduces the amplitude of the wanted pulse at the sample time and thus reduces noise immunity
- Need a way to send pulses without creating ISI

Inter-symbol Interference: ISI

- We could eliminate ISI by matching the spectrum of the pulse to the bandwidth of the channel
- If all energy in the spectrum is within the transmission bandwidth, pulse will not be distorted
- Ideally

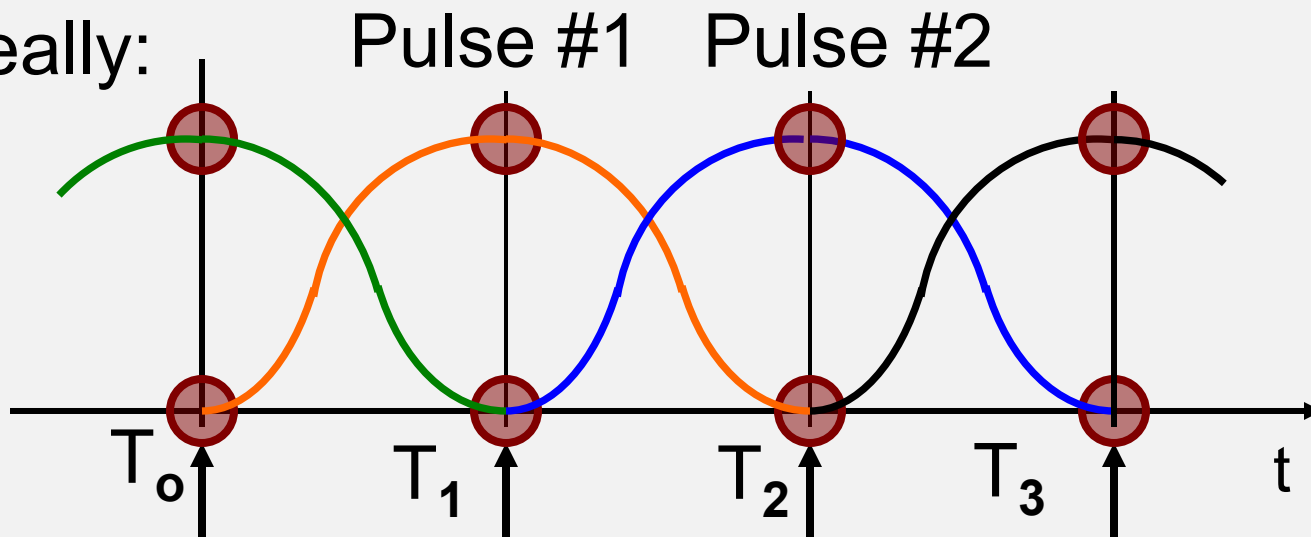


Inter-symbol Interference: ISI

- What we want *cannot* be achieved
- Waveforms cannot be both time *and* frequency limited
- If the waveform is finite in time then the bandwidth is infinite
- We can only approximate the situation of a finite waveform with a finite bandwidth
- However, we *can* design a pulse waveform that produces zero ISI

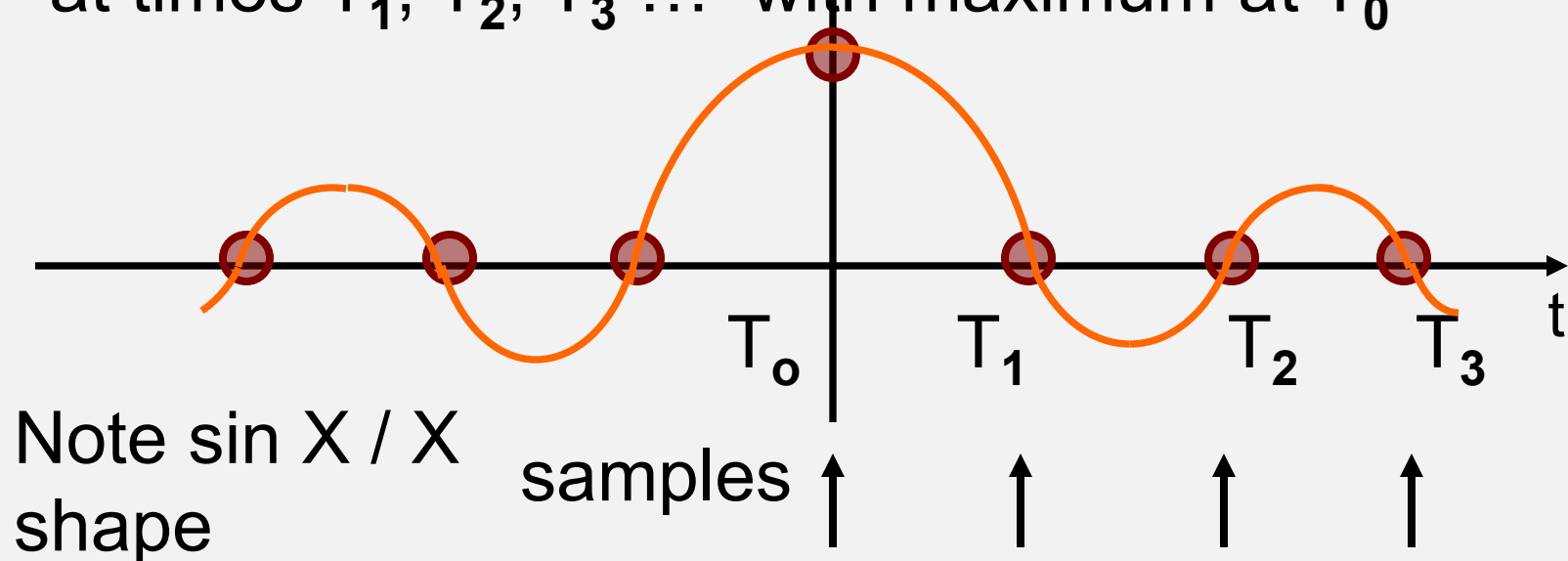
Inter-symbol Interference: ISI

- We want the value of each pulse at the receiver to be zero at every sample time *except* for the value of the target pulse
- Ideally:



Inter-symbol Interference: ISI

- Limited bandwidth in Rx $\Rightarrow$ infinite pulse duration, creating tails
- Ideally – force waveform at receiver to be zero at times $T_1, T_2, T_3 \dots$ with maximum at T_0

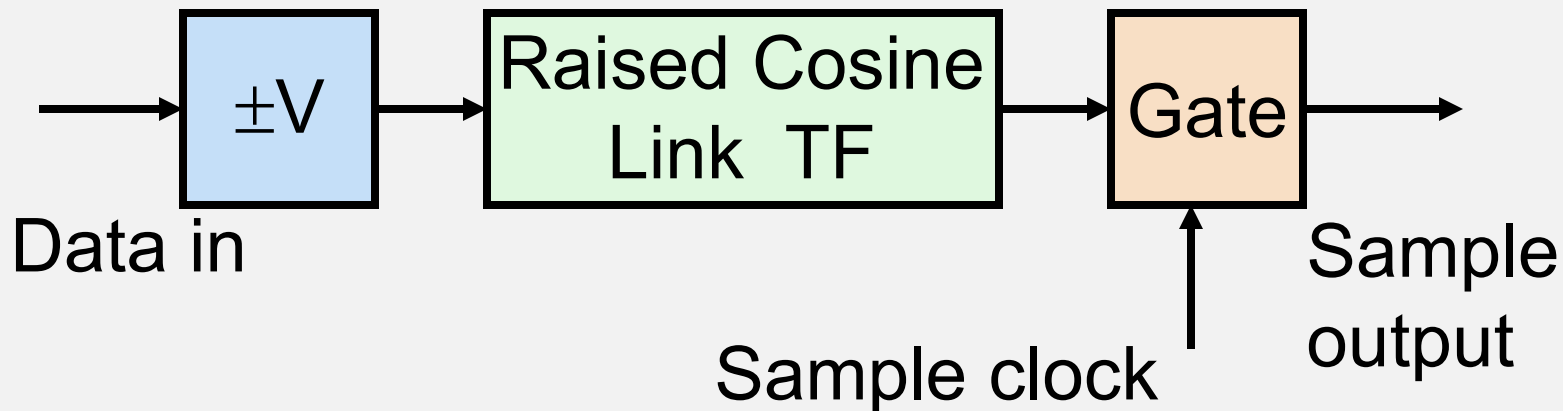


Zero ISI waveform design

- We need a link that can produce the required *zero ISI waveform* at the receiver output
- We must start our digital communication system design at the output, not the input
- Classical solution is to make the entire link transfer function = a *Nyquist raised cosine transfer function*
- The pulse in Rx then has a $\sin X / X$ shape

Zero ISI Digital Transmission System

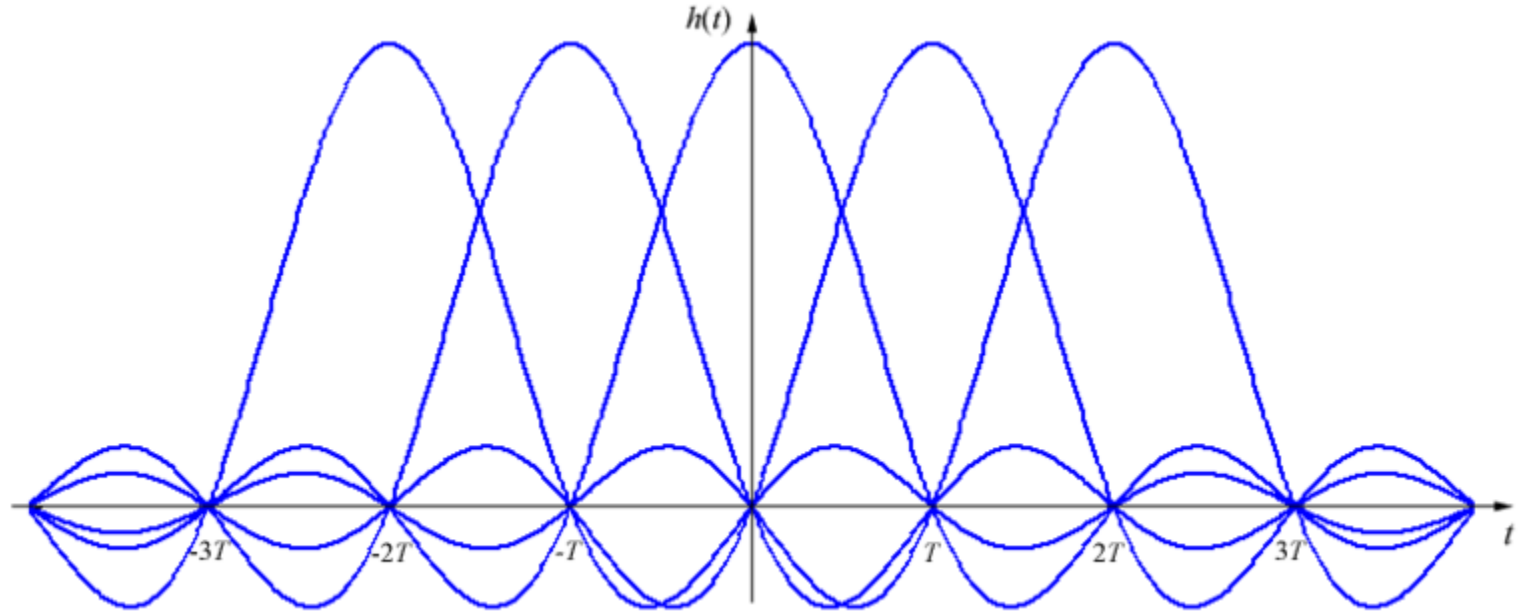
- Block diagram of link including Tx and Rx



Baseband link

Nyquist Transfer Function

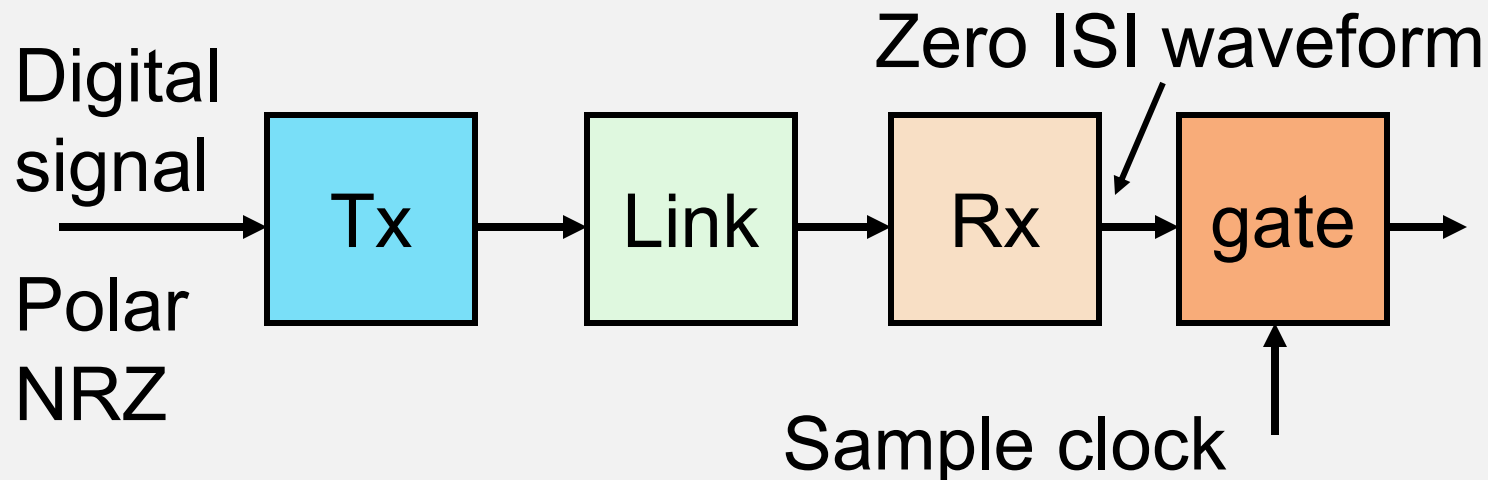
- Pulse period is T_s
- Sample at intervals of T_s
- Pulse has maximum amplitude at $t = 0$
- Zero amplitude at each succeeding interval of T_s
i.e. at $t = T_s, 2T_s, 3T_s \dots NT_s$
- *Nyquist's Zero ISI Principle* (Nyquist 1924)



Raised cosine transfer function (RCTF) across the link meets the Nyquist ISI criterion. At $t = 0$ the middle pulse is at its maximum and the sum of the other impulses is zero

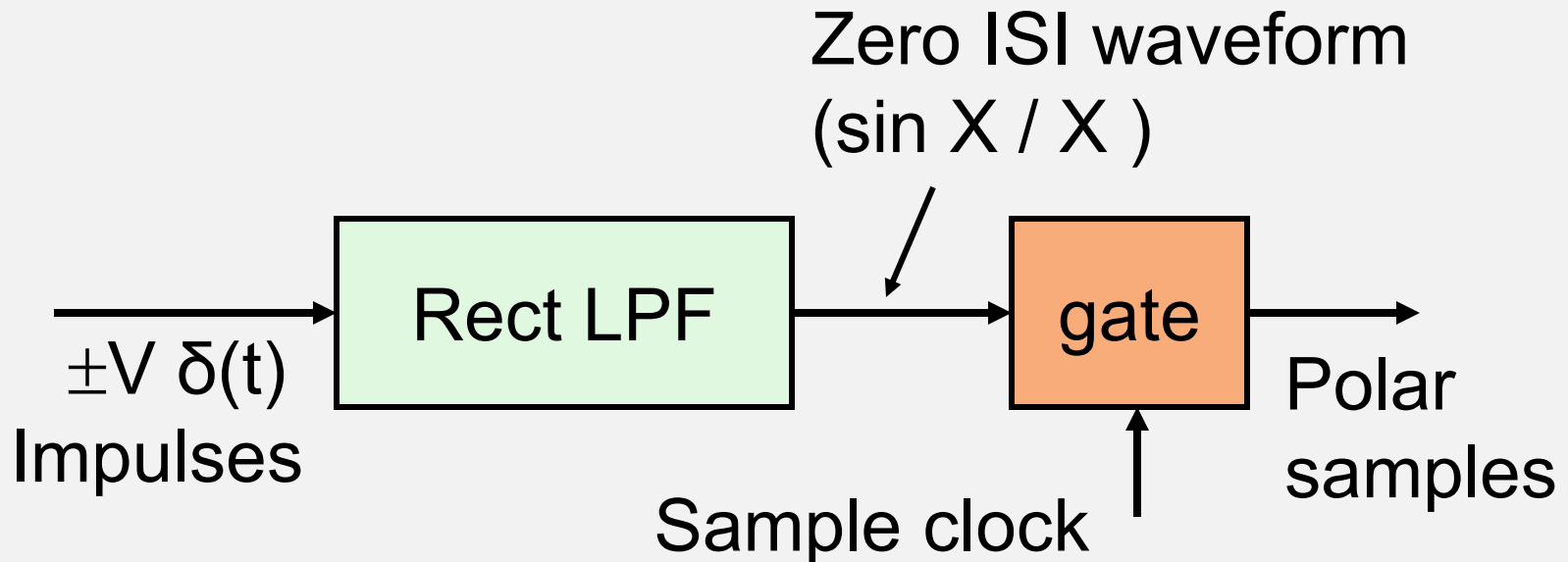
Nyquist Filtering

- Networks which produce the required time waveforms are called *Nyquist Filters*
- None exist in practice but we can get reasonably close



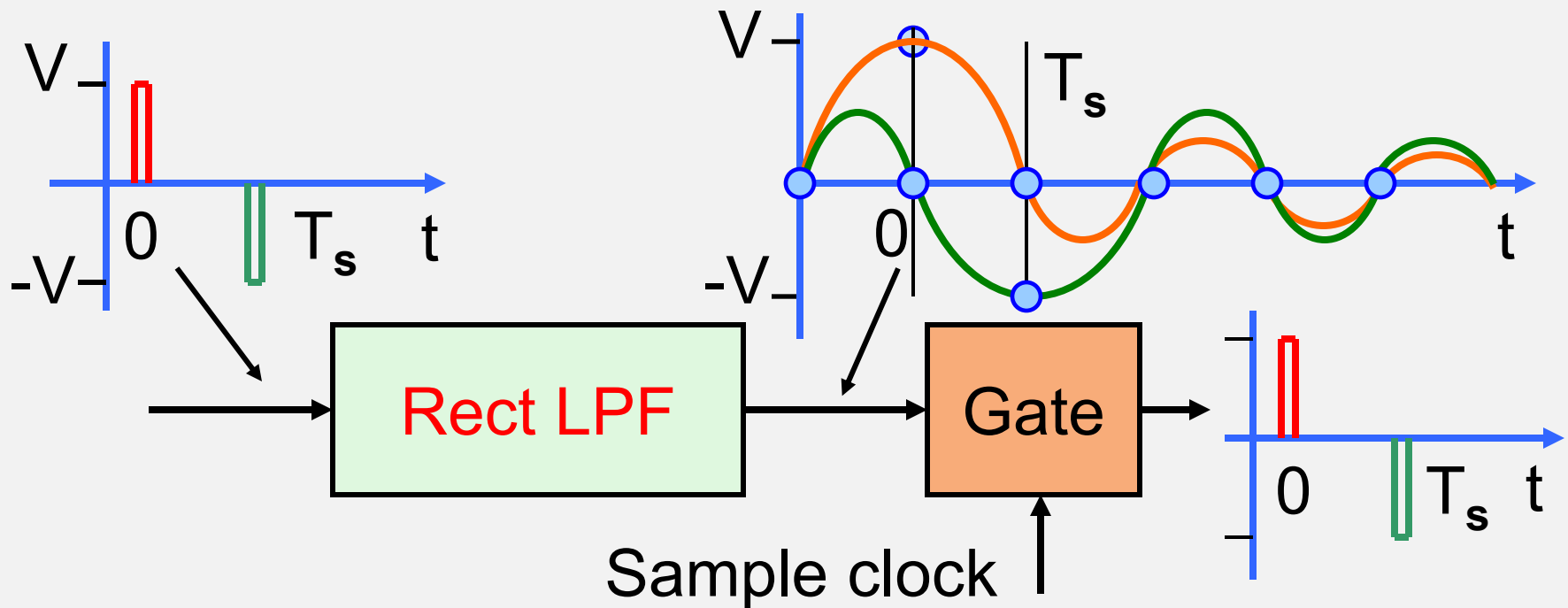
Nyquist Filtering

- Simplest case of a Nyquist filter is a rectangular low pass filter with an impulse as input signal



Rectangular LPF model

- Impulse response of a rectangular low pass filter is a $\sin X / X$ waveform



Analyzing Communication Systems

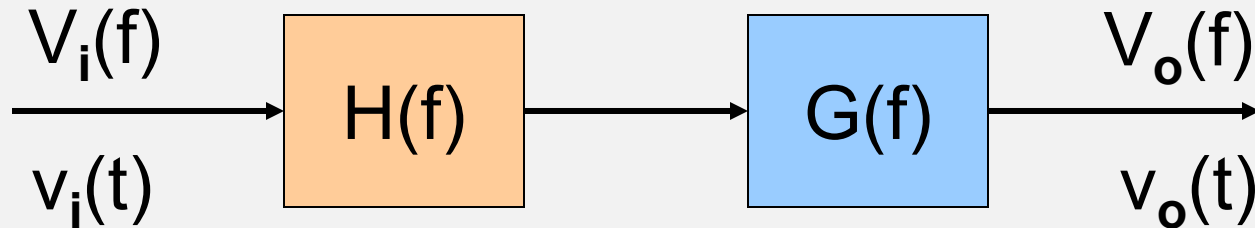
- To understand how the signal transforms across the link, we have two choices:
- Frequency domain
 - Use signal spectra and transfer functions
- Time domain
 - Use signal waveforms and impulse responses

Analyzing Communication Systems

- We move between time and frequency with the Fourier Transform (FT)
- Input to link is an impulse – time waveform
- Output of link is zero ISI time waveform
- We will analyze link in frequency domain
- Use FT to convert input and output waveforms to spectra

Frequency and time

- Time waveforms and frequency spectra



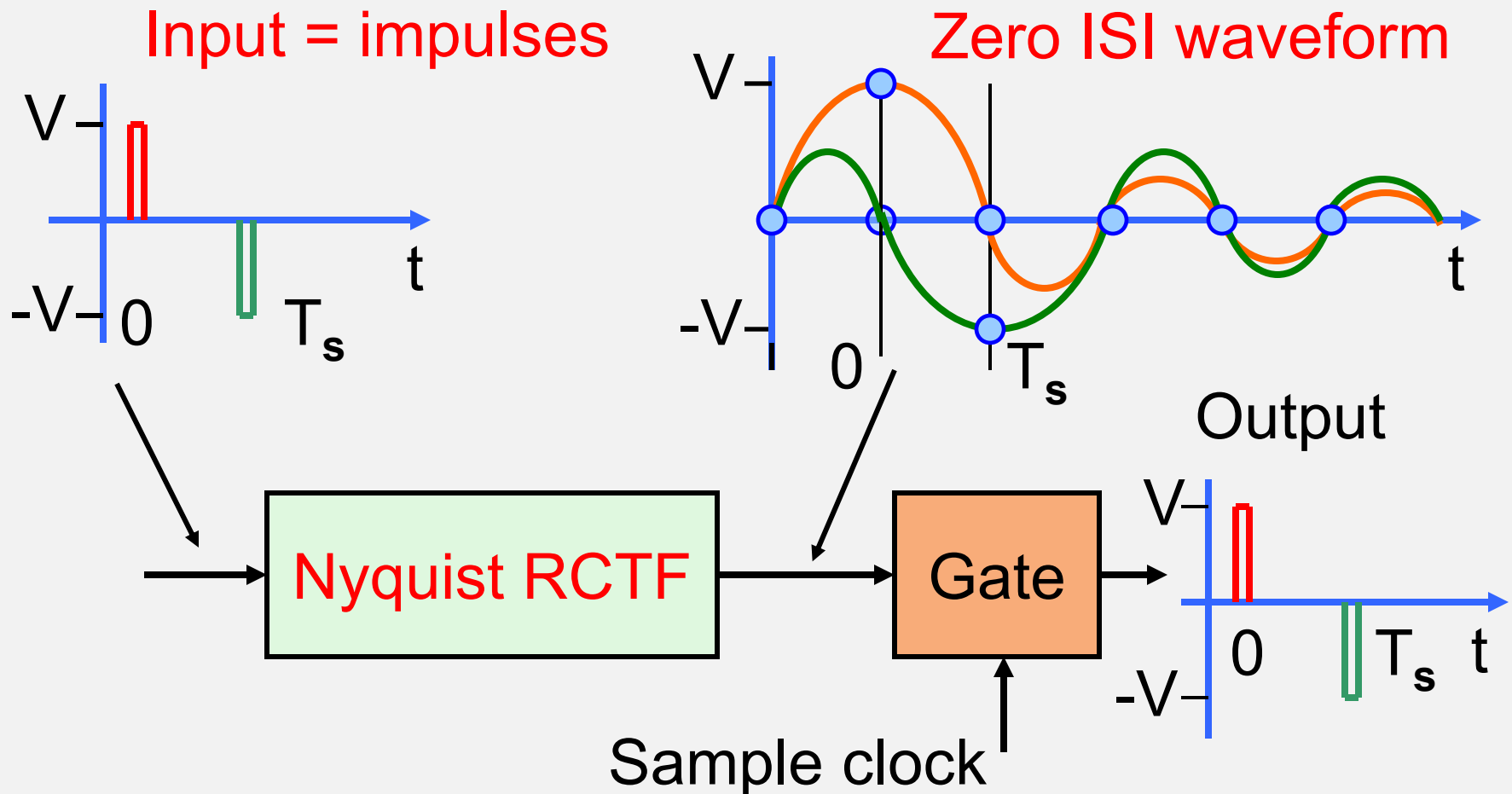
$V_o(f) = V_i(f) \times H(f) \times G(f)$ multiply transfer functions
by input spectrum

$v_o(t) = v_i(t) * h(t) * g(t)$ convolve time waveform
and impulse responses

Zero ISI design principles

- Want the output for an impulse input to be a zero ISI waveform
- Make entire link transfer function (TF) equal to *Nyquist Raised Cosine Filter*, $TF = H_N(f)$
- Then the output of the link to an impulse input is $v_o(t) = h_N(t)$, which is a zero ISI waveform

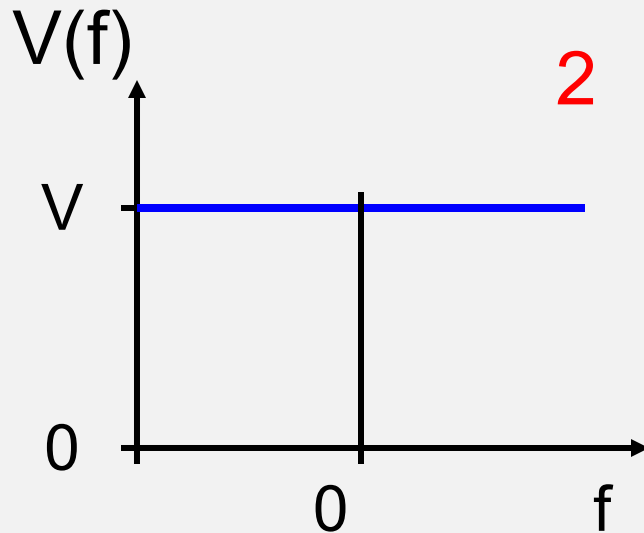
Generic Digital Transmission Link



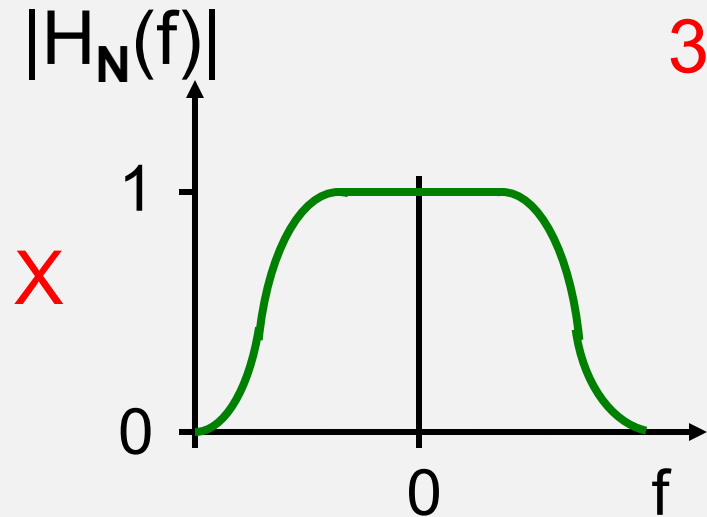
Zero ISI design principles

- In time domain depict impulse as $v_{in}(t) = \pm V \delta t$
- Perform FT to frequency domain to find its spectrum
- The impulse spectrum is described by $V_{in}(f) = V$ volts/Hz, which corresponds to a *flat* spectrum

F
r
e
q

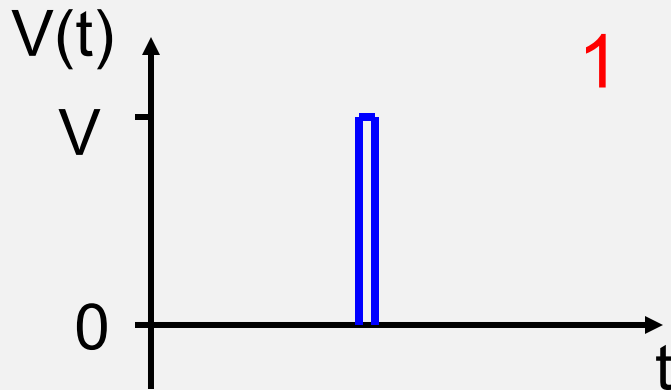


Input pulse spectrum

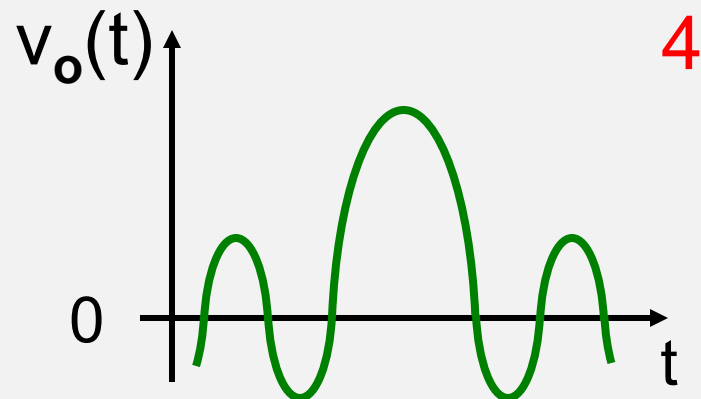


Nyquist RC TF

T
i
m
e



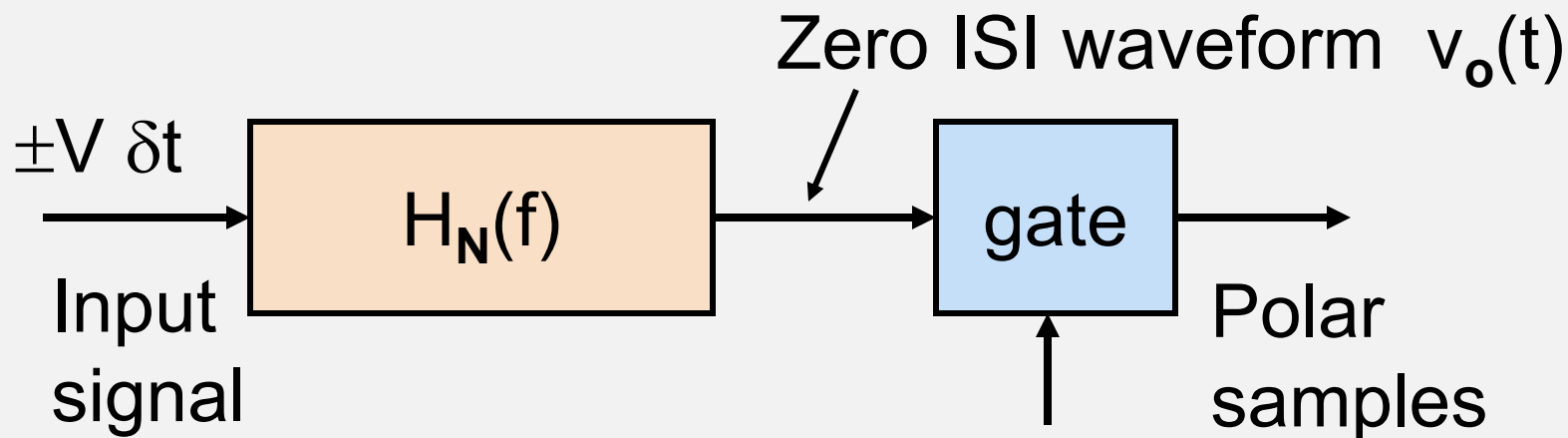
Input pulse $V \delta t$



Output waveform

Zero ISI design

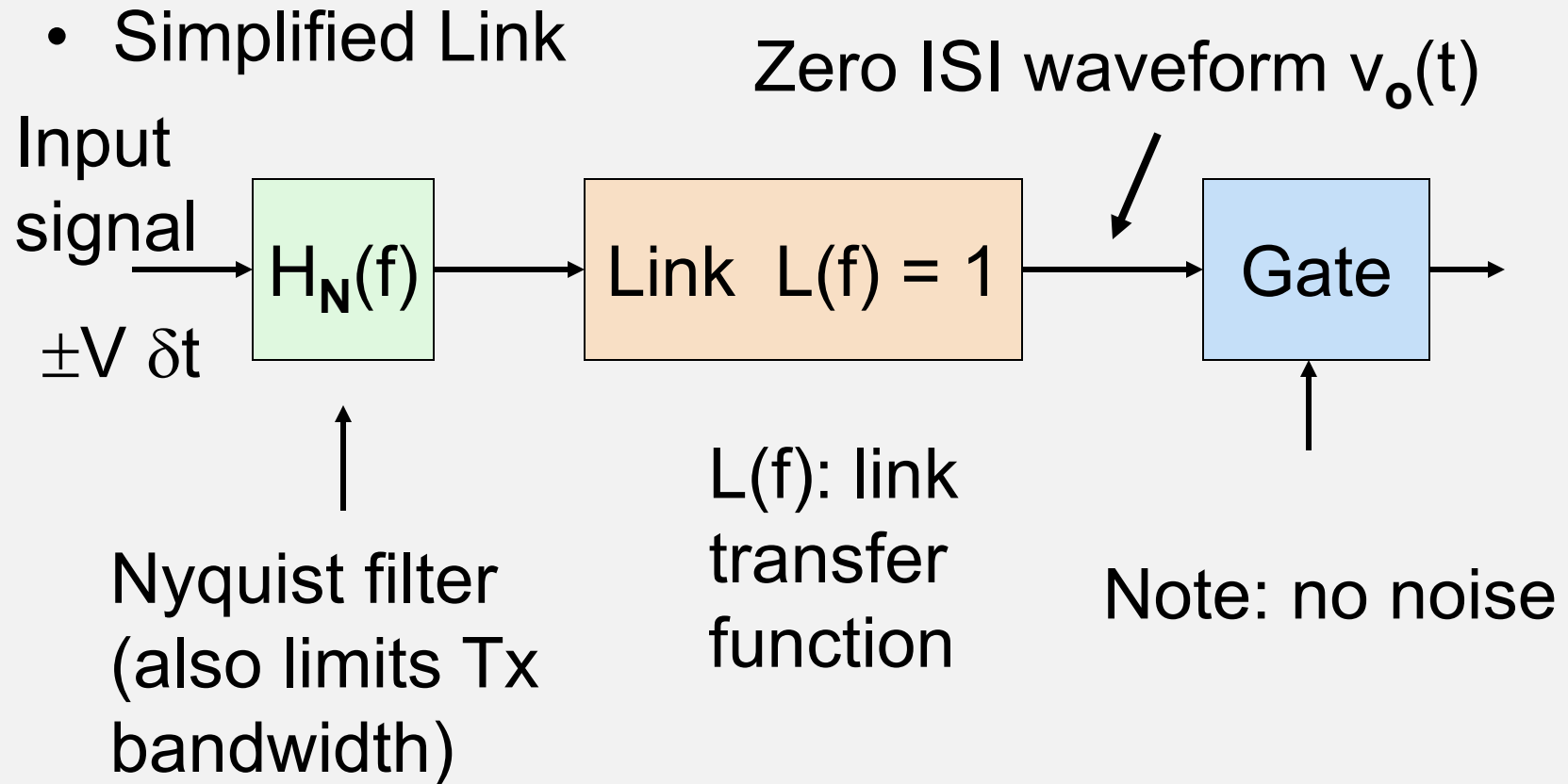
- Output spectrum is
$$V_o(f) = V_{in}(f) \times H_N(f) = \pm V \times H_N(f)$$
- Output waveform is $FT^{-1}[\pm V \times H_N(f)]$
$$v_o(t) = \pm V \times h_N(t)$$



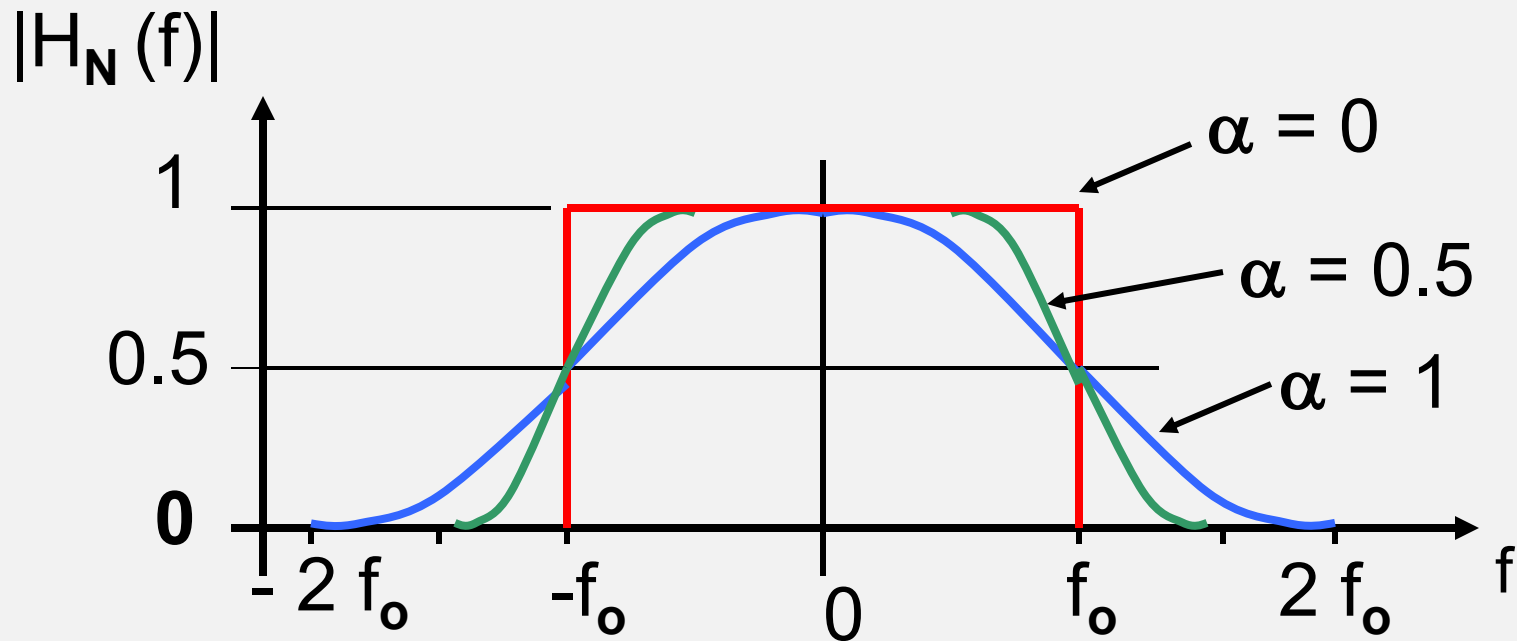
Designing for Zero ISI

- We design for zero ISI spectrum at the input to the sampling gate / output of receiver
- Work in frequency domain to force the output spectrum to be Nyquist TF, $H_N(f)$
- This is the basis for the design of *all* digital transmission systems
- Under ideal link conditions a filter at the transmit end can provide the required TF

Zero ISI design – Ideal link



Nyquist Lowpass TF (two sided)



All Nyquist TFs have voltage magnitude 0.5 at $f = f_o$

Nyquist Transfer Function

- End-to-end transfer function of link must be a Nyquist transfer function
- Minimum transmission bandwidth is f_o Hz when $\alpha = 0$
- TF then has infinitely fast roll-off
- Practical systems must have $\alpha > 0$
- Bandwidth for transmission is $> f_o$

Raised Cosine Transfer Function

- Baseband - Low Pass version (Eq. 5.2)

$$H_N(f) = 1 \quad \text{for } |f| < \frac{1}{2} R_s (1 - \alpha)$$

$$H_N(f) = \cos^2 \left\{ (0.5\pi/\alpha R_s) \times [|f| - \frac{1}{2} R_s (1 - \alpha)] \right\}$$

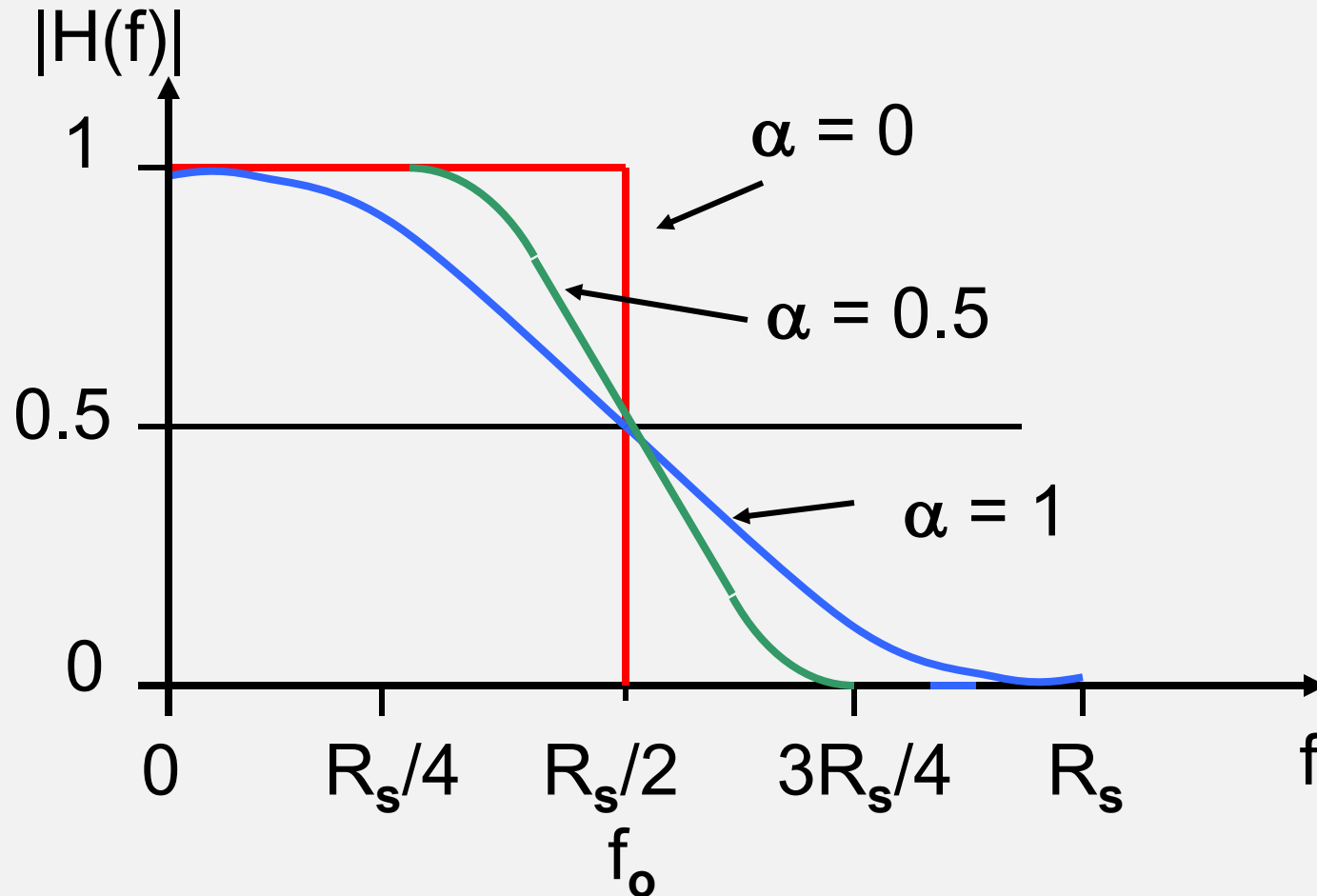
$$\text{for } \frac{1}{2} R_s (1 - \alpha) < |f| < \frac{1}{2} R_s (1 + \alpha)$$

$$H_N(f) = 0 \quad \text{for } |f| > \frac{1}{2} R_s (1 + \alpha)$$

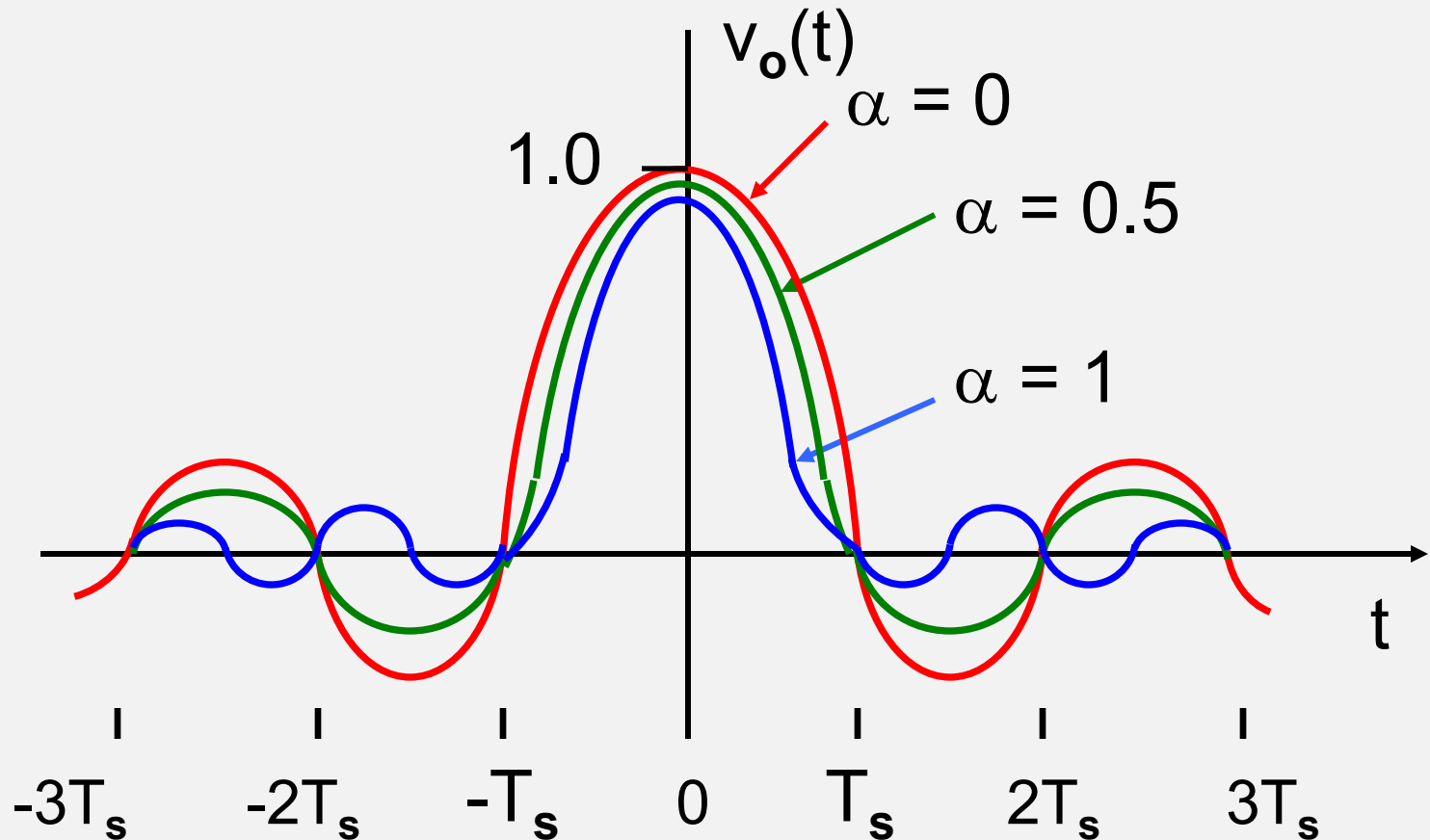
$$\text{and } 0 \leq \alpha \leq 1$$

- $R_s = 1 / T_s$ is the *symbol rate* (pulse rate)
- α is the *roll off factor* of the transfer function
- $f_o = \frac{1}{2} R_s$

Nyquist Lowpass RC TF (one sided)

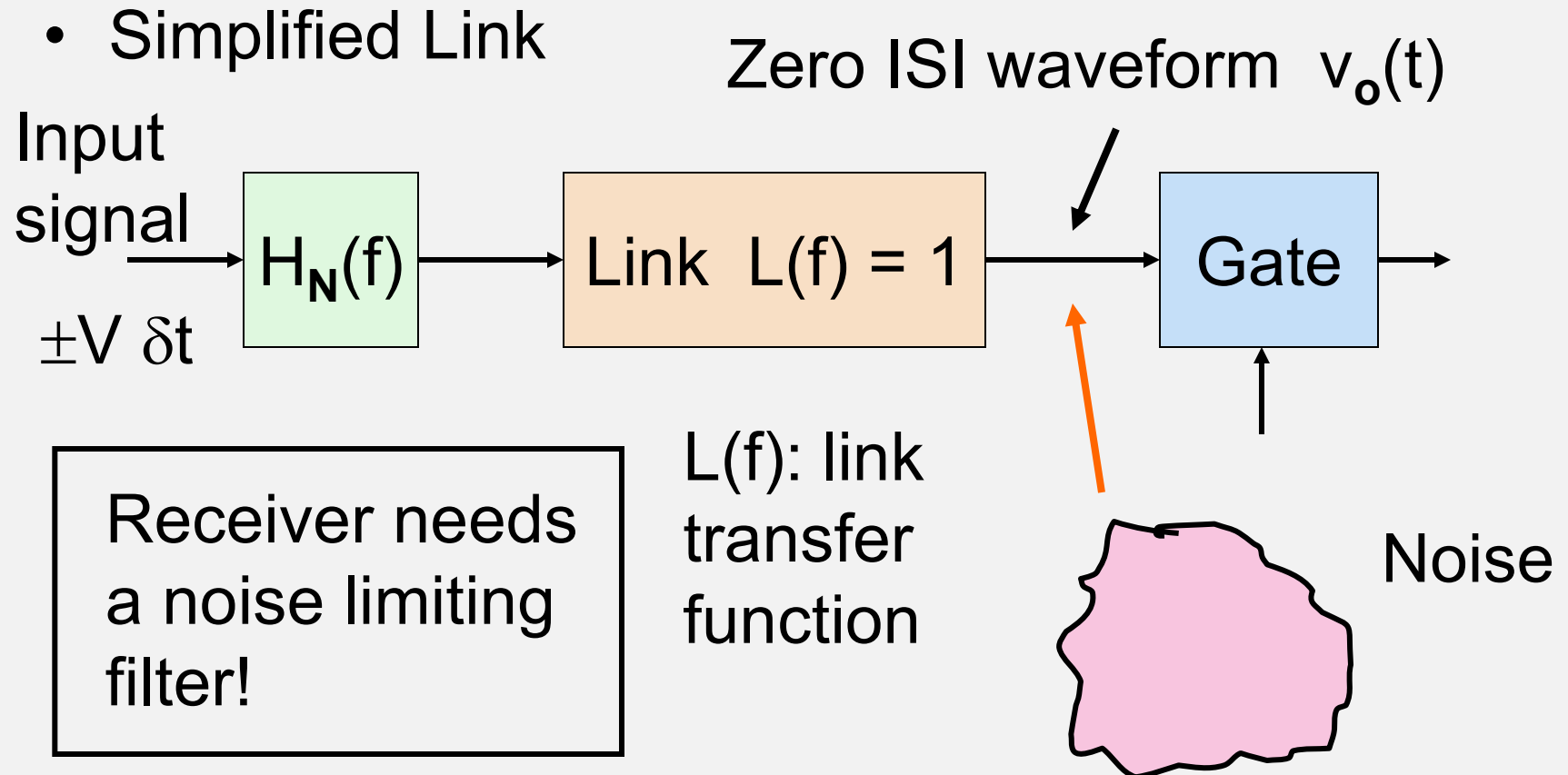


Nyquist TF waveforms



Zero ISI waveforms for Nyquist TF

Zero ISI design – Non ideal link



Zero ISI design

Noise into receiver needs to be minimized

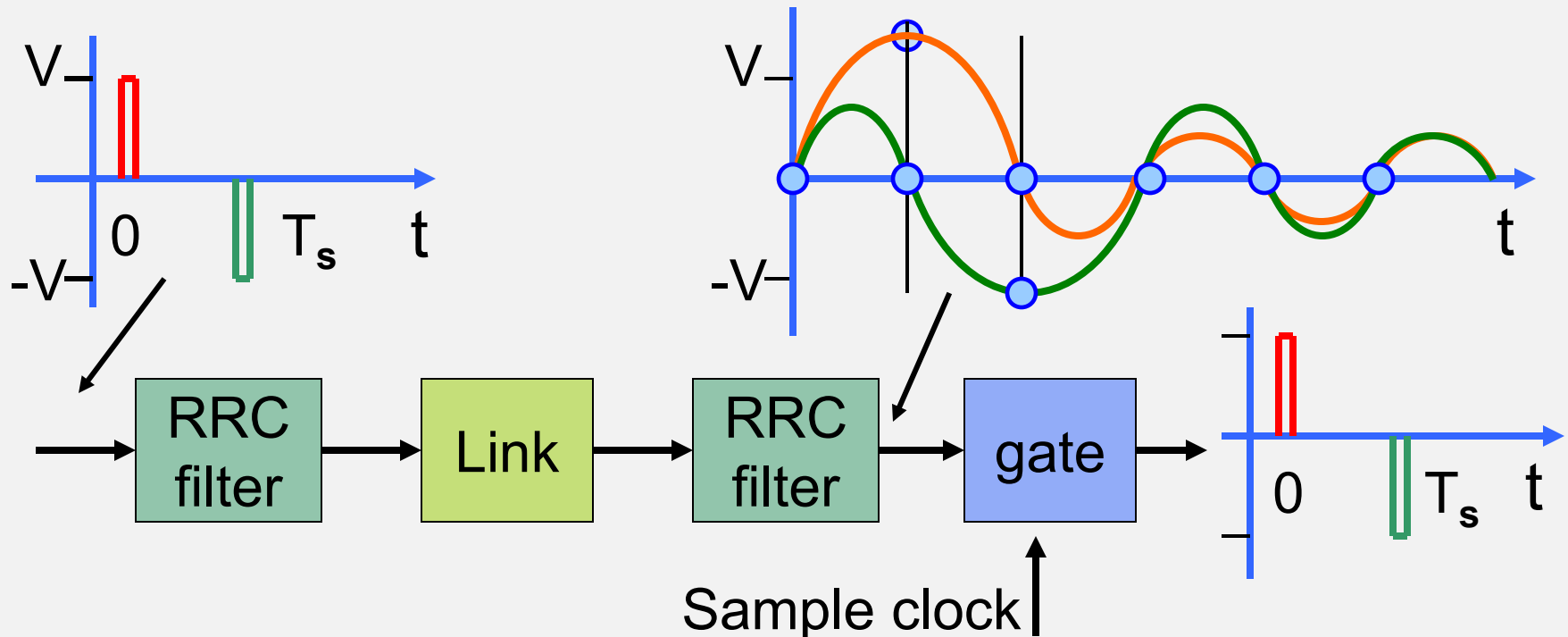
- Solution: Place *half* of Nyquist filter at transmit end of link, *half* at receive end
- Filter transfer function at either end, $H_{\text{rrc}}(f)$, is given by
$$H_N(f) = H_N(f)_{\text{NYQUIST}} = H_{\text{rrc}}(f) \times H_{\text{rrc}}(f)$$
- Hence: $H_{\text{rrc}}(f) = \sqrt{H_N(f)_{\text{NYQUIST}}}$
- *Square Root Raised Cosine Filter* (SRRC filter)

Zero ISI design

- Often shortened to just *Root Raised Cosine Filter* (RRC filter)
- Just remember that *Root* refers to Square Root

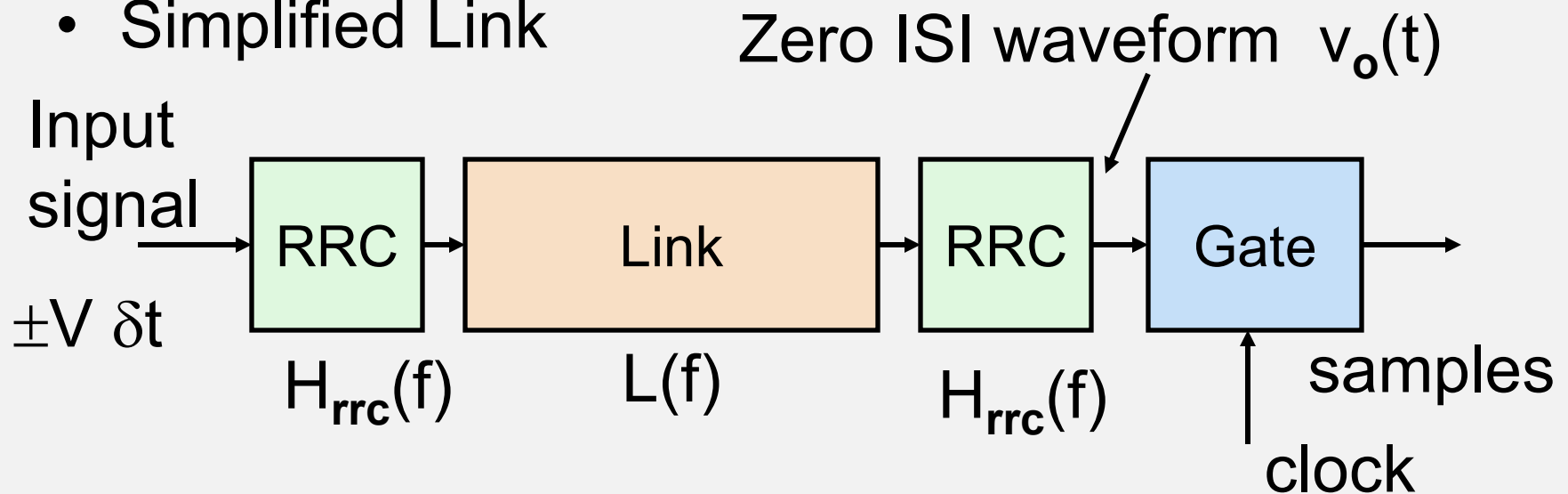
Raised Cosine TF Link

- Transmitter and receiver use RRC filters to obtain zero ISI waveform



Zero ISI design

- Simplified Link



$$V_o(f) = V_{in}(f) \times H_{rrc}(f) \times L(f) \times H_{rrc}(f) = V_{in}(f) \times H_N(f) \times L(f)$$

This gives required zero ISI waveform $v_o(t)$

Zero ISI design – Link transfer function

- If the link transfer function $L(f)$ is flat ($= 1$) over the bandwidth of the RRC filters, then

$$V_{in}(f) \times H_{rrc}(f) \times 1 \times H_{rrc}(f) = V_{in}(f) \times H_N(f)$$

so

$$V_o(f) = V_{in}(f) \times H_N(f)$$

and

$$v_o(t) = v_{in}(t) * h_N(t) = \text{zero ISI waveform at receiver output}$$

- If link transfer function $L(f)$ is not flat, use *equalizer* to flatten it, i.e., $E_q(f) \times L(f) = 1$

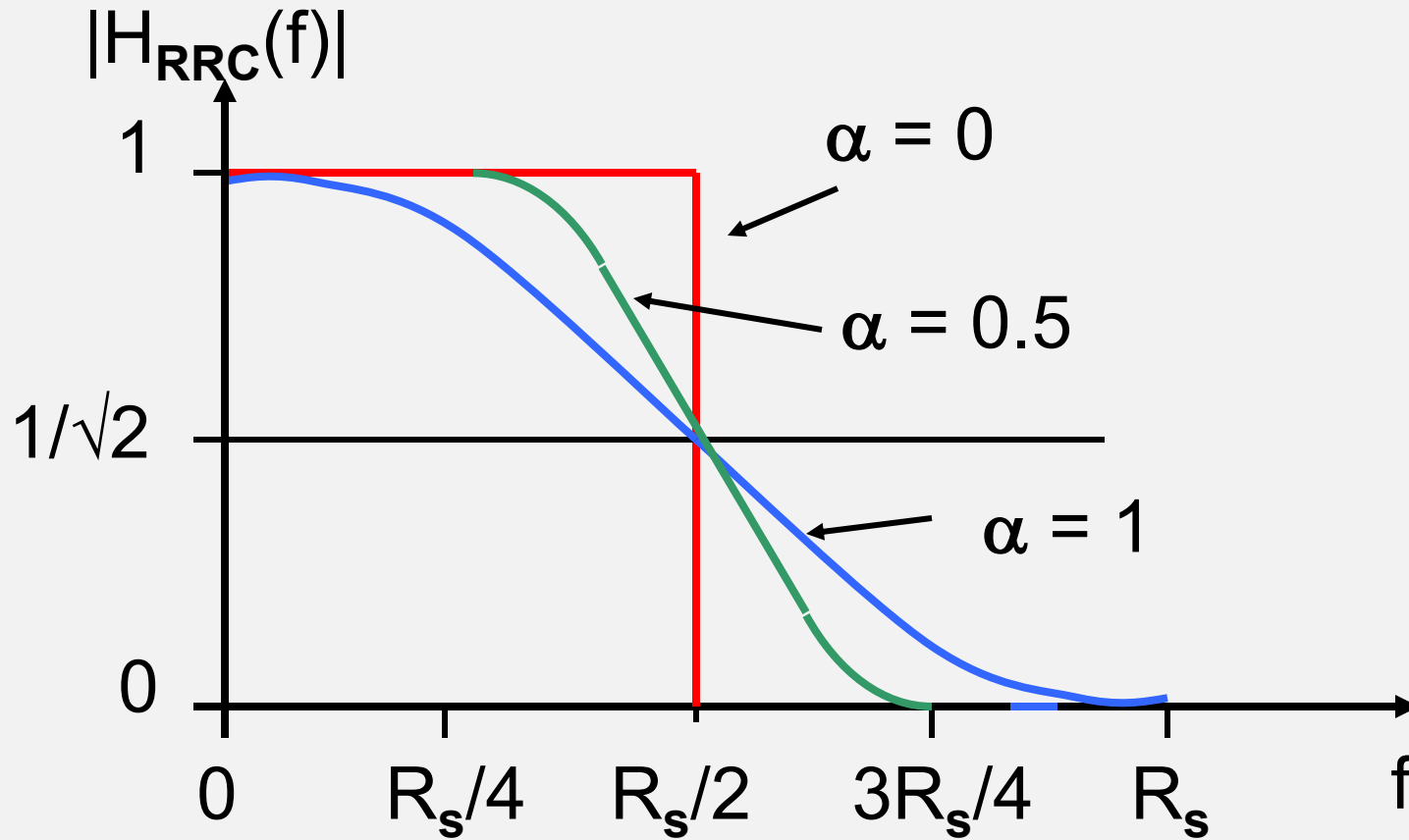
RRC filters are matched filters

- Square root raised cosine filters in transmitter and receiver gives a *Matched Filter* (MF) response
- A link with a matched filters has the best C/N ratio possible
- A matched filter response is a desirable design criterion in communication systems

Bandwidth and Symbol rate

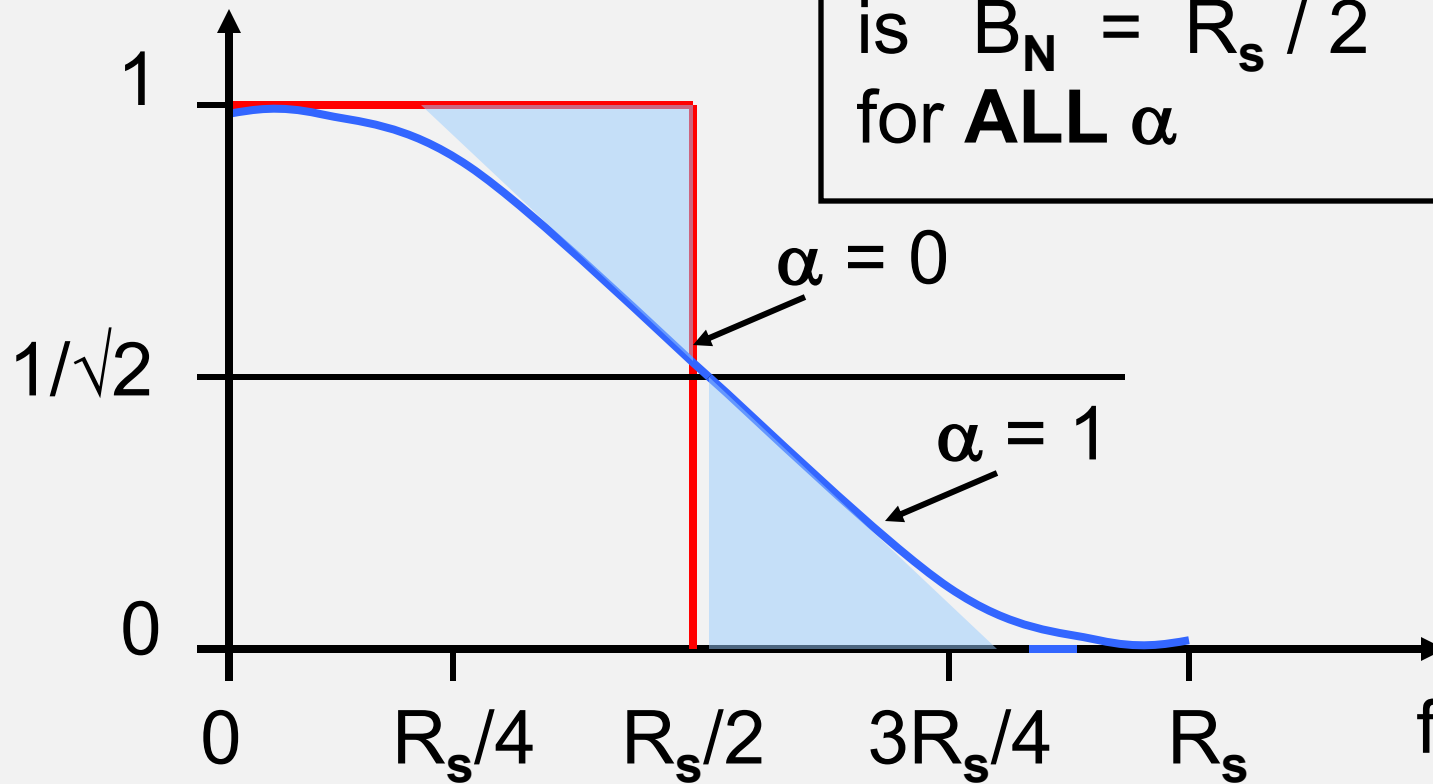
- Roll off factor α sets bandwidth of RRC filter
- Rx waveform gives zero ISI when output is sampled at correct time, with sampling rate of R_s (i.e. at every T_s)
- R_s = symbol (pulse) rate

RRC Lowpass Filter (One-sided)



Noise Bandwidth of RRC Lowpass filter

$$[H_{\text{RRC}}(f)]^2 = |H_{\text{N}}(f)|$$



LPF Noise Bandwidth
is $B_N = R_s / 2$
for **ALL** α

Occupied bandwidth

- Bandwidth required to transmit baseband pulses (on wires) depends on two factors:
 - Symbol (pulse) rate of signal, $R_s = 1 / T_s$
 - Roll off factor of RRC filters, α
- Occupied bandwidth for pulses on wires
$$B_{\text{occ}} = \frac{1}{2} R_s (1 + \alpha)$$
Occupied bandwidth = required channel bandwidth
- Minimum bandwidth is $\frac{1}{2} R_s$ with $\alpha = 0$

Example for Wire Transmission - 1

- Baseband transmission of pulses
- Symbol rate is 5 ksps (5 kbaud)
- Pulse rate is in *symbols per second* (sps) or *baud*
- What is minimum possible bandwidth of link for zero ISI at receiver?

$$B_{\min} = \frac{1}{2} \times R_s = 2.5 \text{ kHz} \quad (\alpha = 0)$$

Example for Wire Transmission - 2

- Baseband transmission of pulses
- Symbol rate is 5 ksps (5 kbaud)
- RRC filters with $\alpha = 0.5$ are used in transmitter and receiver
- What is occupied bandwidth of signal?
- Also referred to as the *channel bandwidth* of the link

Example for Wire Transmission - 2

- Occupied bandwidth of baseband signal is given by

$$B_{\text{occ}} = \frac{1}{2} R_s (1 + \alpha)$$

- Hence

$$B_{\text{occ}} = \frac{1}{2} \times 5 (1 + 0.5) = 3.75 \text{ kHz}$$

- This is also the channel bandwidth
- Note: the link must have a flat TF over the range DC – 3.75 kHz

Course Project Update (1/3)

- Information on team membership and project topic is due by midnight tonight (Wednesday)
- There may be a stage of iteration with the instructor to settle team membership and topic
- These items should be finalized by Thursday midnight
- Let the instructor know by email if you are looking for a team to join

Course Project Update (2/3)

- Next task: Compose an Initial Report (IR)
- Guidance on the preparation of the IR (3+ pages) has been posted to Canvas
- Should consist of three sections: Title page, Introduction, and Background
- Title page, list of references, figures and Tables do not count towards page limit

Course Project Update (3/3)

- See comment on the use of AI in the Guidance document (and sources generally)
- Reading: Familiarize yourself with your selected Satcom system
- Initial Report is due by the start of class on Wednesday, April 8

Digital Radio Links (RF)

- Move now to consideration of satellite transmission link
- We need an RF carrier wave (a wire just won't do)
- Baseband is modulated onto the RF carrier wave with frequency f_c Hz

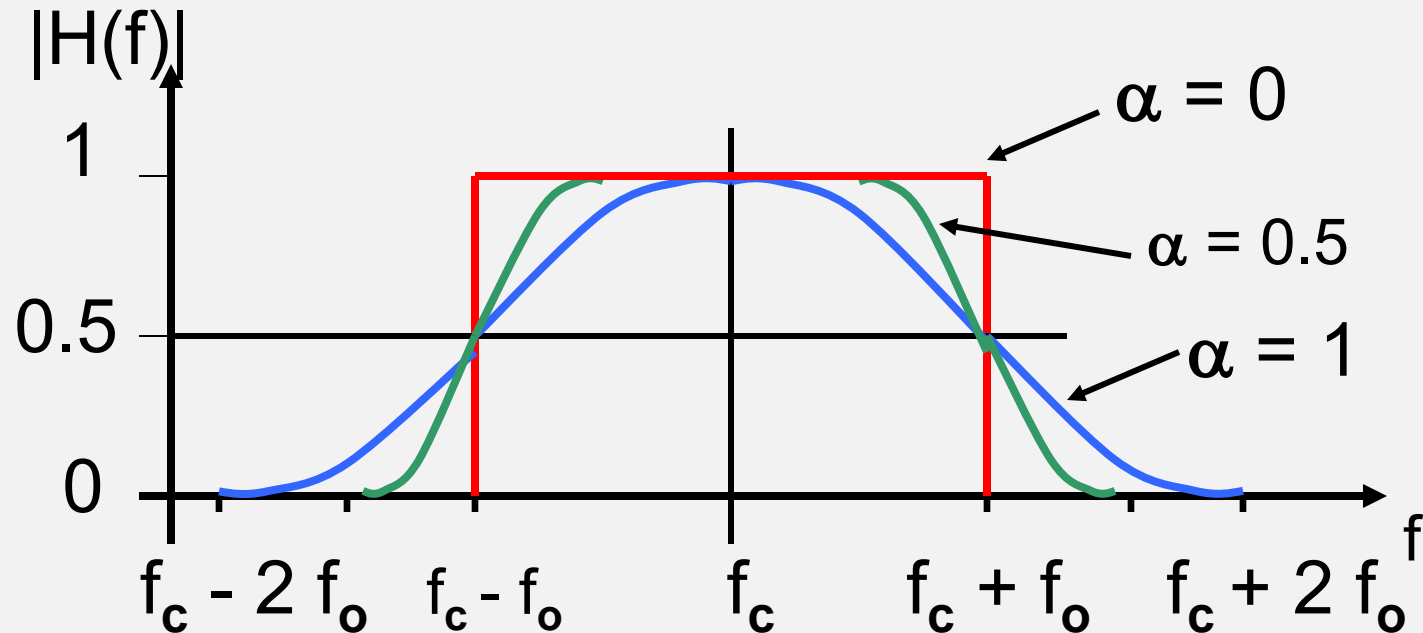
Digital Radio Links (RF)

- Available bandwidth is limited so filters must be applied
- This implies ISI
- Digital radio links also need RRC-type filters to suppress ISI
- Principles are the same as for baseband link

Digital Radio Links (RF)

- Filters must now have a *bandpass transfer function* centered on f_c
- Slide baseband filter center frequency from 0 Hz to f_c Hz
- **Doubles the** bandwidth required to send pulse signals compared to baseband transmission
- Two-sided nature of the TF has become manifest in moving from baseband to bandpass

Nyquist Bandpass TF



Occupied bandwidth is $R_s (1 + \alpha)$

Occupied Bandwidth of Bandpass RRC

- Bandwidth for **baseband** digital signals

$$B_{\text{occ}} = \frac{1}{2} R_s (1 + \alpha)$$

- Bandwidth for **bandpass** (RF) digital signals

$$B_{\text{occ}} = R_s (1 + \alpha)$$

- This is **twice** the LPF bandwidth
- All satellite links are bandpass links
- Noise bandwidth is equal to the symbol rate

$$B_N = R_s \text{ for all values of } \alpha$$

Symbol Rate and Bit Rate

- *Symbol rate* is the number of digital symbols (pulses) sent per second
- *Bit rate* is the number of digital bits sent per second
- Different modulation schemes can transmit different numbers of bits in a single symbol
- Examples: BPSK, QPSK, m-PSK, QAM

Digital Radio Example - 1

- Radio link transmits BPSK signal at 1 Mbps
- Bandpass RRC filters available with $\alpha = 0.4$
- What channel bandwidth is required in RF link?
- BPSK is *binary phase shift keying*
- Transmits one bit per symbol
- Symbol rate $R_s = 1 \text{ Msps}$
- RF bandwidth $= R_s(1 + \alpha) = 1.4 \text{ MHz}$

Digital Radio Example - 2

- Radio link transmits BPSK signal at 1 Mbps
- Bandpass RRC filters available with $\alpha = 0.4$
- What is the noise bandwidth of the receiver?
- Symbol rate is 1 Msps = 1 Mbaud
- Rx noise bandwidth = $R_s = 1.0$ MHz
- Observe that the channel bandwidth is always greater than the noise bandwidth

Exercise 3

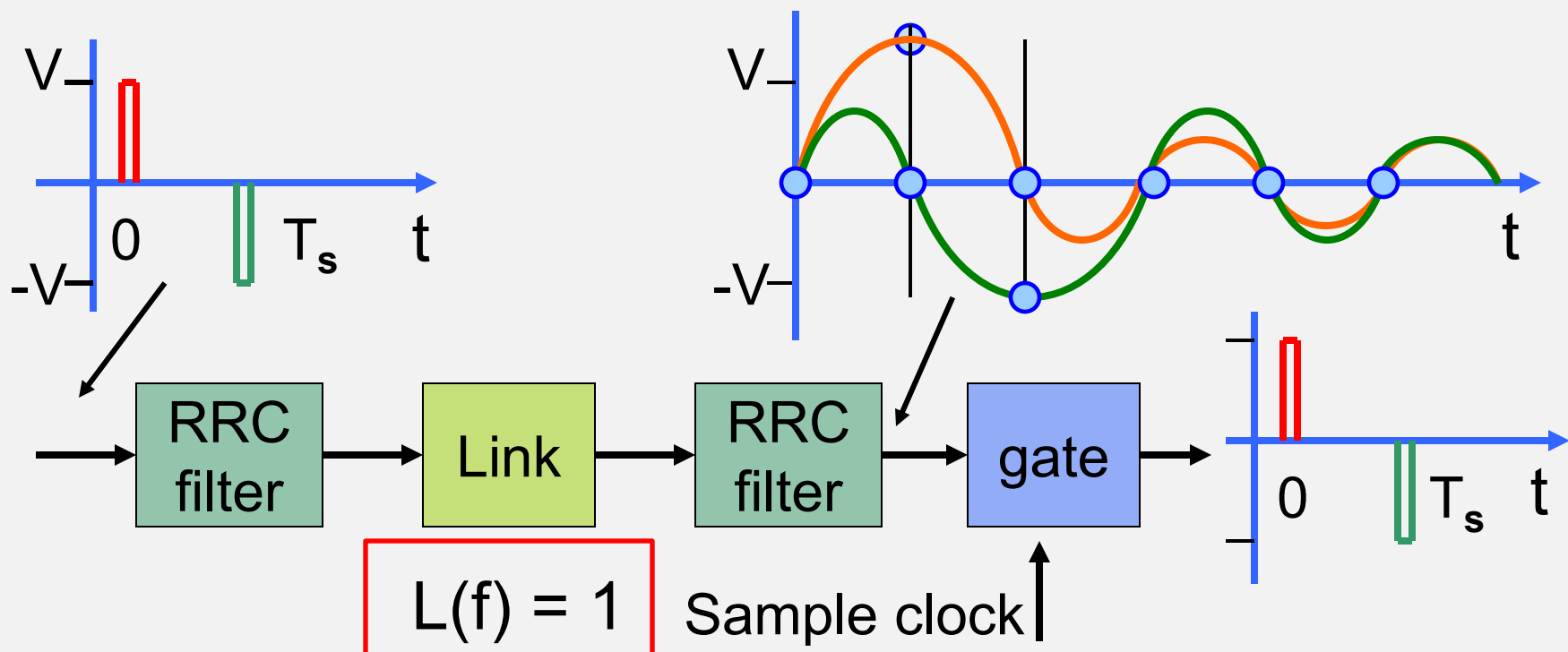
- A satellite communication link transmits QPSK signals at a bit rate of 2 Mbps
- QPSK transmits two bits per symbol
- Tx and Rx have RRC filters with $\alpha = 0.2$
- What is the symbol rate, R_s ?
- What is the bandwidth occupied by the QPSK signal, B_{occ} ?
- What is the receiver noise bandwidth, B_N ?

Exercise 3

- The symbol rate for QPSK is half the bit rate of 2 Mbps , so $R_s = 1 \text{ Msps}$
- The occupied bandwidth is given by $B_{\text{occ}} = R_s(1 + \alpha) = 1.2 \text{ MHz}$
- The Rx noise bandwidth is simply the symbol rate, so $B_N = 1.0 \text{ MHz}$

Raised Cosine TF Link – Link TF

- Transmitter and receiver use RRC filters to obtain zero ISI waveform



Link TF – Re-examine Equalization

- Requirements on the transfer function, $L(f)$:

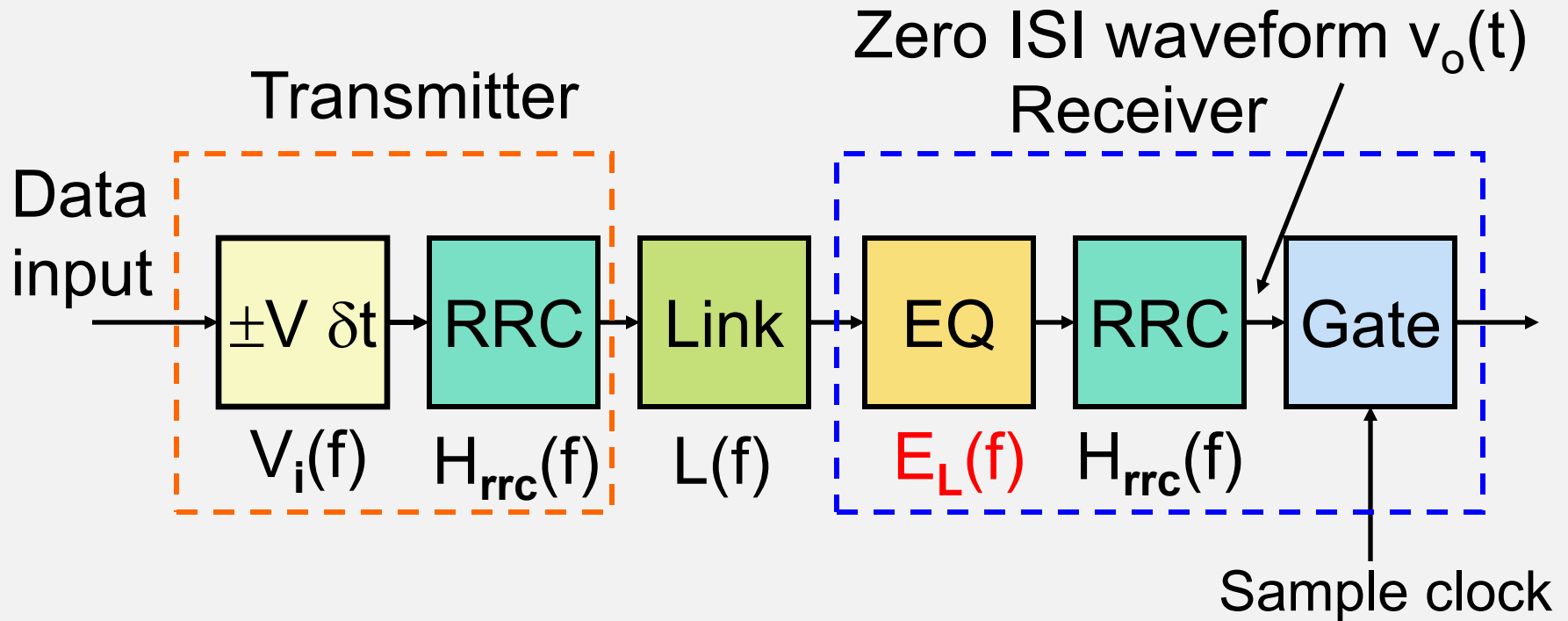
$$|L(f)| = 1 \quad (\text{flat amplitude response})$$

$$\Phi(f) = kf \quad (\text{linear phase response})$$

over bandwidth occupied by link signal

- What if these conditions are not met by the link?
- Insert an equalizer with a compensating transfer function, $E_L(f)$, i.e., $L(f) \times E_L(f) = 1$

Baseband Digital Link

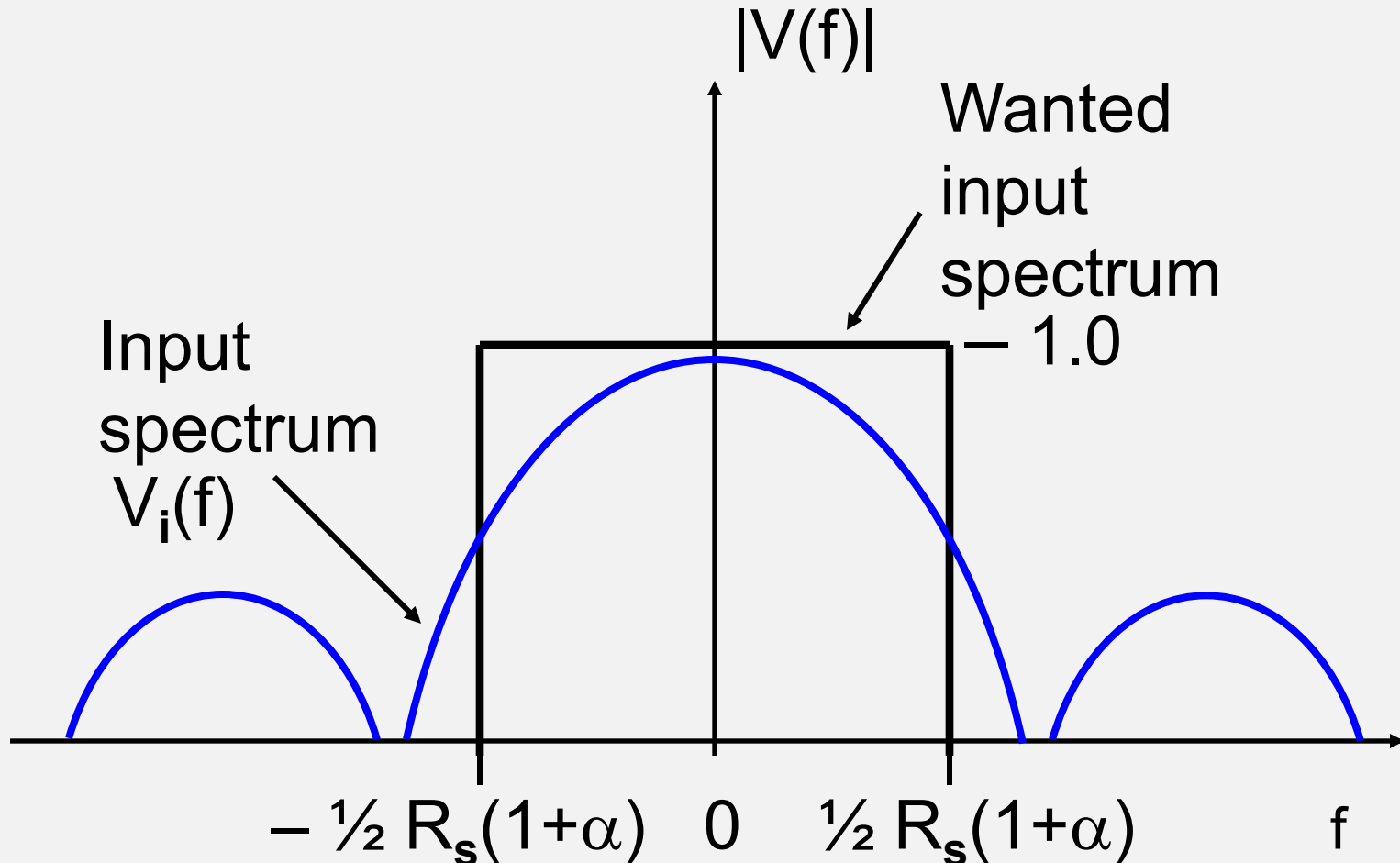


$$\begin{aligned}
 V_o(f) &= V_i(f) \times H_{rrc}(f) \times L(f) \times E_L(f) \times H_{rrc}(f) \\
 &= 1 \times H_N(f) \times 1 = H_N(f)
 \end{aligned}$$

NRZ Input Waveform – Equalization

- With NRZ signal input (square pulses) there is also a need for equalization at the input
- The Nyquist RC transfer characteristic is satisfied only if the input is an impulse ($\pm V \delta t$)
- But with NRZ signal the input consists of rectangular pulses (not impulses)
- Spectrum of NRZ pulses is $\sin X / X$ (not flat)
- We need input spectrum to be $\pm V$ (flat)

Input Spectrum of NRZ Pulse



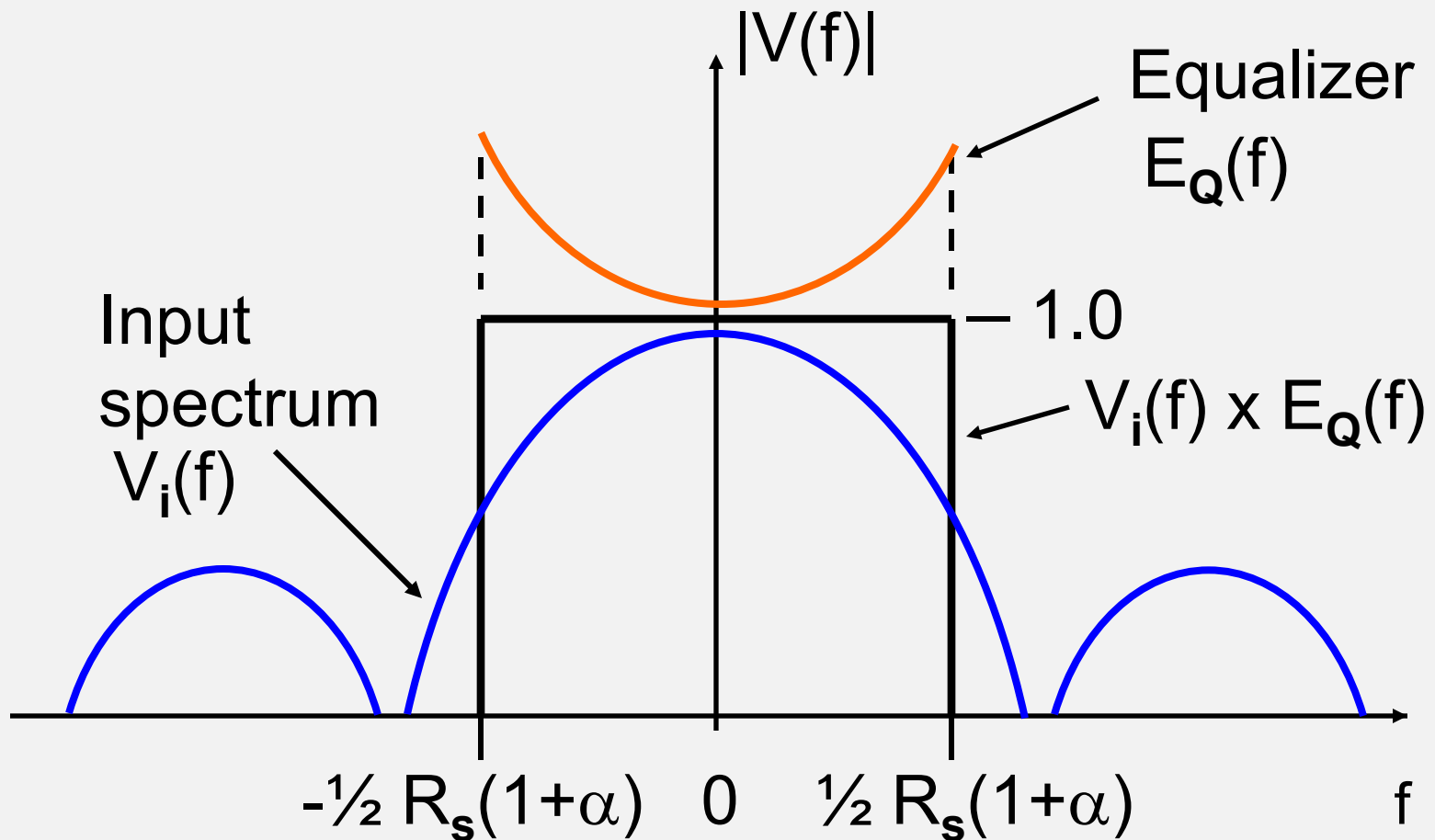
NRZ Input Waveform – Equalization

- Add an input equalizer (i.e, at the Tx end) with frequency response

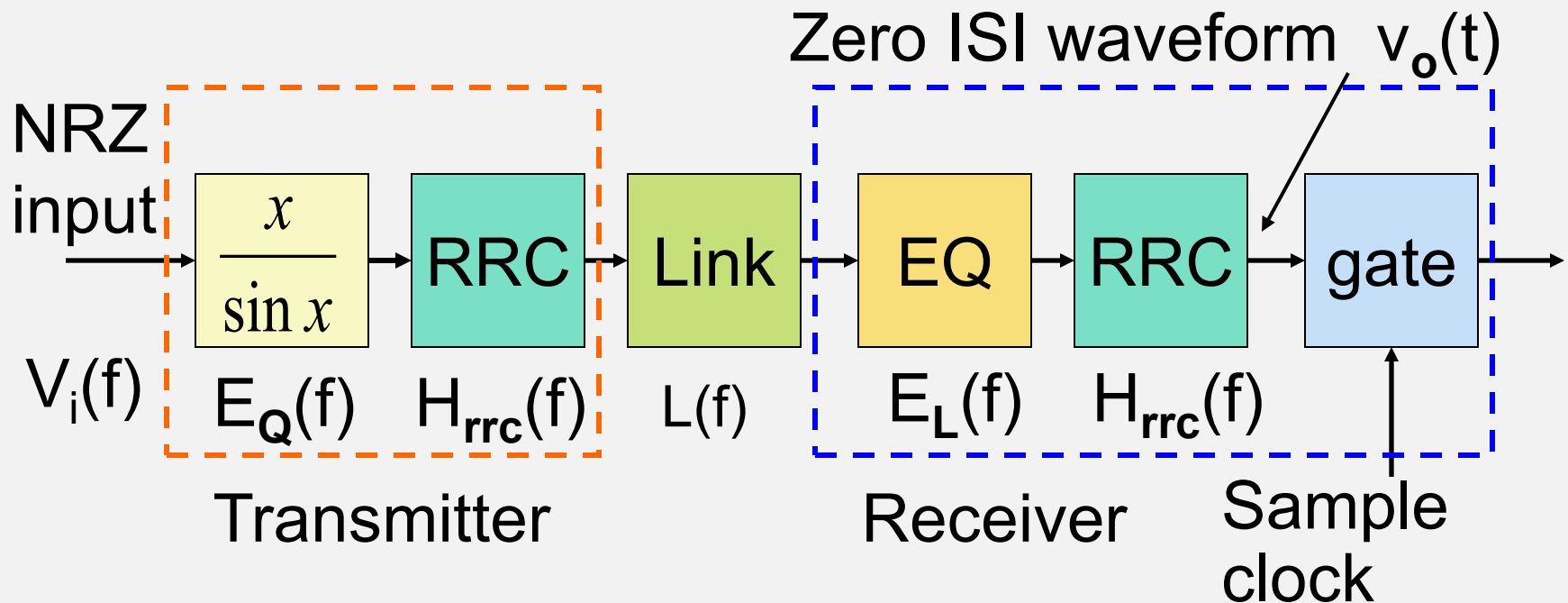
$$E_Q(f) = X / \sin X$$

- Equalizer operates over frequency range 0 Hz to $\frac{1}{2} R_s (1 + \alpha)$ Hz (baseband)
- Make sure $\alpha < 1$ or else $E_Q(f) = \text{infinity}$
- In practice use α values between 0.2 and 0.5

Input equalizer



Generic Baseband Digital Link



$$\begin{aligned}
 V_o(f) &= V_i(f) \times E_Q(f) \times H_{rrc}(f) \times L(f) \times E_L(f) \times H_{rrc}(f) \\
 &= 1 \times H_N(f) \times 1 = H_N(f)
 \end{aligned}$$

Transmitter Filter

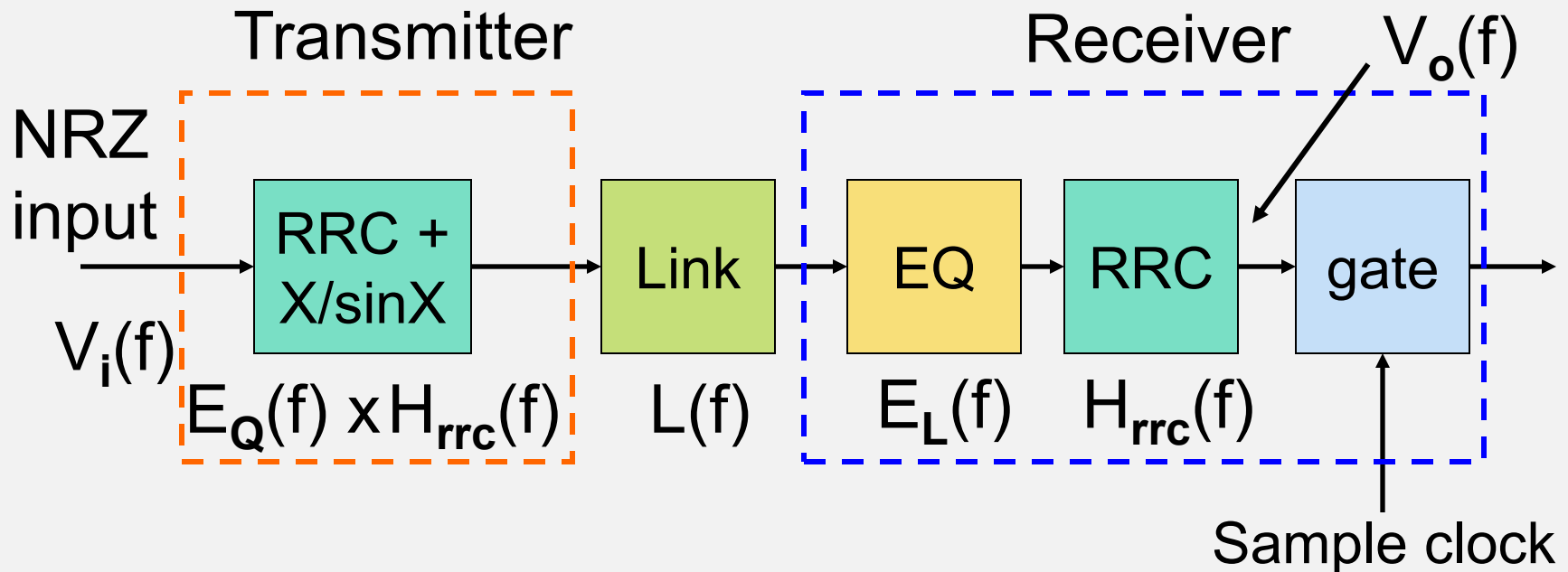
- We can combine the $X / \sin X$ equalizer and RRC filter in the transmitter

- Use a single filter with new transfer function

$$G(f) = E_Q(f) \times H_{rrc}(f)$$

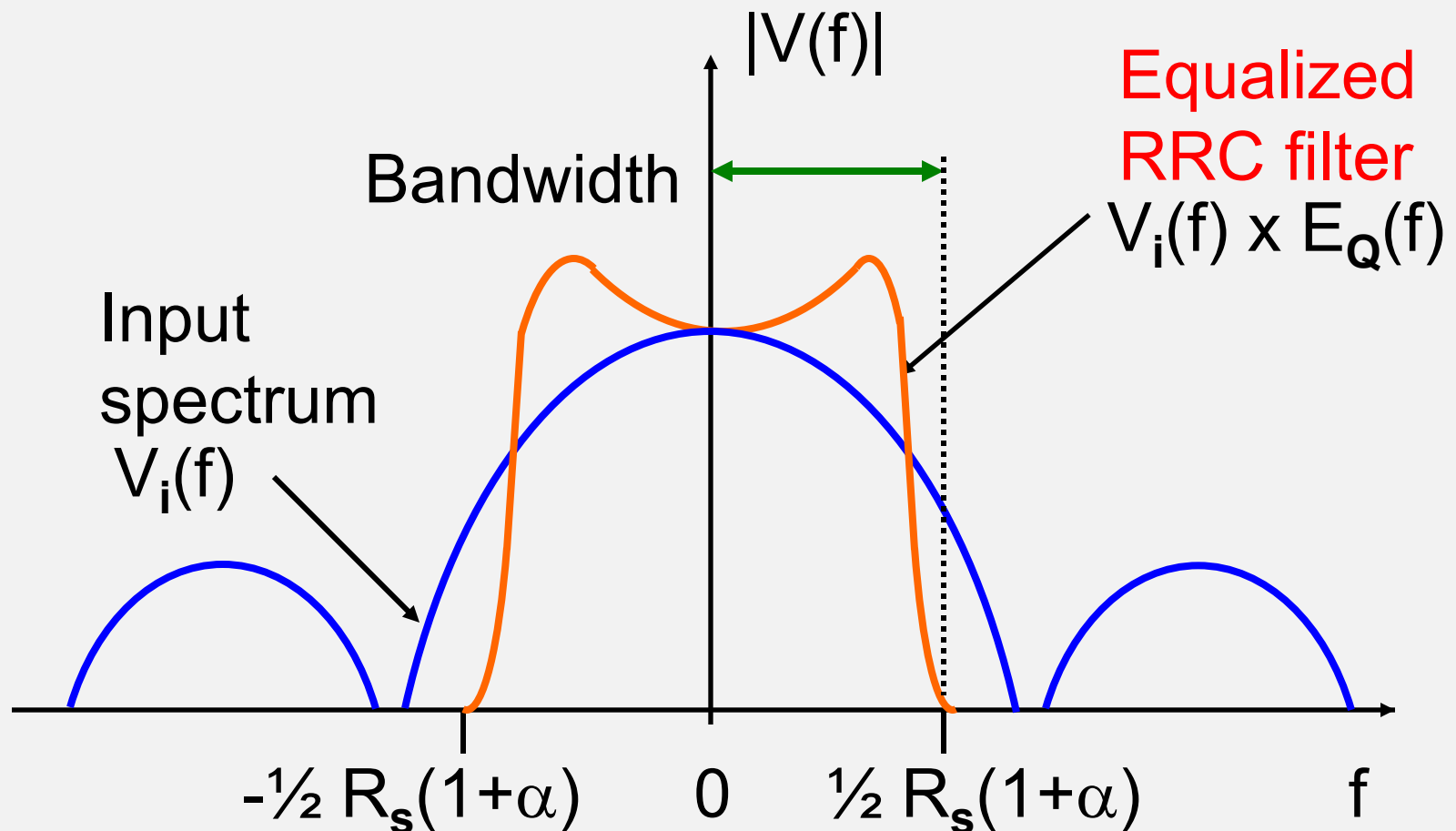
- Chebychev filters can be designed to approximate $G(f)$

Generic Baseband Digital Link



$$\begin{aligned}
 V_o(f) &= V_i(f) \times E_Q(f) \times H_{rrc}(f) \times L(f) \times E_Q(f) \times H_{rrc}(f) \\
 &= 1 \times H_N(f) \times 1 = H_N(f)
 \end{aligned}$$

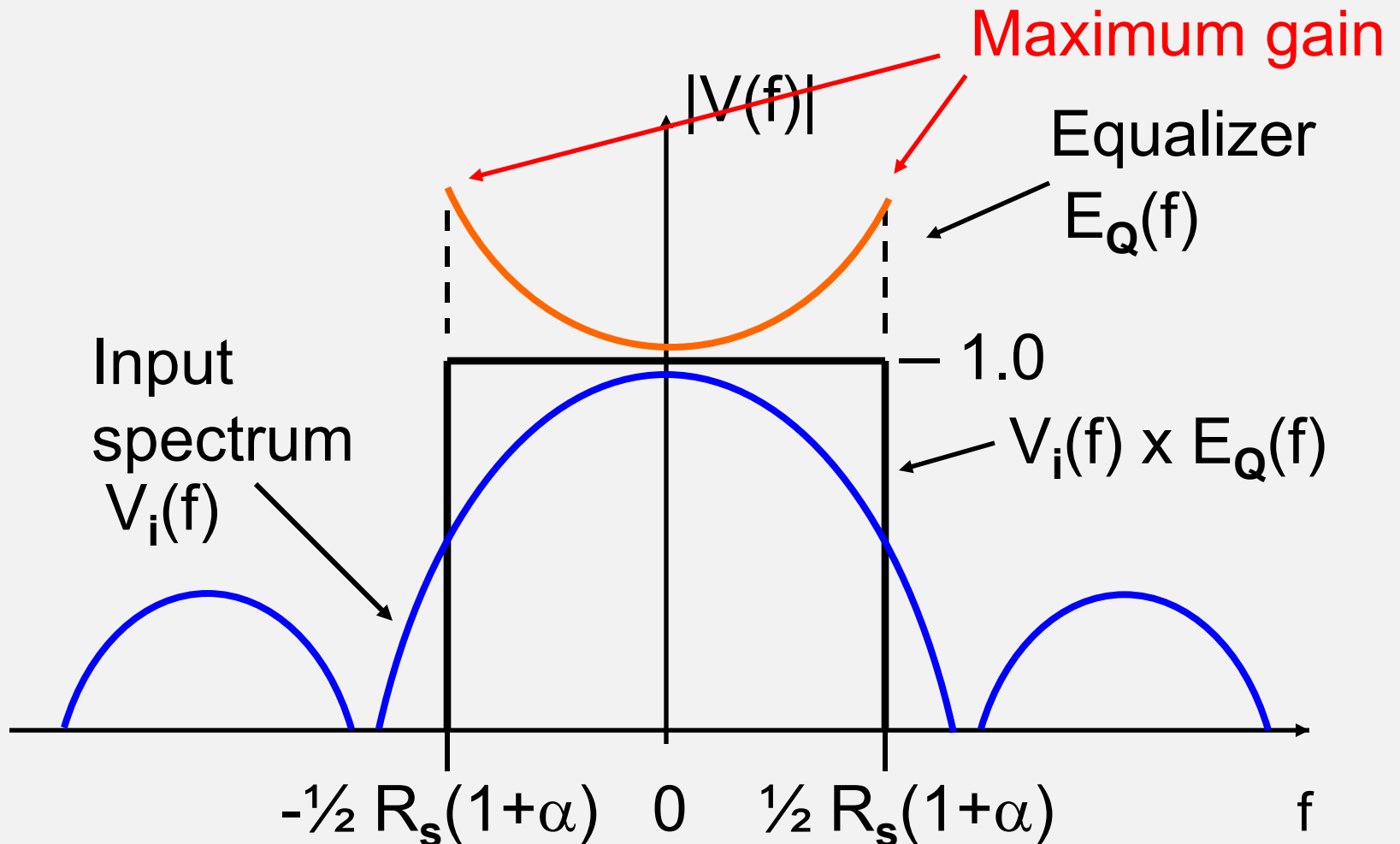
Transmitter Bandpass Filter - Baseband



Exercise 4

- Transmitter has RRC filter with $\alpha = 0.5$
- Transmit equalizer has $|G(f)| = \pi f T / \sin(\pi f T)$
- What is maximum gain of the equalizer (approximately)?

Input equalizer



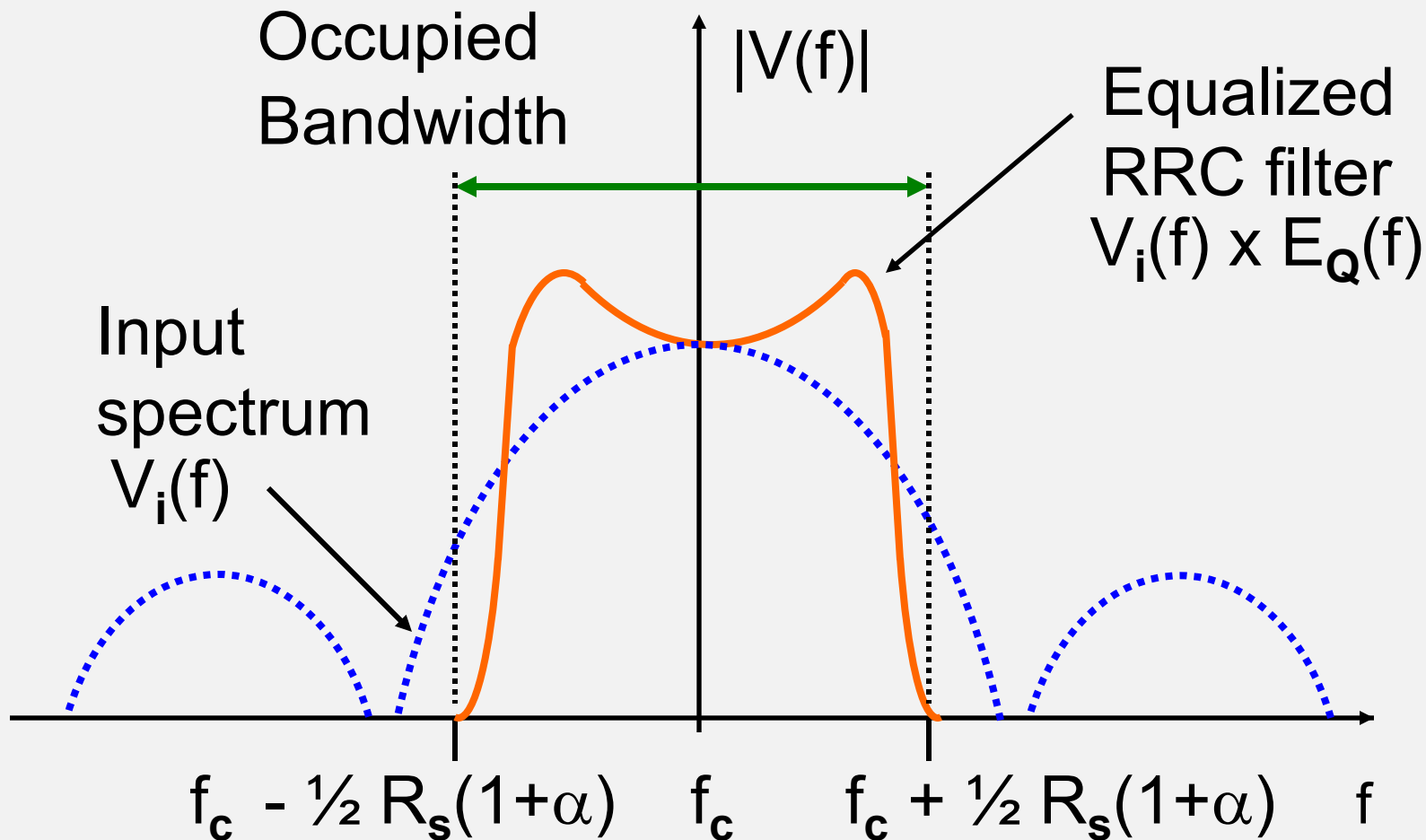
Exercise 4

- Transmitter has RRC filter with $\alpha = 0.5$
- Transmit equalizer has $|G(f)| = \pi f T / \sin(\pi f T)$
- What is maximum gain of the equalizer (approximately)?
- $T = 1 / R_s$
- Edge of transmission band is at $\frac{1}{2} R_s(1+\alpha)$ Hz for baseband link
- So $(\pi f T) = \pi \frac{1}{2} R_s(1+0.5)1/R_s = 0.75 \pi$ and
 $G = 10 \log_{10}[(0.75 \pi / \sin(0.75 \pi))^2] = 10.5 \text{ dB}$

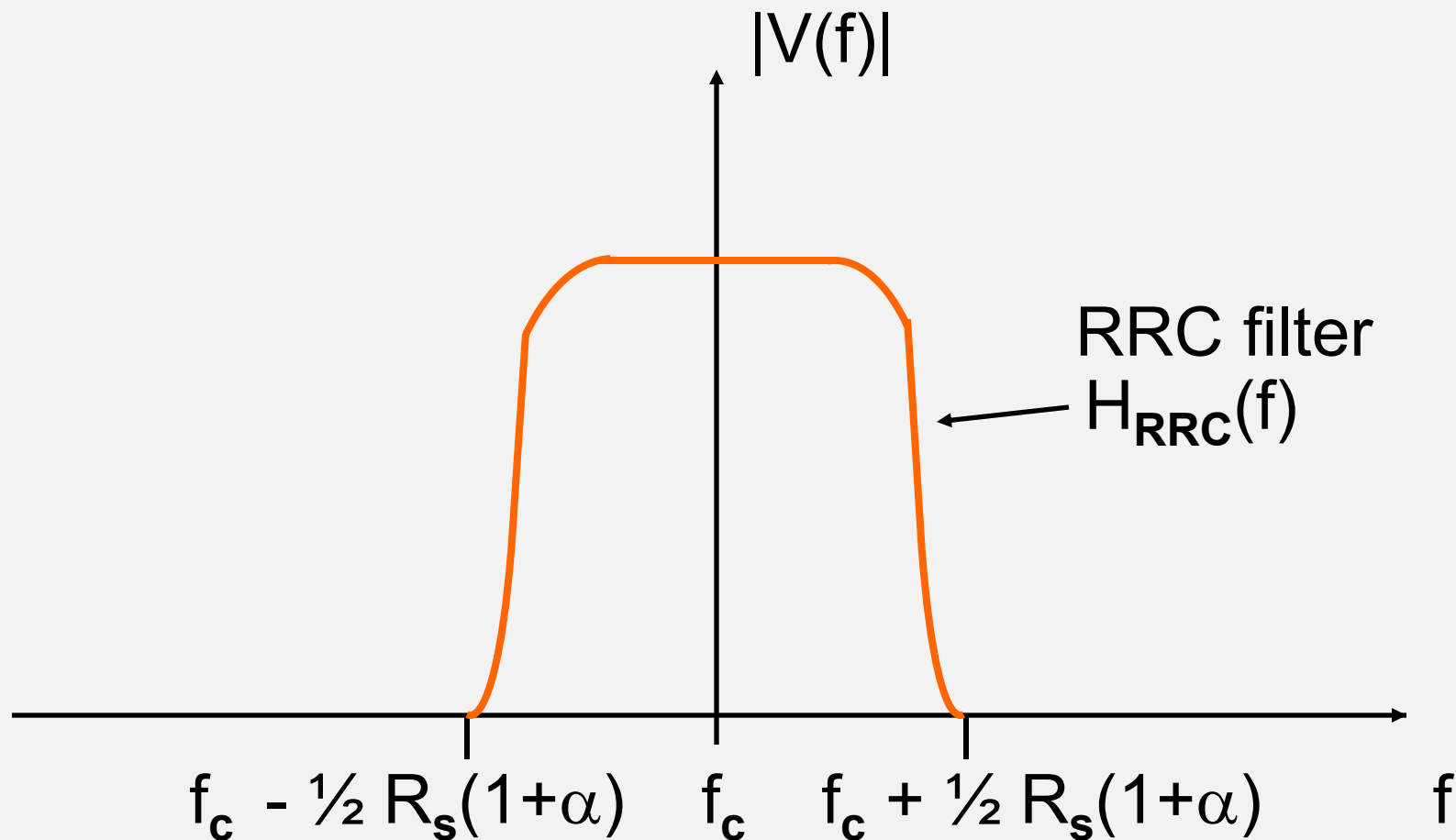
RF data link / RRC filter

- Input spectrum of modulated NRZ symbol is $\sin X / X$
- RRC filter in transmitter requires equalization
- For an RF data link equalization is implemented at intermediate frequency (IF)
- Overall link transfer function still must be a Nyquist Raised Cosine TF

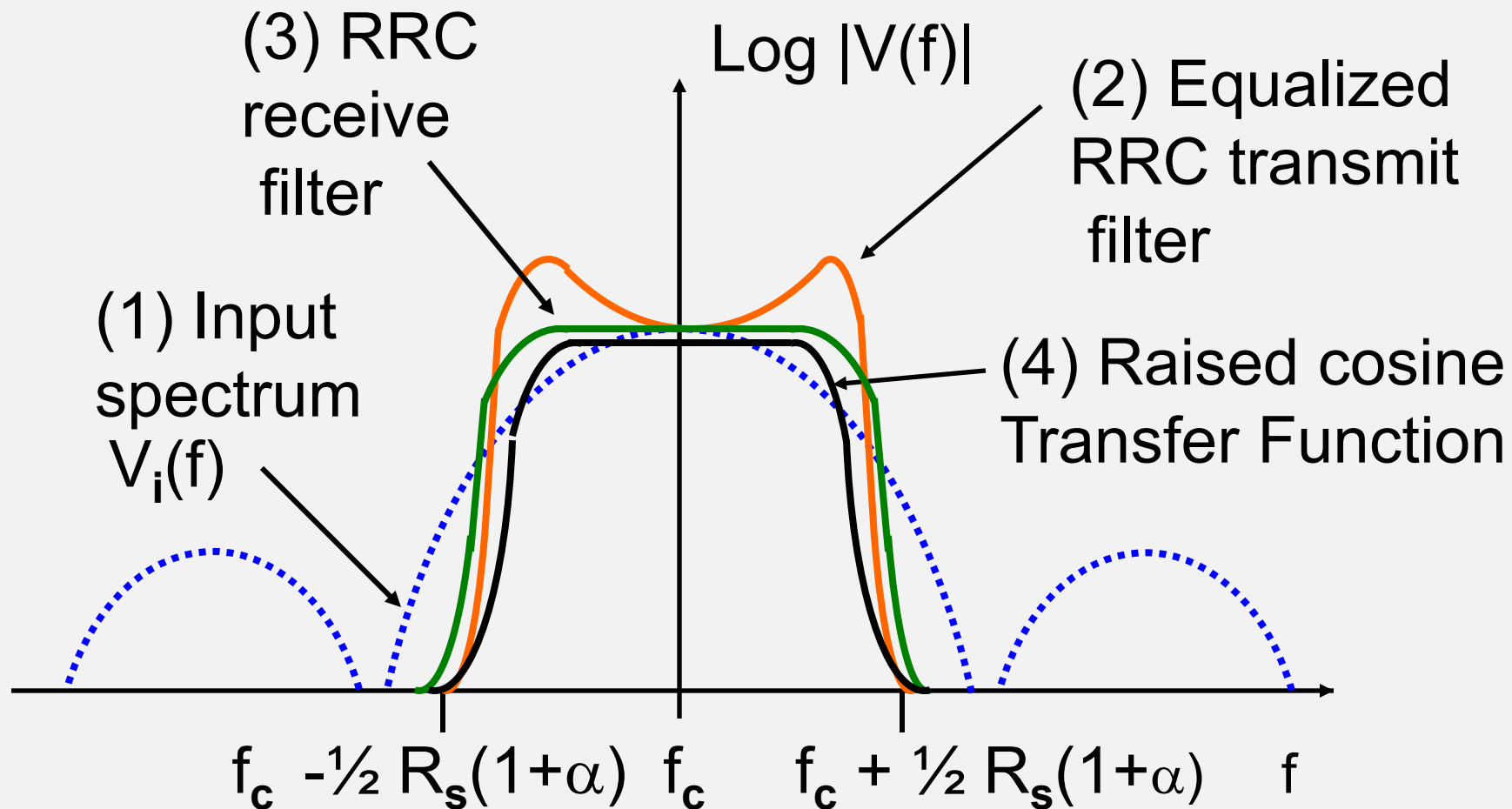
Transmitter Bandpass Filter - RF



Receiver Bandpass Filter



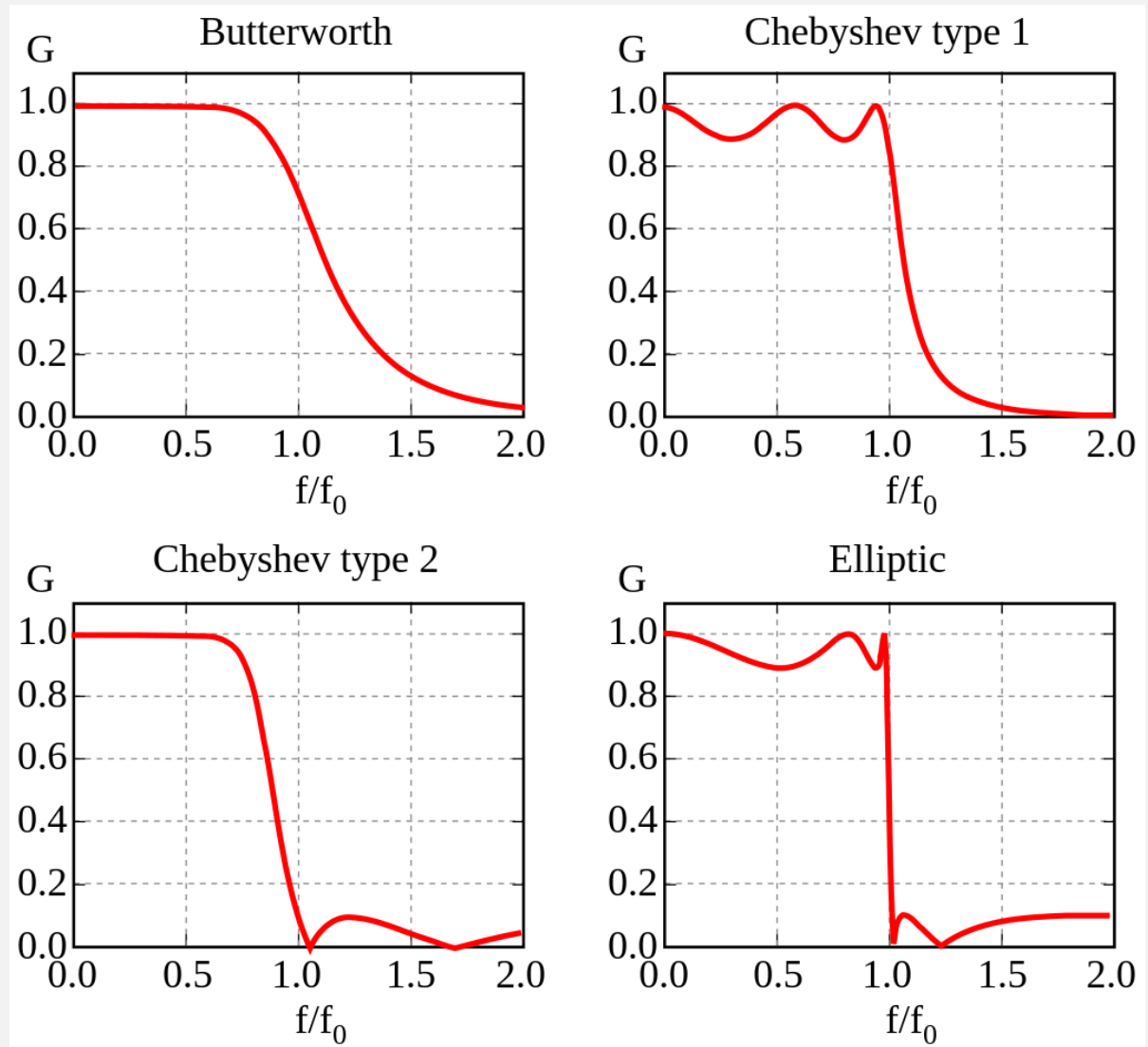
Complete Link Transfer Function



Implementing Zero ISI Designs

- The ideal RRC filter does not exist
- We must use real filters that approximate RRC transfer function
- Butterworth filter with $N = 3$ is good approximation to RRC with $\alpha = 1/2$
- Chebyshev filter has ripples in pass band and can be a good approximation to RRC + $X / \sin X$

Examples of filter response functions



Implementing Zero ISI Designs

- Waveform at receiver output has zeroes displaced from ideal positions = some ISI
- Link will not achieve theoretical BER (Bit Error Rate) - we must add *implementation margin* to theoretical C/N value
- That is, boost the C/N by ~ 1 dB to realize desired BER
- Will need to track and minimize the rate at which bit errors occur on the link due to noise

Summary (1/2)

- All links have finite bandwidth
- Pulse transmission must be designed to give zero ISI waveform at receiver
- Use RRC filters in transmitter and receiver
- Use equalizers to flatten link TF
- Bandwidth occupied by baseband signal is

$$B_{\text{occ}} = \frac{1}{2} R_s (1 + \alpha)$$

- Minimum possible bandwidth with $\alpha = 0$

Summary (2/2)

- Bandwidth occupied by RF (bandpass) signal is

$$B_{\text{occ}} = R_s (1 + \alpha) \text{ Hz}$$

where $0 \leq \alpha \leq 1$

- Noise bandwidth for RF channel is always equal to the symbol rate R_s in Hertz, regardless of value of α .
- Reminder: $B_N = R_s$ for **ALL** RF RRC filters

Digital Transmission on Satellite Links

- Baseband signal must be impressed on the RF carrier using digital modulation
- On satellite links *phase modulation* is almost universal
- Digital phase modulation is called *Phase Shift Keying* (PSK)

BPSK and QPSK

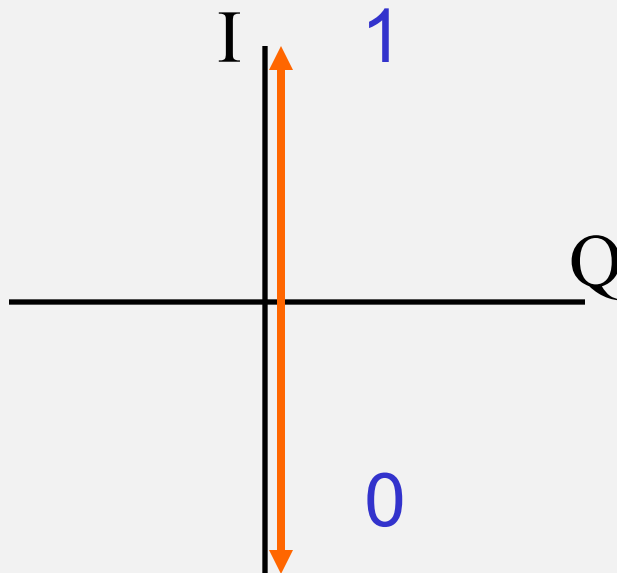
- In PSK a symbol corresponds to a specific phase state of the carrier
- A symbol can carry more than one bit
- BPSK = Binary phase shift keying
- QPSK = Quadrature phase shift keying
- BPSK symbols have 1 bit per symbol
- QPSK symbols have 2 bits per symbol

Digital Modulation

- The baseband input consists of a serial stream of bits (logical states are 0 and 1)
- In BPSK the logical states are converted one bit at a time into one of *two* RF carrier phase states (symbols), e.g., 0° and 180°
- In QPSK the logical states are converted two bits at a time into one of *four* RF carrier phase states (symbols)

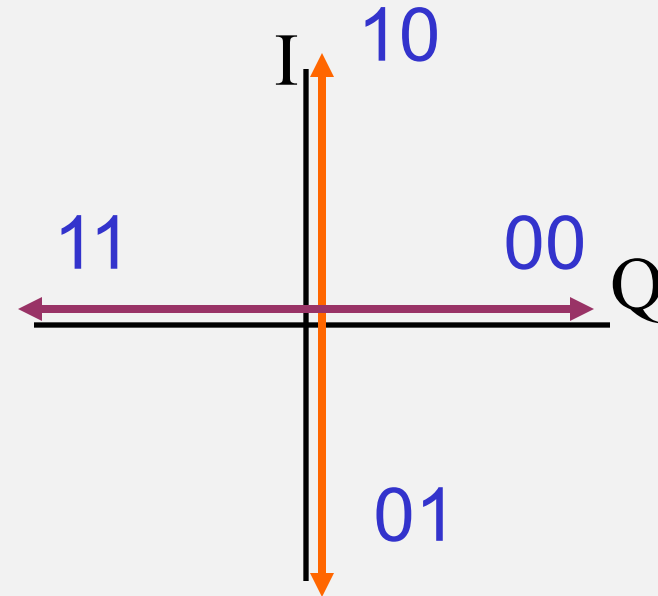
BPSK and QPSK

BPSK



Two phases
= 1 bit / symbol

QPSK



Four phases
= 2 bits / symbol

Digital Modulation

- Note the distinction between bit rate (bps) and symbol rate (sps or baud): $\text{bps} \geq \text{sps}$
- The set of all symbols for a particular type of modulation is called an *alphabet*
- Relation between number of symbols, m , and number of bits carried per symbol, N , is

$$m = 2^N$$

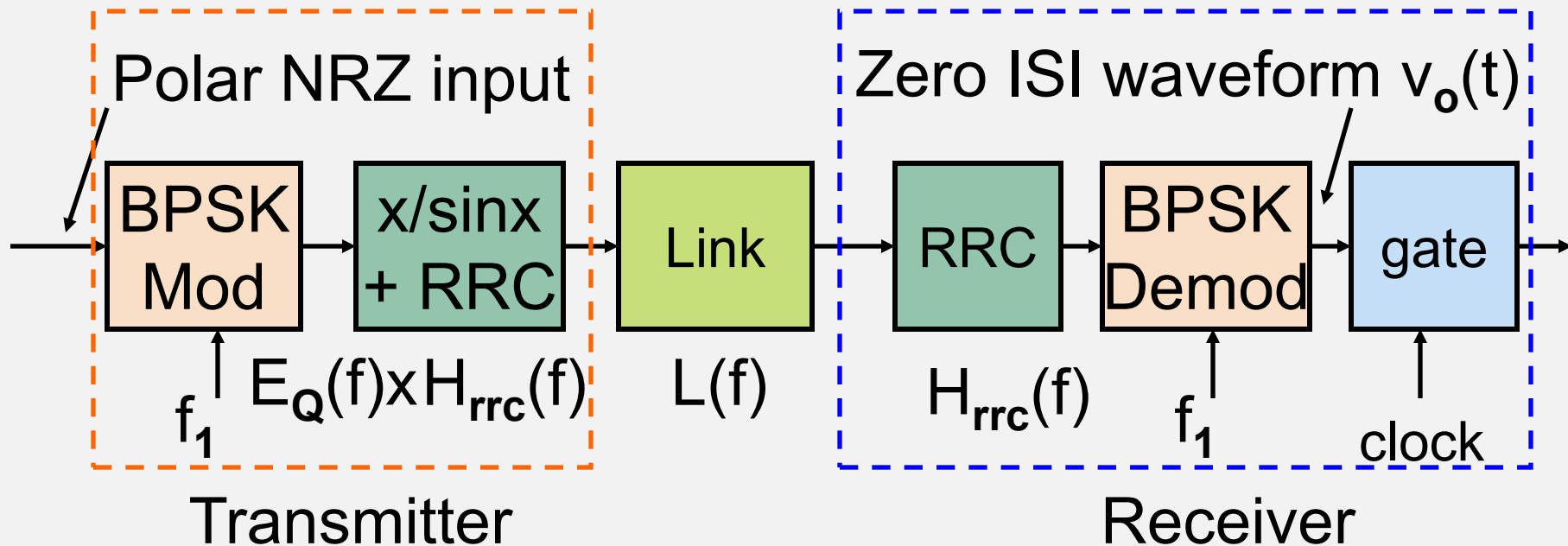
BPSK and QPSK - Overview

- BPSK has lowest C/N requirement for a given BER (Bit Error Rate)
- That is, we can achieve the desired BER with the least powerful (and least expensive) link design
- QPSK sends twice as much data as BPSK in same bandwidth - but requires 3 dB higher C/N for same BER
- QPSK is a popular choice for satellite links

BPSK Modulation Design

- Begin with bi-phase (BPSK) modulation
- Need to impose the modulation on the carrier wave (could be imposed at IF stage) at the transmit end of the link
- Need to demodulate the signal at the receive end of the link
- Add a BPSK modulator to the Tx end and a BPSK demodulator to the Rx end

Generic BPSK Link



Note: BPSK modulator and demodulator are multipliers

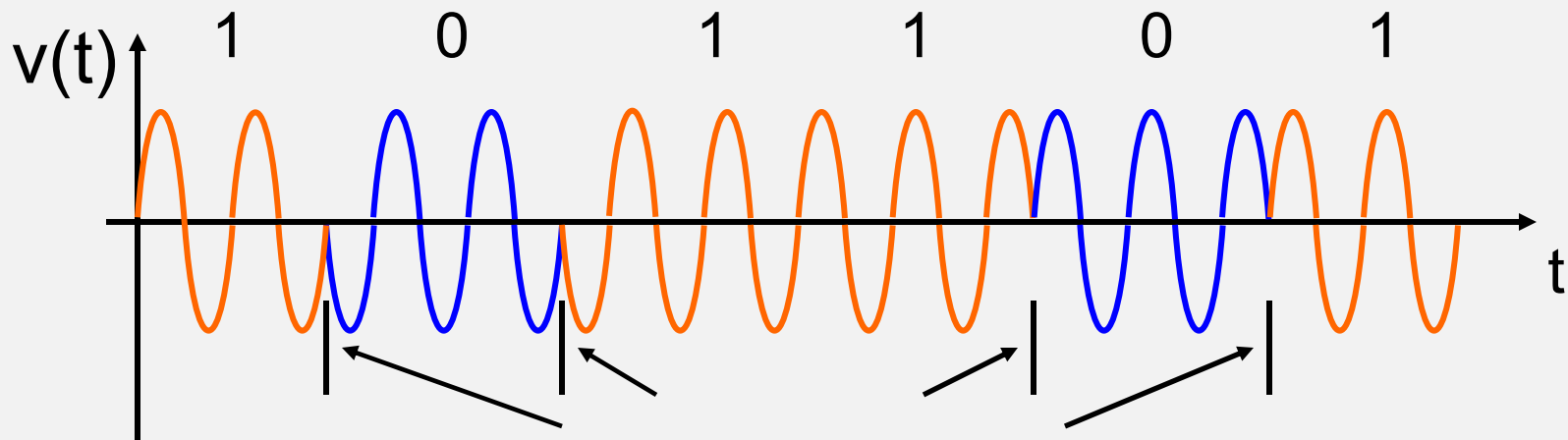
BPSK

- Modulated IF (and RF) signal has form (unfiltered)

$$v(t) = U_i(t) \times V_{IF} \sin \omega t$$

$$\omega = 2 \pi f = \text{angular IF frequency}$$

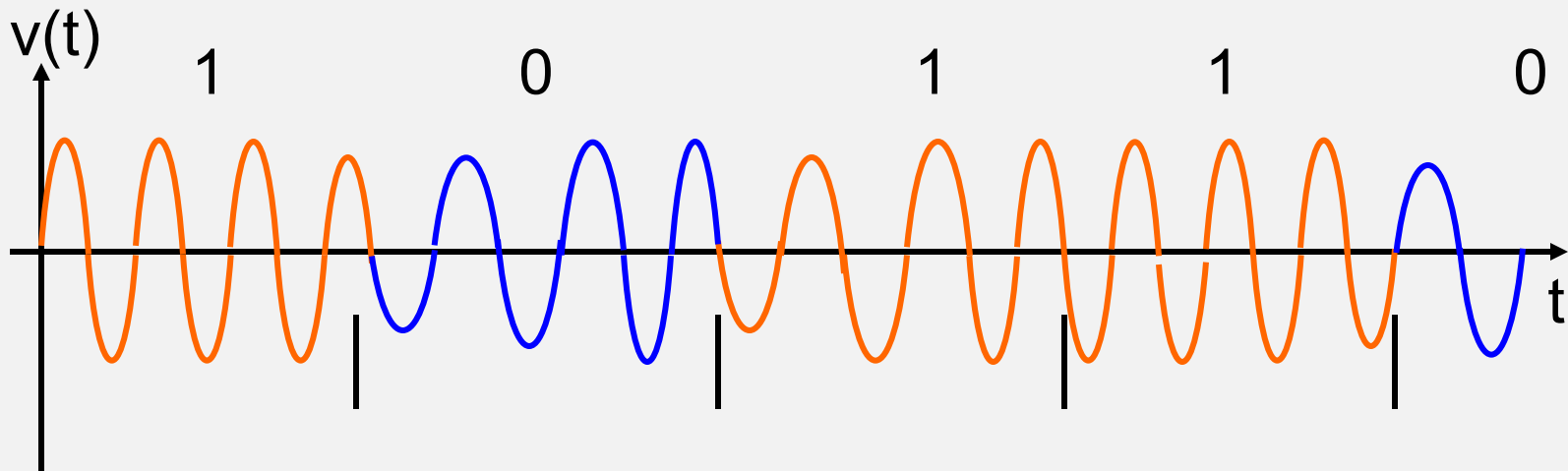
$$U_i(t) = +1 \text{ or } -1 \text{ (modulating bit)}$$



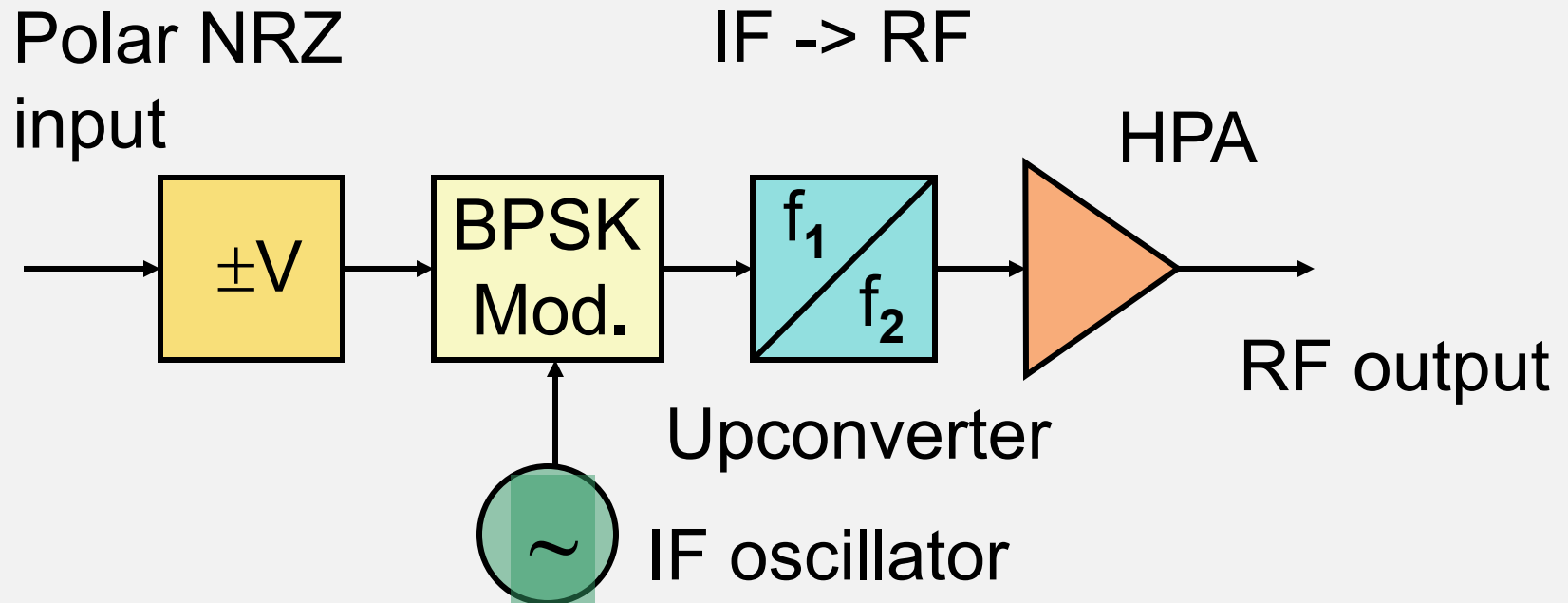
Phase reversals

Filtered BPSK

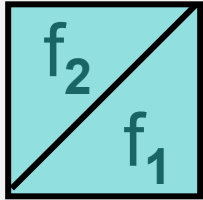
- Filtering removes high frequencies associated with sudden jumps in phase
- Waveform changes phase over a symbol period
- Some AM is introduced



BPSK Modulator

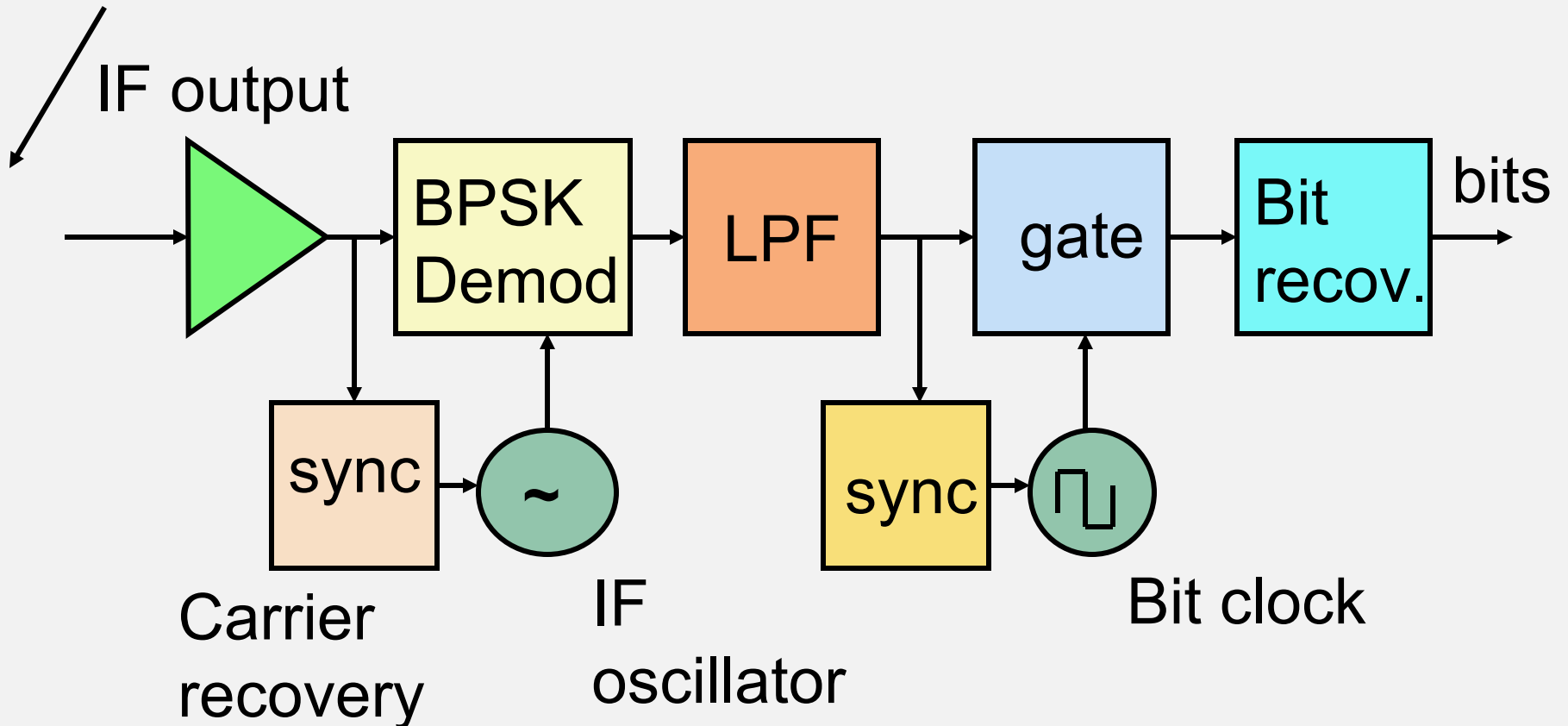


BPSK Modulator is a multiplier usually implemented at IF



BPSK Demodulator

Downconverter (RF $\rightarrow$ IF)



BPSK Demodulation

- BPSK IF signal has form

$$v_{\text{IF}}(t) = U_i(t) \times V_{\text{IF}} \sin(\omega t) \quad (\sim \text{Eqn 6.5})$$

$$\text{where } U_i(t) = +1 \text{ or } -1$$

- Multiply in receiver with in-phase local signal

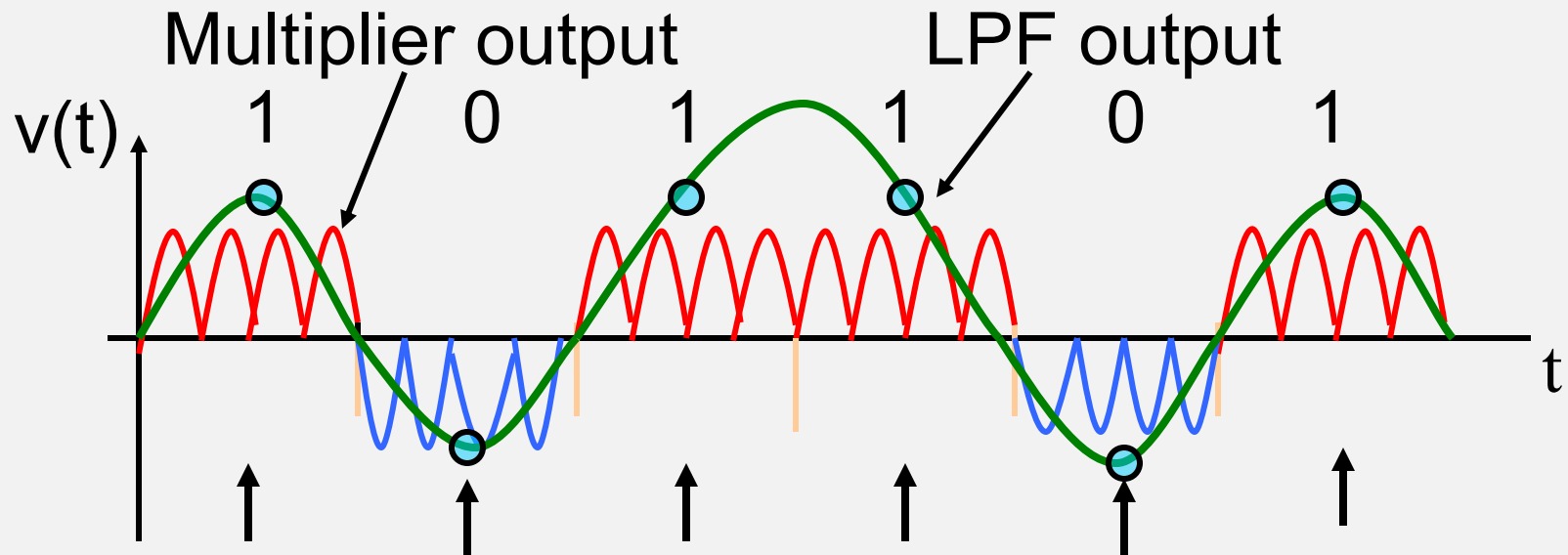
$$v_{\text{LO}}(t) = \sin(\omega t)$$

- Product is ($\sim$ Eqn 6.6)

$$\begin{aligned} v_o(t) &= U_i(t) \times V_{\text{IF}} \sin(\omega t) \times \sin(\omega t) \\ &= U_i(t) \times V_{\text{IF}} \times \frac{1}{2} [1 - \cos(2\omega t)] \end{aligned}$$

BPSK Demodulation

- Signal after multiplier / mixer has form
$$v_o(t) = U_i(t) \times V_{IF} \times \frac{1}{2} [1 - \cos(2\omega t)]$$
$$\omega = 2\pi f = \text{angular IF frequency}$$



Digital Modulation

- After the LPF we have

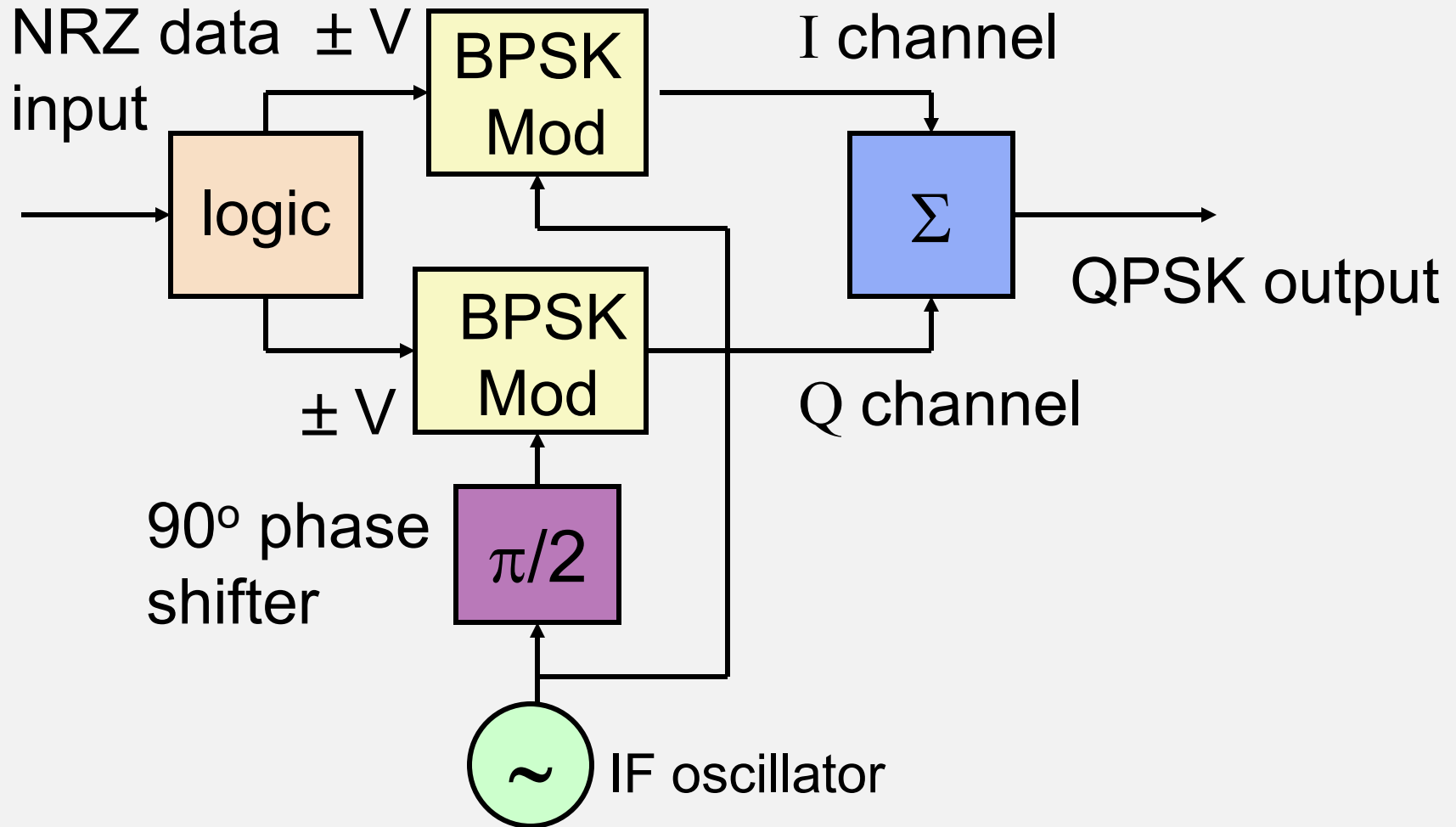
$$v_r(t) = \frac{1}{2} U_i(t) \times V_{IF}$$

- We have recovered the baseband signal, $U_i(t)$
- The bit recovery circuitry that follows the LPF will output the actual values of the bits
- Subject to the impact of ISI and noise (we might get bit errors)

QPSK Modulation

- BPSK has two phase states (0 and π rad) = 1 bit symbol
- QPSK has four phase states = ($\pi/4$, $3\pi/4$, $5\pi/4$, $7\pi/4$ rad) = 2-bit symbol
- QPSK signal can be generated from two BPSK channels that have carrier waves shifted in phase by 90° , i.e., in phase quadrature
- The two channels are distinguished as I-channel (*In-phase*) and Q channel (*Quadrature*)

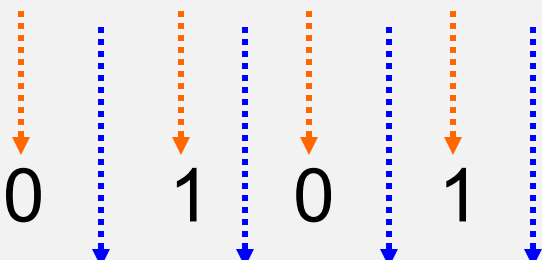
QPSK Modulator



QPSK Modulator

- Logic unit takes two bits at a time
- First bit goes to I modulator (first channel)
- Second bit goes to Q modulator (second channel)
- IF signal in Q channel is shifted by $\pi/2$ rad from IF signal in I channel
- Signals from the two (orthogonal) channels are then summed for transmission

QPSK Modulator

- The bits are sampled alternatively by the I and Q modulators
 - And are transmitted simultaneously
 - Data sequence 0 0 1 1 0 1 1 0 at R_b bps
 - I channel 0 1 0 1 at $R_b/2$ bps
 - Q channel 0 1 1 0 at $R_b/2$ bps
- 

QPSK Modulation

- A general equation for QPSK is (Eqn 6.11)

$$v_{\text{QPSK}}(t) = V \sqrt{2} \cos(\omega_c t - \phi)$$

where ϕ has values of $\pi/4$, $3\pi/4$, $5\pi/4$ or $7\pi/4$ rad

- Expanding into Eqn 6.12:

$$v_{\text{QPSK}}(t) = V\sqrt{2} [\cos \omega_c t \cos \phi + \sin \omega_c t \sin \phi]$$

- First term is in phase with carrier, second is in phase quadrature (shifted by 90°)

QPSK Modulation

- Hence we can write QPSK waveform as

$$v_{\text{QPSK}}(t) = U_I(t) V \cos \omega t + U_Q(t) V \sin \omega t$$

where

$$U_I = \sqrt{2} \cos \phi \quad \text{and} \quad U_Q = \sqrt{2} \sin \phi$$

are the modulating bits for the in-phase and quadrature channels (values: +1, -1)

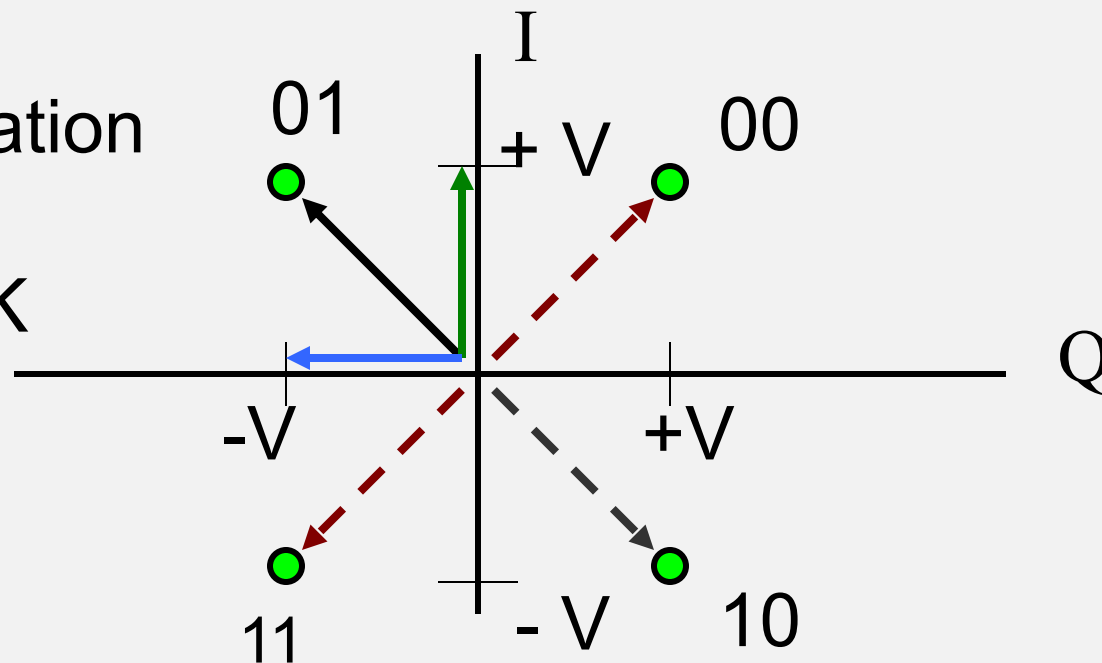
QPSK Modulator

- Input data 0 0 1 1 0 1 1 0
- I channel bits 0 1 0 1
- I Output (volts) +V -V +V -V
- Q channel 0 1 1 0
- Q Output (volts) +V -V -V +V
- QPSK Phase $\pi/4$ $5\pi/4$ $3\pi/4$ $7\pi/4$
- Can be seen from constellation diagram

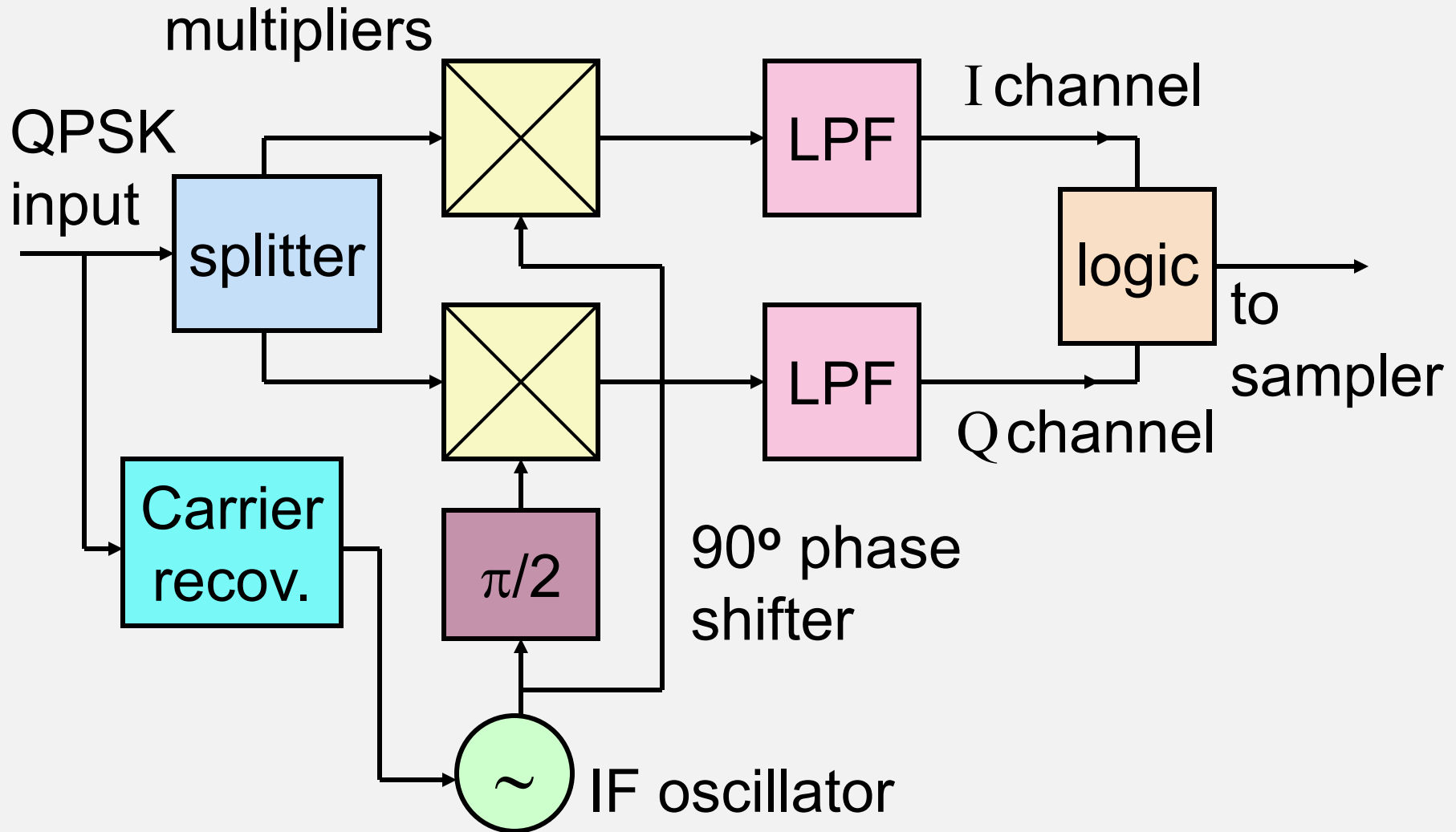
Constellation Diagrams

- Constellation diagram shows the tip of the I and Q phasor

Constellation
Diagram
for QPSK



QPSK Demodulator



QPSK Demodulation

- QPSK signal at IF stage is

$$v_{\text{QPSK}}(t) = U_I(t) V_{\text{IF}} \cos \omega t + U_Q(t) V_{\text{IF}} \sin \omega t$$

- Apply same signal to two multipliers
- I channel multiplier is driven by in-phase LO

$$v_{\text{ILO}}(t) = \cos \omega t$$

- Q channel multiplier driven by quadrature LO

$$v_{\text{QLO}}(t) = \cos (\omega t + \pi/2) = \sin \omega t$$

QPSK Demodulation

- I channel multiplier output is

$$\begin{aligned}v_I(t) &= [U_I(t) V_{IF} \cos \omega t + U_Q(t) V_{IF} \sin \omega t] \times \mathbf{\cos \omega t} \\&= U_I(t) V_{IF} \cos \omega t \cos \omega t \\&\quad + U_Q(t) V_{IF} \sin \omega t \cos \omega t \\&= U_I(t) V_{IF} \frac{1}{2} (1 + \cos 2\omega t) \\&\quad + U_Q(t) V_{IF} \frac{1}{2} \sin 2\omega t\end{aligned}$$

LPF removes double frequency terms

QPSK Demodulation

- LPF output of I channel is

$$v_I(t) = \frac{1}{2} U_I(t) V_{IF}$$

This is the original I channel input $U_I(t)$ multiplied by $\frac{1}{2} V_{IF}$

- We have recovered $U_I(t)$

QPSK Demodulation

- Q channel multiplier output is

$$\begin{aligned}v_Q(t) &= [U_I(t) V_{IF} \cos \omega t \\ &\quad + U_Q(t) V_{IF} \sin \omega t] \times \mathbf{\sin \omega t} \\ &= U_I(t) V_{IF} \cos \omega t \sin \omega t \\ &\quad + U_Q(t) V_{IF} \sin \omega t \sin \omega t \\ &= U_I(t) V_{IF} \frac{1}{2} \sin 2\omega t \\ &\quad + U_Q(t) V_{IF} \frac{1}{2} (1 - \cos 2\omega t)\end{aligned}$$

- I channel LPF removes double frequency terms

QPSK Demodulation

- LPF output of Q channel is

$$v_I(t) = \frac{1}{2} U_Q(t) V_{IF}$$

This is the original I channel input $U_Q(t)$ multiplied by $\frac{1}{2} V_{IF}$

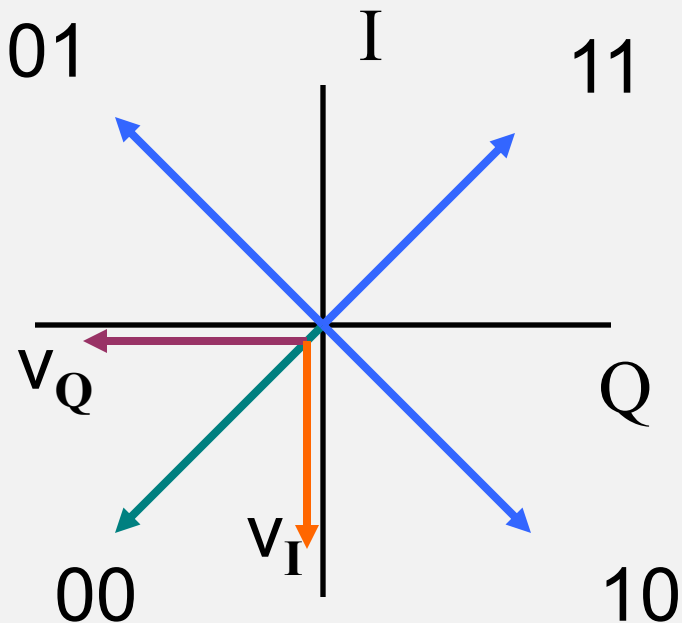
- We have recovered $U_Q(t)$
- Summing: we have recovered the values of the modulating bits (two at a time)

QPSK Modulation in practice

- QPSK signal is generated as sum of I and Q channel signals
- Logic circuit drives *both* I and Q modulators for each pair of input data bits
- E.g., 0 0 input drives I and Q channels with -1 signals and

$$\begin{aligned}V_{\text{QPSK}} &= -\cos \omega t - \sin \omega t \\ &= -\sqrt{2} \cos (\omega t + \pi/4)\end{aligned}$$

QPSK generation in practice



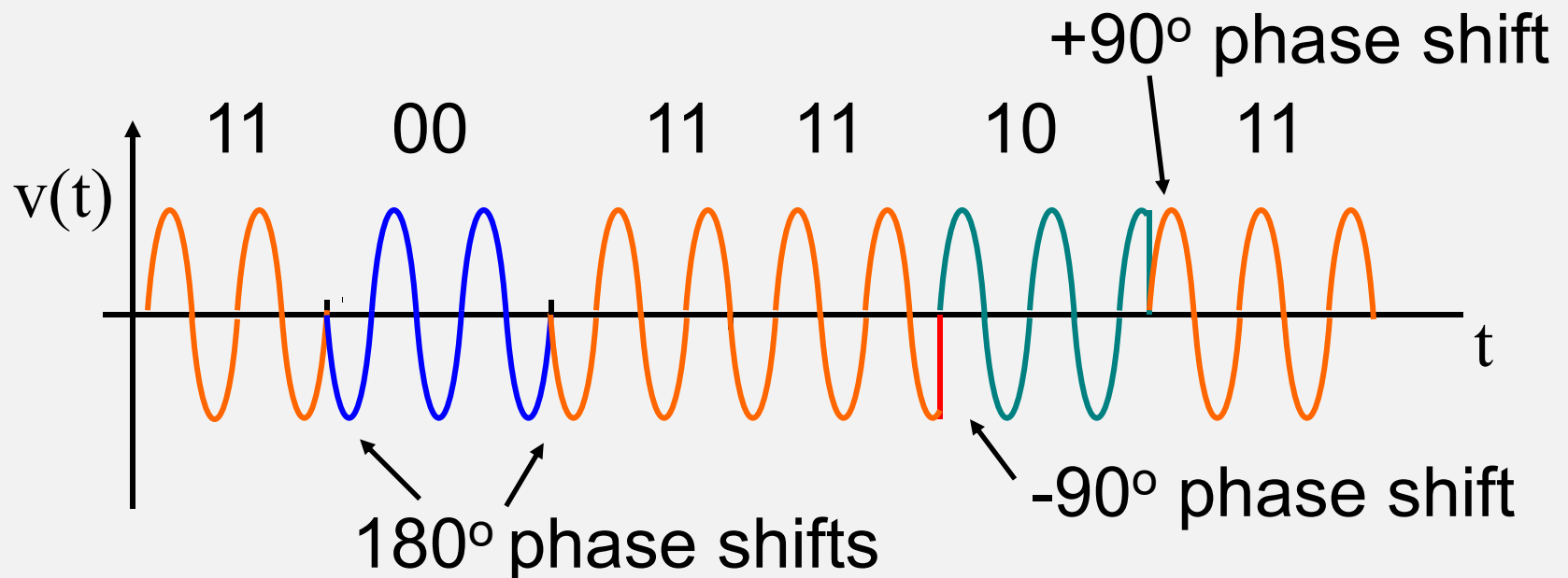
Each phasor is generated by summing I and Q signals
E.g., 00 is generated by $U_I(t) = -1$, $U_Q(t) = -1$

QPSK Waveform

- QPSK signal has form (unfiltered)

$$v_{\text{QPSK}}(t) = U_I(t) \cos \omega t + U_Q(t) \sin \omega t$$

$$U_I(t) \text{ and } U_Q(t) = +1 \text{ or } -1$$

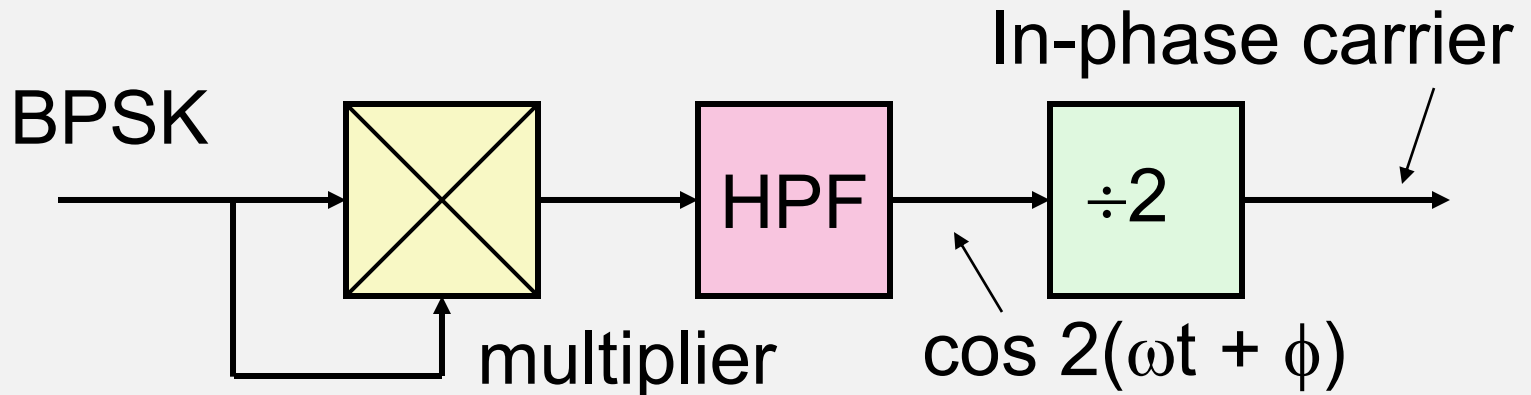


Carrier recovery

- Local oscillator in demodulator needs to be synchronized to received BPSK or QPSK carrier
- Must be derived from the received signal using a *carrier recovery circuit*
- For BPSK apply the received signal to both inputs of a mixer
- Obtain the square of the BPSK signal

Carrier recovery - BPSK

- Squaring circuit



$$\begin{aligned} v_o(t) &= [U_I(t) \times V_{IF} \sin(\omega t)]^2 \\ &= U_I(t)^2 \times V_{IF}^2 [\sin(\omega t)]^2 \\ &= 1 \times \frac{1}{2} V_{IF}^2 [1 - \cos(2\omega t)] \end{aligned}$$

HPF output is $\frac{1}{2} V_{IF}^2 \cos(2\omega t)$

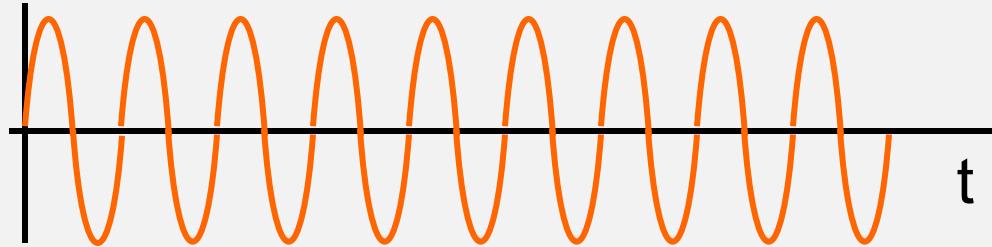
Carrier recovery

- For BPSK, square the signal and divide by two (using a PLL) to obtain in-phase carrier
- For QPSK, square the signal twice and divide by four to obtain in-phase carrier
- Dividing by two in BPSK carrier recovery circuit leads to ambiguity of 180°
- Dividing by four in QPSK carrier recovery circuit leads to ambiguity of $n \times 90^\circ$, $n = 0$ to 3

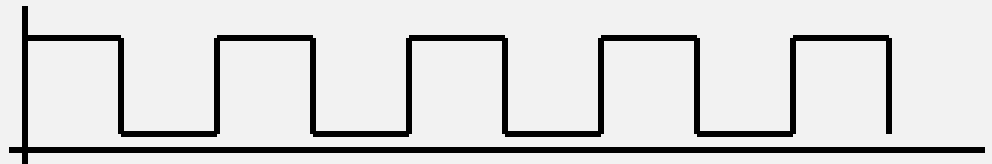
Divide by two ambiguity

With BPSK ambiguity...

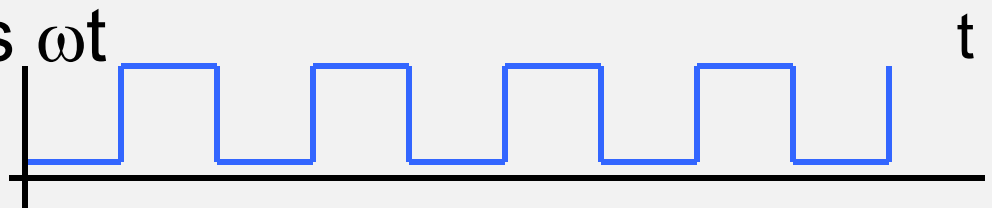
- $\cos 2\omega t$



$\div 2$

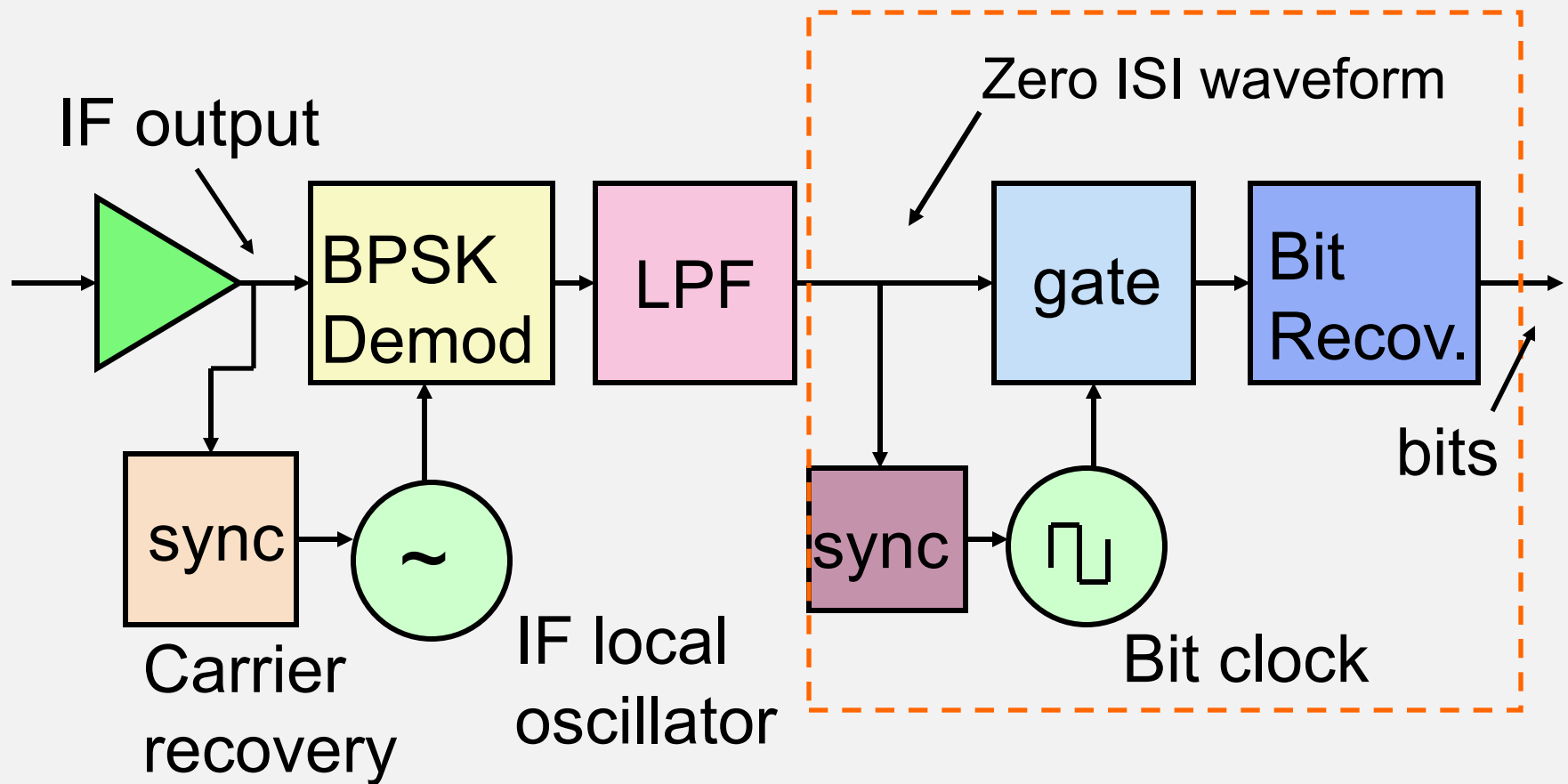


$\Rightarrow \cos \omega t$



...the complement of the bit stream^t could be produced (resolve with code words in bit stream)

Sampling and Bit Recovery



Summary (1/2)

- Digital transmission via satellite can use BPSK or QPSK
- Input to transmitter is NRZ rectangular pulses
- BPSK modulator is a multiplier
- BPSK demodulator multiplies by in-phase carrier
- Carrier and bit recovery is needed in receiver

Summary (2/2)

- QPSK transmits two BPSK signals in phase quadrature
- QPSK signal carries two bits per symbol
- Can send same data rate as BPSK in half the bandwidth (but requires twice the power)
- QPSK demodulator is two BPSK demodulators in phase quadrature
- Carrier and bit recovery is needed in receiver

